

The European nuclear landscape

27 March 2026

Nuclear power today and ahead



Richard Bexelius
+46-8 763 97 24
richard.bexelius@seb.se

Contents

	Page
Introduction	3
Executive summary	3
Global outlook for nuclear power	6
Outlook	7
The evolution of nuclear power: Cycles of promise and setback	11
Key drivers and enablers for new nuclear – a European perspective	12
What are the challenges?	14
Key hurdles for developing new nuclear power and solutions	16
Technology development	19
Small modular reactors (SMR)	22
Levelised cost of energy (LCOE)	25
Nuclear as a sustainable investment.....	29
The fuel supply chain	30
Decommissioning and waste management.....	32
Outlook by geography	35
Nuclear optimism in the Nordics.....	37
Concluding remarks	57

Introduction

Is this the moment of nuclear power's revival?

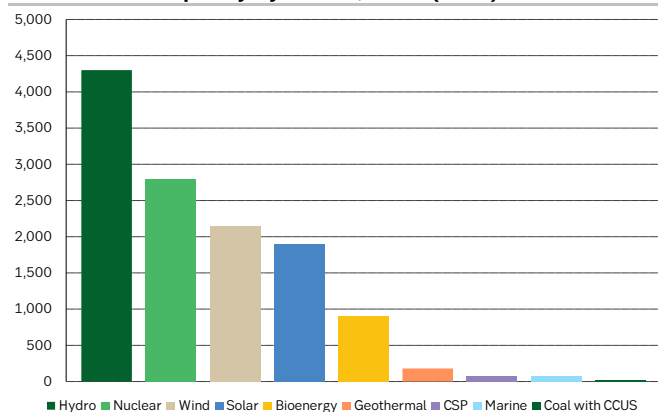
The energy landscape has undergone myriad changes following geopolitical strife, rapid technological development and deployment, as well as in consumption patterns, and we have decided to take a closer look at the nuclear power option. Is the revival of nuclear power now at hand, or is this another false dawn, and what challenges would need to be overcome to grow nuclear capacity, even as its share of overall power output has fallen in recent years? These are the questions we contemplate in this report.

Executive summary

Key drivers for new nuclear power are energy security, stable baseload capacity, low-emission policies and the age of the current nuclear fleet

The European nuclear outlook is shifting out of a period of stagnation towards a cautious revitalisation phase, but one defined far more by execution, regulatory clarity and financing models than by technological promises. The macro case is well-known: energy security after Russia's invasion of Ukraine, decarbonisation targets, and structurally rising electricity demand driven by electrification and the exponential growth in data centres. Against that backdrop, nuclear retains a clear systemic role providing firm, low-carbon capacity with a compact footprint and very high-capacity factors.

Low-emission capacity by source, 2020 (TWh)



Source: IEA

In the near term, Europe remains a replacement and life-extension market. Half the world's reactors are over 30 years old and Europe's fleet skews older still, with the EU average around the high-30s and several countries managing 40-50-year old assets. Life extensions are therefore the cheapest carbon abatement option available to many systems and, from a credit perspective, the most attractive: low capex intensity, known assets, established operating teams, and immediate revenue continuity. Numerous governments have quietly moved to extend unit lifetimes; France's "Grand Carénage" provides the template, while Belgium, Finland, Spain, the Netherlands, Hungary, Romania and others are enacting or studying similar measures. These programmes

reduce near-term supply risk and defer large replacement capex. The primary credit watchpoints are outage duration, maintenance scope creep, and emergent materials issues, France's stress corrosion experience is a reminder that aging fleets can still surprise on both cost and availability. Even so, the levelised cost of energy (LCOE) for life-extended assets typically sits well below the market clearing price in fossil-importing Europe and well below new-build alternatives, preserving healthy interest coverage if outages are managed.

New initiatives are added to new capacity under construction

New build projects are re-emerging in Europe, but investors will price projects largely off delivery risk rather than technological ambition. The empirical record is plain: the first wave of Generation III+/EPR/AP1000 projects in the West suffered long construction durations and significant cost overruns. Flamanville 3's grid connection in December 2024, more than 12 years over its original target and at a final cost north of EUR 13bn, in addition to new grid connections and sharply increased financing costs, underscores that schedule risk translates directly into interest-during-construction and leverage strain. Regulatory frameworks are evolving towards models that remove a substantial portion of this risk from utility balance sheets and place it within more appropriate public or shared-risk structures. The UK's pairing of a contract for difference (CfD) at Hinkley Point C with a Regulated Asset Base (RAB) model for Sizewell C, and Sweden's proposed state loans covering up to 75% of capex combined with CfD-style revenue floors, indicate a trend: public co-investment, price stabilisation and consumer-funded risk sharing are prerequisites for bankability. When these safeguards exist, new nuclear can attain investment-grade quality on its own or within a ring-fenced vehicle; in their absence, utilities face construction and merchant risks that place undue pressure on their balance sheets.

Risk sharing is essential to enabling new nuclear capacity

Small modular reactors (SMRs) may shape the next decade, but they should be approached conservatively in credit models until first-of-a-kind units validate both schedule reliability and the expected standardised cost reductions.

SMRs show promise but have yet to be implemented

The theoretical advantages – factory fabrication, modularity, lower site risk, and better scalability – map well to financiers' preferences: smaller ticket sizes, repeatable scope, and the prospect of learning curves steeper than for bespoke gigawatt-scale plants. Yet the learning-rate dividend materialises only with multi-unit series and streamlined licensing. In Europe, credible first movers exist, such as the UK SMR programme, Poland's industrial offtake-led pipeline, and Sweden's Ringhals shortlist, and several large technology buyers are entering into long-tenor PPAs to anchor projects. Through a credit lens, the key is not technology variety but revenue visibility and counterparty strength. Long-dated, inflation-linked PPAs with investment-grade hyperscalers or utilities can materially de-risk cash flows. However, until module manufacturing,

multi-site licensing and fuel supply (e.g., HALEU for some advanced designs) are demonstrably bankable, SMRs should be modelled with conservative CODs, contingency on EPC, and state backstops.

The competitiveness picture is more nuanced than simple LCOE comparisons suggest, and a system-value perspective is essential. While new European nuclear typically sits above wind and solar on headline cost metrics, and utility-scale PV, onshore wind, and batteries continue to fall in cost, renewables-heavy systems are already experiencing some disadvantages in the current form. Negative price periods are growing, wind capture rates are deteriorating in some saturated zones, while grid and land constraints are tightening, all of which complicate the dynamics of the energy markets.

Nuclear's high-capacity factor and firm output provide system value not captured in LCOE tables. In carbon-priced Europe, where gas can still clear marginally above EUR 100/MWh, even capital-heavy nuclear can sit "in the money" under CfD or RAB regimes. For credit, the takeaway is that nuclear competitiveness is policy-driven when firm low-carbon capacity is remunerated through regulated returns or CfDs, thus nuclear assets can deliver stable cash flows despite unfavourable merchant LCOE comparisons.

Different countries have varying strategies, if at all, to operate nuclear power

European countries are taking different paths in nuclear energy, yet the overall direction remains broadly supportive. France has renewed its commitment through the EPR2 programme and efforts to streamline regulation, although the performance of its aging fleet, and the pace of refurbishments, will continue to shape near-term cash generation. The United Kingdom is also reinforcing its nuclear strategy, allocating further funding to Sizewell C and advancing a structured SMR programme, even as it openly manages the gradual retirement of its older advanced gas-cooled reactors (AGRs).

Across Central and Eastern Europe, countries such as Czechia, Slovakia, Hungary, Romania, and particularly Poland view nuclear power as essential to replacing coal and strengthening energy security. Their emerging financing models rely heavily on state-backed credit, export-credit agency support, and long-term offtake arrangements to underpin project economics.

The Nordic region shows a wide spectrum of approaches. Sweden is positioning itself for up to 5GW of new capacity, supported by state financing tools and contracts-for-difference, with Ringhals likely to become one of the first European sites to host an SMR. Finland, meanwhile, has highlighted the value of large-scale nuclear through the stabilising effect of Olkiluoto 3, while at the same time reducing its exposure to Russian suppliers following the cancellation of the Hanhikivi project. Switzerland faces a different set of challenges. It continues to operate one of Europe's oldest reactor fleets, while political debate persists over whether to lift its long-standing ban on new construction.

Fuel sourcing and supply chains to be adapted to reduce sensitivities to energy supply, ensuring energy security

Fuel supply, waste, and decommissioning are no longer peripheral but core credit topics. Since 2022, European utilities have accelerated diversification away from Russian enrichment and fuel services, leaning on Orano, Urenco and North American supply to reduce geopolitical concentration. For new projects, particularly SMRs that may rely on higher-assay fuels, a robust, multi-decade fuel-supply strategy will be essential, as lenders and rating agencies will treat it as a critical prerequisite for financing. On waste, Europe has moved from principle to practice. Sweden and Finland's deep geological repository programmes have materially reduced policy and tail risk perceptions, and France and Switzerland are advancing their own solutions. The existence of funded decommissioning schemes, with ring-fenced assets and regulator-approved accruals, supports credit quality by capping long-tail cash outflows; however, cost inflation and extended lifetimes argue for periodic top-ups and stress-testing of fund adequacy. For utilities, transparent, conservative provisioning remains a hallmark of strong governance.

Nuclear construction and operating capabilities to be ramped up as it has been a low priority for decades

Supply-chain and workforce limitations are both tangible and highly relevant from a credit perspective. Europe's nuclear construction capabilities have eroded over decades, and rebuilding the specialised manufacturing base, welding and quality assurance/control capacity, and the project-management expertise required for nuclear-grade work, will take time. Governments are beginning to respond. France's plan to recruit around 10,000 workers a year over the next decade illustrates the scale of the challenge, but repeated "stop-and-go" policy cycles would risk stranding investments and weakening contractors' financial resilience. For project-level credit analysis, serial construction at a single vendor and genuinely standardised designs are not optional, they are likely the only reliable way to deliver units on schedule and to shift risk from project sponsors to EPC contractors and insurers at acceptable premiums. Lenders will gravitate towards projects with fully secured supply chains, credible timelines (rather than optimistic five-year build assumptions) and liquidated damages structures that are both meaningful and financially absorbable.

Regulatory framework for nuclear operations to be modernised to provide certainty along with funding strategies

Regulatory clarity has strengthened in recent years. The EU's decision to classify nuclear as a transitional activity within its taxonomy, subject to stringent "do no significant harm" criteria, has eased perceived regulatory risk and opened the door for nuclear-related investment to qualify for green or transition-labelled financing where the requirements are met. This matters because the cost of capital is the single most decisive variable in nuclear economics: reducing the weighted

average cost of capital by even 200-400bp through RAB or CfD models, or through taxonomy-aligned financing, can materially reshape project valuations and debt-service metrics. The implication is that nuclear financing is, by design, a public-private undertaking. Government guarantees, export-credit agency support and multilateral participation are not peripheral features, they are the essential mechanisms that close funding gaps and draw private lenders in on terms they can accept.

Development of power demand affects outlook. Demand surge expected, but how large will it be?

A final point concerns demand and offtake. Europe's power demand outlook has meaningful upside risk. AI-driven data centre load growth, rail and road electrification, new electro-intensive industry, and early-stage hydrogen production are all pointing to tighter systems in the early-to-mid-2030s. Several large technology companies are now willing to sign very long-dated, firm, carbon-free PPAs and even to co-invest in nuclear. For credit, these offtakes can provide exactly the duration and counterparty quality that balance sheets and project lenders need, especially if paired with state support mechanisms. The most bankable nuclear projects in Europe over the next decade may be those with a blended revenue stack: regulated base returns, indexed CfD floors, and top-up corporate PPAs that secure upside and hedge curtailment or availability risk. However, the proof is in the pudding, and although all forecasts indicate strong demand growth, it has yet to materialise in a decisive way.

There are several risks and challenges to address

What then are the principal risks from a creditor's perspective? First, construction risk remains the defining variable; even with RAB/CfD structures, schedule slippage compounds financing costs and reputationally impairs future series builds. Second, policy durability must be scrutinised country by country. Nuclear is a multi-decade asset class; shifts in government can inject uncertainty into tariff design, cost recovery, or programme pacing, and that translates into refinancing and covenant risk. Third, supply chain and workforce gaps could cap throughput just as programmes scale, increasing unit costs and extending timelines. Fourth, fuel and technology geopolitics haven't vanished: projects reliant on Russian technology or services face sanctions and delivery risks; those requiring advanced fuels need credible Western production. Fifth, public acceptance remains a latent risk, particularly around siting, lifetime extensions, and waste repositories; delays here are indistinguishable from cost overruns in the credit model.

Lifetime extensions are low risk at a competitive cost of production and will be the most prevalent phenomenon in the foreseeable future

Set against those risks are material opportunities. Life extensions across Europe provide low-risk, high-return decarbonisation and capacity, supporting solid free cash flow and dividends where regulation is stable. If the next generation of projects adopts modern risk-sharing structures, combining RABs, CfDs, government loans, and support from export-credit agencies and multilateral banks (MDBs), they can deliver quasi-regulated returns and reach investment-grade status, provided construction is managed professionally and repeated at scale. SMRs, if they cross the execution threshold, could open a broader investor base thanks to smaller checks, replicability, and industrial offtakes. Finally, nuclear's role as a hedge against renewable capture risk and gas price volatility provides portfolio value to integrated utilities: balanced fleets with hydro, nuclear, and flexible assets can stabilise earnings in systems where merchant renewables face pricing cannibalisation and curtailment.

The most important parts of the outlook, therefore, are less about whether nuclear "wins" against renewables and more about how Europe structures nuclear to do its job, and in combination with other generation sources. In Europe, we believe that credit-positive nuclear will be built around standardised designs rather than bespoke one-offs, supported by public risk-sharing rather than exposed to full merchant risk, and anchored in long-dated revenue commitments from governments or investment-grade corporate off-takers. It will sit within credible, long-term fuel-supply and waste-management strategies, backed by ring-fenced decommissioning funds and transparent governance. And it will rely on supply chains scaled for repeatable, serial construction rather than isolated, heroic builds.

In sum, the European nuclear narrative has flipped from phase-out to pragmatic reintegration. For creditors, the investment outlook looks stronger if the risk picture is balanced. The most compelling opportunities over the next decade lie in life extensions, in a handful of flagships regulated new builds, and in the first commercially disciplined SMR references backed by strong counterparties. The sector's challenges, construction reliability, policy steadiness, supply chain depth, and long-tail liabilities are well known, and the mitigants are also clear. Execution will decide whether this cycle becomes a durable credit story or another false dawn.

Global outlook for nuclear power

With the backdrop of an aging fleet, new nuclear is again discussed and anticipated

Nuclear power has long provided stable, carbon-free electricity and once stood as a symbol of national energy independence. However, a series of accidents – Three Mile Island in 1979, Chernobyl in 1986, and Fukushima in 2011 – severely eroded public confidence, triggering phase-outs in countries such as Germany and Italy. In the years that followed, escalating construction costs, project delays, and competition from cheap gas and renewables weakened the sector, particularly in OECD economies. In addition, mature nuclear markets saw competition from other power generation alternatives as well as plateauing demand following the initial decades-long growth phase, while current nuclear capacity growth is concentrated primarily in the Chinese market.

Energy security and rising demand

Today, the geopolitical landscape is reviving interest in nuclear energy. Growing concerns over energy security, surging electricity demand from AI-driven data centre and transport electrification, and new private-sector investment are reshaping the debate. Small modular reactors (SMRs) in particular promise reduced costs and project risks, though the industry remains dependent on strong government oversight and regulatory stability.

Global nuclear generation is projected by the IEA to increase by around 57% by 2035, before moderating as older plants retire. A new growth wave is expected from the late 2040s onward, driven by electricity demand from AI, transport, and hydrogen production, along with the commercial rollout of SMRs. By 2060, global nuclear output could reach 6,500TWh, about 150% higher than today, keeping pace with global electricity demand growth.

Near-term capacity to come from large-scale reactors

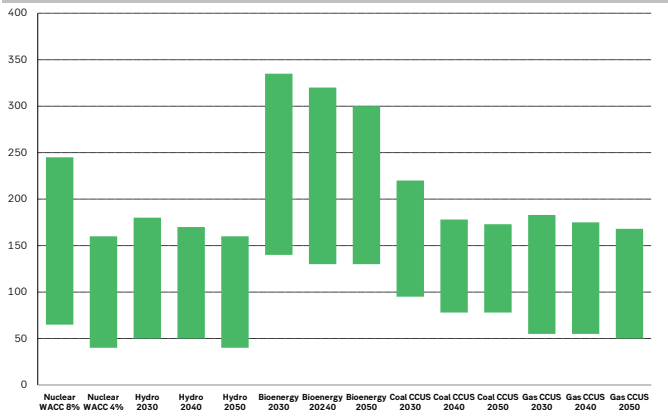
In the near term, most new capacity will come from large, site-built reactors already in development. From 2040 onward, we expect growth to increasingly shift towards SMRs, reaching 70% of new additions between 2050 and 2060, as factory-built designs achieve cost reductions through learning and scale.

Generation costs vary for each region and recurrence affects unit costs

An IEA (International Energy Agency) cost analysis shows significant regional variation in generation costs. SMRs have higher initial capital costs, about 60% above conventional large-scale nuclear, but steeper learning rates (10% cost reduction per capacity doubling versus 4% for traditional plants). Cost convergence between SMRs and large reactors is

expected only after 2040, when mass production reduces unit costs.

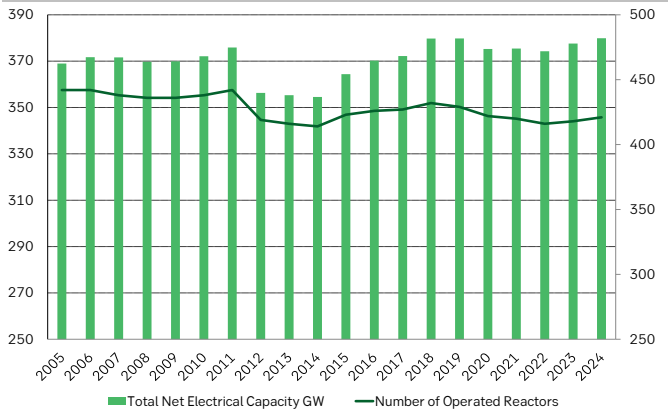
LCOE range and learning curve estimates (USD/MWh)



Source: BNEF

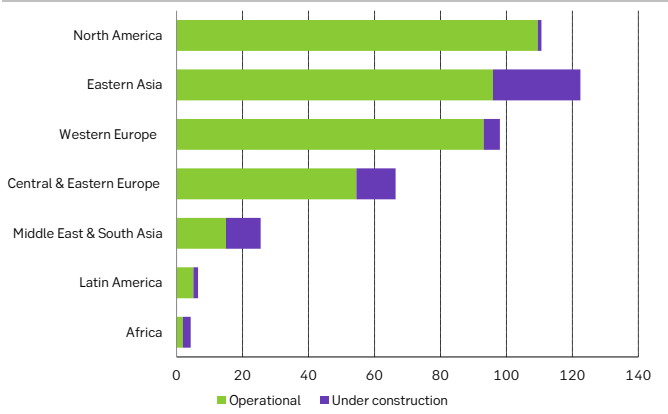
Geopolitical alignment will also shape technological choices. OECD nations are expected to rely on domestic or allied suppliers, while Russia and China maintain their own technology blocs. This fragmentation limits global learning effects and leads to divergent cost trajectories.

Global nuclear capacity and active reactors 2005-2024



Source: IAEA-PRIS (2025)

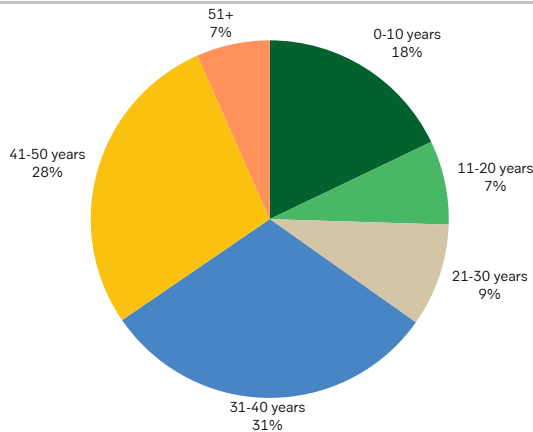
The nuclear power world (number of nuclear plants), 2025



Source: IEA

Regionally, North America, Europe, Greater China, and Northeast Eurasia dominate current nuclear generation. Within a decade, China is likely to overtake both Europe and North America in output. Japan and South Korea plan to double generation by 2030, while emerging regions such as India, Southeast Asia, the Middle East and North Africa will each grow to about 300TWh by 2060. Countries like Bangladesh, Turkey, Egypt and the UAE are commissioning their first reactors, while others are reactivating dormant sites. Overall, about 85GW of new capacity is firmly planned, with a further 880GW expected through 2060.

Age distribution of global nuclear power fleet, 2025



Source: WNISR

An aging nuclear power fleet facing retirements will drive life-time extensions and investments in new output

Half of today's reactors are more than 30 years old, and although many are being extended for energy security reasons, retirements will nearly offset new builds through the 2030s. After 2045, however, the balance shifts and total global capacity rise towards 1,200GW by 2060, with about half of new capacity (410 GW) coming from SMRs.

Nuclear's long-term relevance will be underpinned by four main drivers:

- Energy security: Rising geopolitical instability strengthens the appeal of fuel-secure, domestically managed nuclear fleets.
- Electrification and AI demand: Data centres and digital infrastructure will drive up to 30% of OECD electricity load growth in the early 2030s, reinforcing the need for firm low-carbon generation.
- Land-use and grid advantages: Nuclear's compact footprint and high-capacity factor make it an attractive complement to renewables in land- or grid-constrained regions.
- Supportive financing mechanisms: Long-term power purchase agreements (PPAs) and contracts for difference (CfDs) are helping to de-risk new nuclear investment.

Despite persistent challenges, high upfront costs, long construction times, waste management, and workforce shortages, nuclear energy is poised for a renewed phase of global expansion. By mid-century, the sector will likely evolve into a dual structure: large reactors providing stable baseload, and SMRs offering modular, flexible, and lower-risk options for both mature and emerging markets.

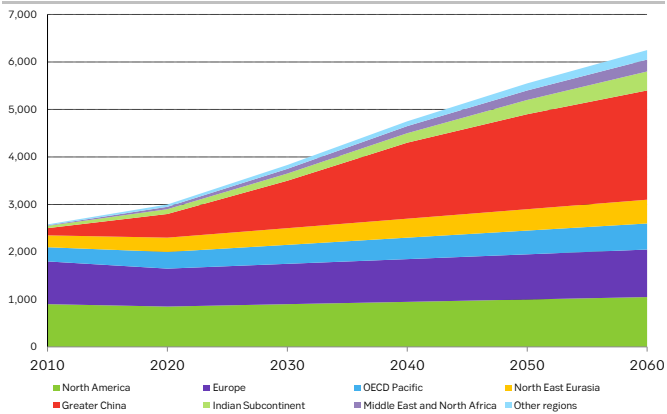
Outlook

Growth in new nuclear capacity is primarily in China

Recent performance

Nuclear power has ridden out a turbulent 15 years for energy markets. The Fukushima accident in 2011 and intensified political resistance in parts of Europe triggered a step down in nuclear's global share of electricity from c.12% to c.10%, largely due to Japan's temporary shutdowns. Since then, output has recovered steadily, reaching a new all-time high in 2024, surpassing the previous peak in 2010, with further growth in output anticipated in the coming years.

Nuclear power projection BNEF (GW)

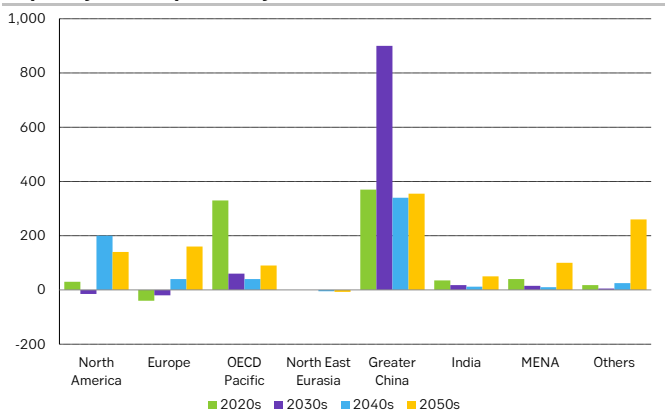


Source: BNEF

Medium-term trajectory

We expect nuclear generation to grow further, but not fast enough to lift its global share materially. By 2035, nuclear output is projected at c. 3,500TWh, with Asia providing close to half of production. The share of nuclear in total power is likely to stabilise around 9-10% through the next decade as renewables expand even faster.

Capacity development by continent and decade in GW



Source: BNEF

Project delivery and near-term cadence

Execution slipped again in 2025, when only two reactors entered commercial service through November and global nuclear capacity declined, reversing earlier expectations of a large wave of completions. Momentum is now shifting into 2026, with about 15 reactors, adding nearly 12GW of capacity, expected to start up and deliver a strong rebound in annual additions

China growing capacity, while Western markets must replace aging capacity

Regional balance

Asia anchors the upturn, led by China’s large pipeline that could lift its operating capacity to c. 140GW by 2035. Europe and North America look broadly stable as new builds offset planned retirements. The US outlook remains uncertain given few new large plants, though policy interest and corporate demand could add upside. However, in the second half of the 2030s, absolute capacity is set to grow according to the International Energy Agency, reflecting more projects and firmer government support, with China and Europe seeing the largest upgrades.

Context vs the 1980s

Despite renewed interest, we think the sector will not replicate the 1980s renaissance. Four decades of capacity attrition across Europe, North America and Japan, together adding up to c. 190GW, combined with public opposition post-Chernobyl and Fukushima, limit how quickly the fleet can expand. In addition, alternatives provide a competitive landscape.

New markets and diversification

China remains the principal growth engine, likely to add c. 46GW in the 2020s (global additions c. 100GW). Interest is also building in the Middle East, Africa and South America; in the Middle East, the UAE, Iran and Türkiye are the most advanced. Regional divergence will persist while the global share trends flat to down; countries adding capacity will see rising local shares.

Country snapshots

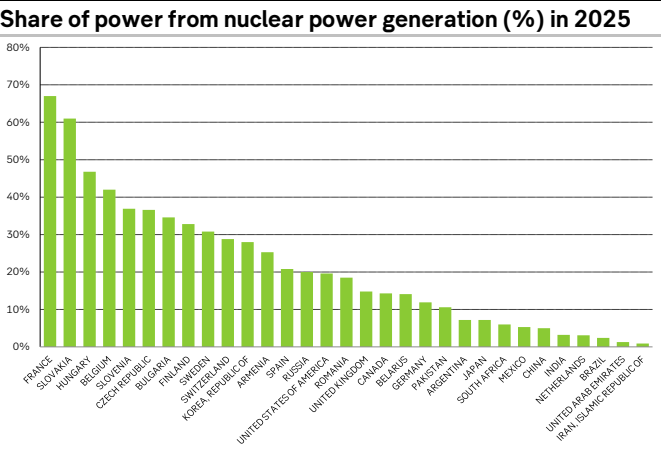
- **China:** Strategy focuses on a rapid expansion of large plants. About 39GW is under construction, narrowing the gap with the US. The supply chain is largely domestic; only a handful of c. 100 planned reactors use Russian designs. After a slower 2019-24 period (installed capacity c. 58GW at end-2024; only one start-up in 2024), four units totalling c. 4.5GW were started in 2025. Nuclear’s share of Chinese generation, at approximately 5% recently, should, based on signalled projects, rise to more than 7% by 2035. The current growth trajectory for Chinese nuclear indicates that it may become the largest nuclear power generator, overtaking the US’ position by the mid-2030s.

Western outlook focuses on SMR technologies

- **US:** Still the largest installed base, but only two reactors have started in the past three decades (Vogtle 3 in 2023; Vogtle 4 in 2024). For the first time in 12 years, no large reactors are under construction. Activity focuses on life-extension and small modular reactor (SMR) development (notably GE Hitachi, Holtec and TerraPower). Given minimal large new build, nuclear’s share in the US generation mix is set to edge down towards 2030, with SMRs unlikely to change totals materially in the early part of the next decade.

Growth in nuclear capacity will come from markets outside traditional nuclear power home markets

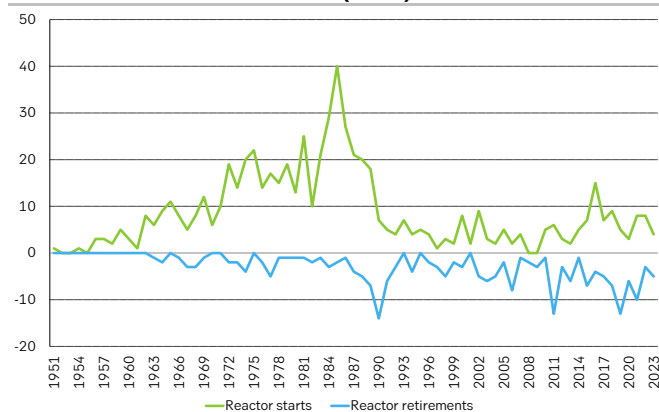
- **Other builders:** India, Egypt and Türkiye also have sizeable programmes underway. Türkiye’s Akkuyu (VVER-1200; VVER means Water-Water Energetic Reactor) has reached completion, its first nuclear plant; Bangladesh’s first reactor (Rooppur, Russian design) is under construction.



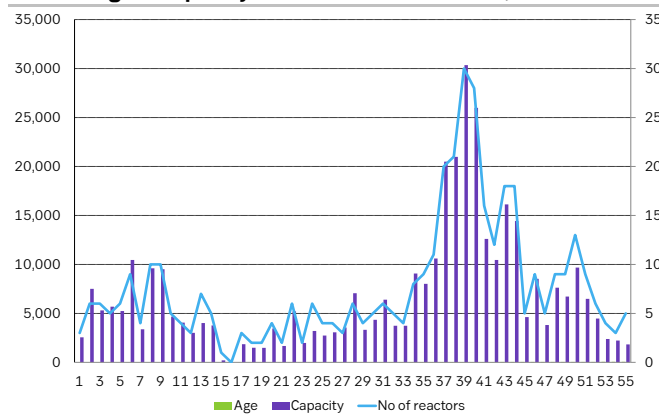
Source: IAEA

Retirements: Europe and Japan account for most decommissioned capacity, reflecting early adoption and aging fleets. Germany tops the list after its 2011 phase-out decision, completed in 2023. The US trend is different: intensive licence renewal has extended some plants towards 80 years. In 2024, only c. 3GW was decommissioned, the lowest since 2016, but retirements should rise as European and Russian fleets age. Belgium will retire three of five remaining units by 2025 (including Doel 1), with no replacements presented; nuclear was 24% of Belgian generation in 2024. Taiwan will close its last reactor in 2025, increasing reliance on oil and gas. In Russia, much retiring capacity will be replaced, though Bilibino's remaining units are due to shut in 2025.

Nuclear starts and retirements (units)



Current age & capacity and number of reactors, 2025



Technology and suppliers

Reactor sourcing typically favours domestic suppliers for reasons of energy security and regulatory control, especially in North America and Russia. Europe is more heterogeneous: historical domestic or US-derived designs gradually gave way to Russian types in parts of the east until Chernobyl curbed adoption; Asian designs are not yet operating commercially in Europe, but several are planned. Asia has moved from importing designs to deploying domestic ones, with China now a leading technology provider. Since the 1990s, global additions have been modest and stable, with market share shifting from Western to Asian designs (e.g., Hualong One,

APR1400, CFR600), while Western efforts have focused increasingly on SMRs.

Technology suppliers mostly regional established supplier with local/regional supply chains

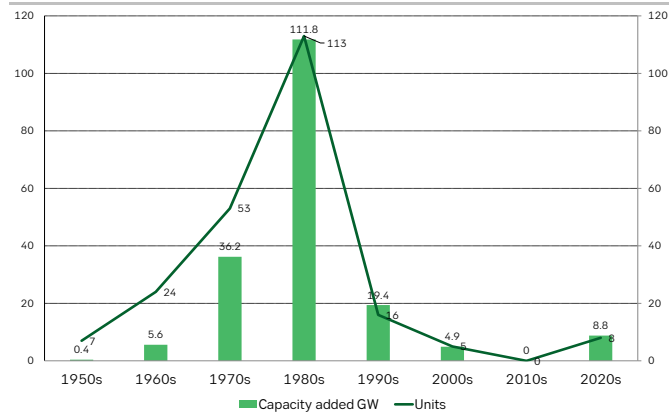
Plant supplier landscape

In the early decade covered, Russian state corporation Rosatom installed more capacity than any other vendor (c. 8GW across China, Russia, Czechia and Ukraine) and expanded volumes in the 2010s, including into Iran and India; it has the second-largest 2020s pipeline. Chinese CNNC and CGN delivered the most in the 2010s and have by far the largest 2020s pipeline, typically collaborating on Chinese designs. KEPCO, the South Korean utility and power plant developer, has been the most successful exporter among other vendors, with new contracts in Poland and Czechia.

French fleet raising its output and new European markets about to construct or plan new nuclear power

Europe's path

Nuclear's share of European generation peaked at 31% in 1997 and stood at c. 21% in 2025. The 2022 dip, when much of the French fleet was offline for urgent maintenance, has partly reversed, although the share may soften again from 2026 as scheduled retirements are currently outpacing new additions. However, considering ongoing projects and plans, the share of nuclear power looks set to regain some territory post-2035. Nuclear expansion plans are advanced in France and the UK, with Central and Eastern Europe (Czechia, Slovakia, Hungary, Romania, Bulgaria) also viewing nuclear as central to transition strategies. Germany and Lithuania are the only completed phase-outs. Poland, with multiple projects in advanced planning and a first large-scale reactor targeted for the early 2030s, is the standout among current non-nuclear countries; rapid deployment is pivotal to cutting coal dependence. Italy is discussing a return, but it remains preliminary. Versus last year, Europe's 2020s outlook improves modestly on delayed UK shutdowns and slightly higher availability assumptions (notably in France). The late-2030s/early-2040s step-up is now stronger on new announcements, especially in Poland, and additional life extensions, including in France, with incremental upgrades in Sweden and Hungary.

European starts of new reactors

Source: IAEA

New complete plants in 2024-25: French Flamanville 3 has suffered significant delays**Milestones in 2024-25**

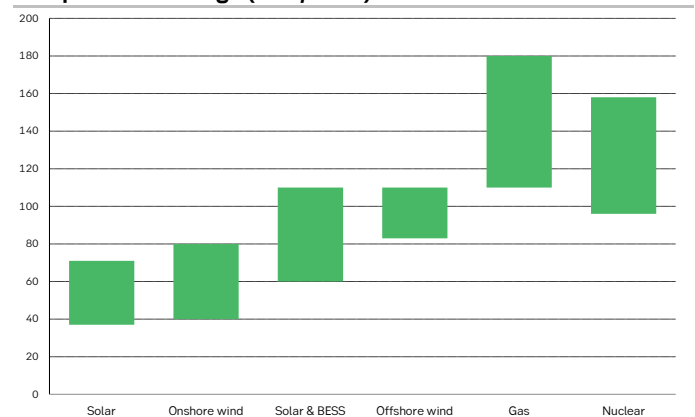
Five plants started in 2024, including Barakah's APR1400 (first new South Korean design deployed abroad), Fangchenggang's first Hualong One in China, and Vogtle 4, the last US large plant under construction, entering service. In 2025, two countries (Türkiye and Bangladesh) joined the club with their first reactors. China brings two new domestic types: CAP1400 (a variant of AP1000) at Shidaowan Guohe and the first commercial-scale fast breeder, CFR600, at Xiapu. Net additions look favourable: c. 22GW of new capacity in 2024-25 versus c. 6GW of retirements.

Construction of the Flamanville 3 EPR nuclear reactor began in December 2007. After years of delays and significant cost overruns, the project reached several major milestones between 2024 and 2025. Fuel loading was completed in May 2024, followed by connection to the national grid on 21 December 2024. Nearly a year later, on 14 December 2025, the reactor reached full power operation at 100% capacity. Industrial commissioning is expected to be completed in the first quarter of 2026, with a major maintenance outage scheduled for late 2026. Originally planned to enter service in 2012, the project has ultimately taken more than 12 additional years to complete and has exceeded its initial budget, with total costs surpassing EUR 13bn, excluding financing costs and grid connection.

Cost overruns and delays are frequent unless the lessons of a supportive framework and a focus on improving output are learned**Costs and competitiveness**

Nuclear's principal challenge remains the high upfront capital expenditure. Projects starting in 2025 display wide cost dispersion, from approximately USD 1.6bn (Xiapu, China) to approximately EUR 13.7bn (Flamanville 3, France). Average capex intensity across the sample is c. USD 3.7 m/MW; Flamanville, at c. USD 8.35 m/MW, after lengthy delays from an original c. USD 3.4bn estimate, is an outlier. Chinese and Russian projects typically come in at around USD 2m/MW, supported by lower labour and material costs, greater standardisation, large-scale serial production and, in some

cases, more relaxed safety requirements. Even so, the capital intensity of nuclear power remains more than twice the global average for solar PV, which is about USD 0.91m/MW. On a levelised basis, however, nuclear can compete as low opex offsets capex, with BNEF estimates ranging from c. USD 50/MWh for 2025 starters to USD 104/MWh. In Europe, fossil fuel plants face high input prices and carbon costs, often pushing LCOE above USD 200/MWh, making even costlier nuclear feasible by comparison. In gas-importing markets without carbon costs (e.g., China, India), new CCGT LCOE can be c. USD 120/MWh, still materially above several nuclear projects. Versus renewables, nuclear remains costlier (utility-scale solar PV can be c. USD 20-60/MWh depending on location), but it offers a dispatchable low-carbon baseload.

European LCOE range (EUR/MWh)

Source: BNEF (2026)

With a rise in large-scale demand, new customers may play an increasingly active role in enabling new nuclear power**SMRs and corporate demand**

R&D spending and corporate PPAs are catalysing emerging nuclear technologies. Major technology firms (Amazon, Google, Microsoft) signed nuclear PPAs in 2024-25 to backstop rising, round-the-clock demand with carbon-free electricity. The SMR pipeline has expanded rapidly since 2020, by roughly 22GW if one assumes c. 300MW per site, although most remain early-stage. China operates the only SMR in service; Russia has granted additional construction permits. Europe and North America host the bulk of proposals, but few have moved beyond initial phases. In 2024, US regulators approved two further SMR designs (NuScale's VOYGR and Kairos Power's SMART). We expect further pipeline growth in the foreseeable future and some progress through national licensing regimes, but acceleration in Europe and North America is essential to meet aspirations for first-of-a-kind SMR operations by 2030 and broader rollout in the early 2030s.

Our conclusion

Nuclear generation capacity is rising again, with a strong focus on Asia, particularly China, driving capacity growth. While some Western markets face the retirement of early-stage nuclear capacity and in some cases initiating life extensions, overall nuclear power net capacity growth looks modest in the near term. The nuclear power sector's global share of power output will likely hover around 9-10% as renewables scale faster, but medium-term risks are now more balanced: recent delays are expected to be offset by a notably stronger cohort of projects coming online, while longer-term prospects are improving as momentum builds from supportive policies, reactor life-extension programmes, and a growing pipeline of small modular reactors.

The evolution of nuclear power: Cycles of promise and setback

First wave of nuclear capacity from late 1950s and onwards

The first dawn: Post-war enthusiasm (1940s-1960s)

In the decades following World War II, nuclear energy emerged as a symbol of progress and modernity. The "atoms for peace" vision captured the imagination of governments and scientists who viewed atomic power as an almost limitless source of clean electricity. The 1950s and 1960s marked the dawn of the nuclear age, with the commissioning of the first commercial plants at Obninsk in the Soviet Union, Calder Hall in the United Kingdom, and Shippingport in the US.

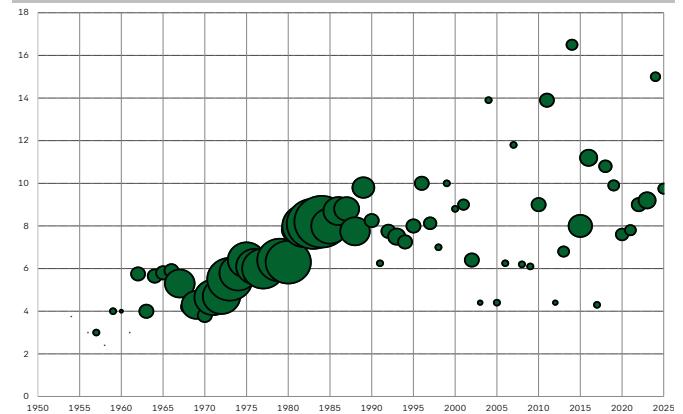
The mood was one of unbridled optimism. Industry advocates famously claimed nuclear power would make electricity "too cheap to meter." The technology proved far more complex and costly than anticipated. Yet, despite high costs and technical challenges, the sector grew rapidly across major economies, driven by the belief that nuclear energy could transform the global power landscape.

Setbacks and public concern (1970s-1980s)

The momentum of the early years carried into the 1970s, when the oil crises strengthened nuclear's appeal as a secure alternative to imported fossil fuels. Governments in the US, Europe and Japan ordered dozens of new reactors, anticipating an era of energy independence.

However, optimism soon gave way to growing concern. Costs began to spiral as projects suffered from long construction timelines and tightening safety regulations. Two major accidents profoundly changed public perceptions: Three Mile Island (1979) in the US and Chernobyl (1986) in the Soviet Union. These events shook public confidence and triggered powerful anti-nuclear movements, particularly in Western Europe. Environmental organisations also rallied against nuclear power, raising concerns over radioactive waste and long-term safety.

Average age from start to grid connection (years and units)



Source: IAEA-PRIS

By the late 1980s, the global expansion of nuclear power had slowed dramatically. Many planned projects were cancelled mid-construction, and the technology became politically divisive in much of the Western world.

The first nuclear renaissance fizzled out

"Nuclear renaissance" hopes (1990s-2000s)

As climate change entered the global policy agenda in the 1990s, nuclear energy began to regain attention as a potential low-carbon solution. The industry promoted new Generation III reactor designs, which promised improved safety, lower costs, and standardised construction methods. Policymakers in countries such as Finland, France, the United Kingdom and the US embraced talk of a "nuclear renaissance."

However, reality once again fell short of expectations. Flagship projects, including Olkiluoto-3 in Finland, Flamanville-3 in France, and Vogtle-3 and 4 in the US, faced years of delays and budget overruns. By the 2010s, the early optimism had faded, and enthusiasm for large-scale nuclear expansion cooled once more.

Nuclear accidents have had adverse effects on new and existing nuclear capacity

Post-Fukushima retreat (2011-mid-2010s)

The Fukushima Daiichi disaster in Japan in 2011 marked another turning point. Triggered by an earthquake and tsunami, the accident reignited global fears about nuclear safety. Germany responded by accelerating its nuclear phase-out, while Japan shut down its entire reactor fleet, with only slow and limited restarts in the years since.

Worldwide, new construction slowed as public and political resistance resurfaced. By the mid-2010s, nuclear's share of global electricity generation had fallen from about 17% in the 1990s to around 10%, reflecting both retirements of aging plants and a lack of new capacity.

Nuclear power as a non-emitting power source

The current debate: Climate urgency and SMR hopes (2020s-)

Today, nuclear power is once again part of energy conversation, this time driven by climate imperatives and the push for net-zero emissions. Governments and industry see nuclear as a potential complement to renewables, capable of delivering firm, low-carbon baseload power. The development of Small Modular Reactors (SMRs) and Generation IV technologies has renewed optimism by promising safer, cheaper and more flexible deployment options.

Countries such as China, Russia and the United Arab Emirates are actively expanding their nuclear fleets, while others, like Sweden, the UK and the US, are re-evaluating their nuclear strategies. Yet, major challenges remain: Construction costs remain high, timelines uncertain, and public scepticism persistent. Meanwhile, the rapid decline in renewable energy costs continues to reshape the competitive landscape.

Despite these uncertainties, there is cautious optimism that if technological and financial barriers can be overcome, nuclear power could play a meaningful role in the global transition to a low-carbon energy system. However, many observers warn that without tangible improvements in cost and delivery performance, this renewed optimism could prove to be yet another false dawn.

Key drivers and enablers for new nuclear – a European perspective

The resurgence of nuclear energy in Europe is driven by multiple converging factors:

- Policy and regulatory support:** The evolving policy landscape is creating favourable conditions for nuclear energy to maintain, and in some places expand, its role in power systems around the world. More governments are acknowledging nuclear power as a low-emissions source of reliable electricity and heat, one that can meaningfully support clean energy transitions while also strengthening energy security by adding stability and resilience to national grids.

This growing policy support can be seen in recent decisions across several countries to approve extensions to the operating lifetimes of existing reactors. Over the past five years, such decisions have been made for 64 reactors in 13 countries, representing roughly 65GW of capacity, or about 15% of the total capacity of today's global nuclear fleet.

European countries are again looking to nuclear power to meet future demands as well as to supply non-emitting baseload power

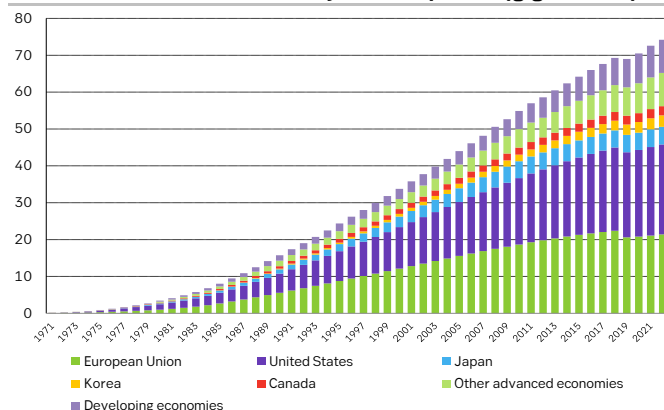
European countries have shifted towards embracing nuclear power due to energy security and climate goals. The resurgence of nuclear energy is fundamentally driven by a convergence of policy, technological and market factors. In Europe, the policy landscape has shifted dramatically in recent years. After decades of stagnation and even phase-out policies in some countries, the energy crisis triggered by Russia's invasion of Ukraine, coupled with the urgent need to decarbonise, has led to a renaissance in nuclear policy support. Over 40 countries globally, including many in Europe, now have explicit plans to expand nuclear capacity or are actively considering it.

France focusing on large-scale plants; other European countries looking to SMR plants

The EU's inclusion of nuclear in its sustainable taxonomy, establishment of the European SMR Industrial Alliance, and national initiatives in France, Sweden, Finland, Czechia, and Poland illustrate this trend. Governments offer financial support, loan guarantees, and streamlined regulations, particularly for small modular reactors (SMRs) and advanced designs. France has confirmed its plans to proceed with the Grand Carénage programme, including a 2021 decision to extend the operating lives of all 20 of its 1,300-megawatt reactors. Other European countries have taken similar steps: Belgium, Hungary, the Czech Republic, Finland, Spain, Romania and the Netherlands have all announced extensions for reactors representing capacities ranging from 0.5 to 2.2GW.

In the US, the Inflation Reduction Act expanded clean-energy tax incentives to include nuclear power, significantly improving the economic outlook for all operating reactors. The current administration has announced a focus on expanding nuclear power capacity to meet future needs. In the past five years, 22 reactors have applied for lifetime extensions. By 2024, every US reactor that had been running for at least 30 years had applied for an additional 20-year licence, and more than a fifth of them had already sought a second 20-year extension.

Avoidance of CO2 emissions by nuclear power (gigatonnes)



Source: IAEA

Japan revised its Electricity Business Act in 2023 as part of its Green Transformation initiative, allowing reactors to operate beyond 60 years by discounting periods when they were offline for reasons outside operators' control. That same year, the country introduced a new investment support mechanism, the Long-Term Decarbonised Capacity Auction, which guarantees a fixed income for 20 years. This mechanism is designed to cover fixed operating costs and provide predictable revenues, encouraging investments in decarbonised power sources, including nuclear energy.

Beyond Europe, Mexico and South Africa have both announced new lifetime-extension decisions covering reactors that together account for half of each country's total nuclear capacity.

Nuclear power can support energy security

- **Energy security and geopolitics:** The energy crisis following Russia's invasion of Ukraine exposed Europe's vulnerability to gas supply disruptions, prompting a pivot to domestically controlled, dispatchable, low-carbon energy sources. Nuclear power's high-capacity factor and independence from weather make it a strategic asset for energy security and electricity price stabilisation. From an EU perspective, with 98GW of installed capacity, nuclear currently supplies around 24% of EU electricity and forms a stable baseload that supports system resilience. Further geopolitical instability and dependence on energy sourced from afar may affect the ability to make independent decisions as well as the ability to compete in certain disciplines.

Regardless of cycles in policies to phase out emitting power and heat sources, the process to reduce fossil fuels in the energy system continues

- **Decarbonisation and climate commitments:** Nuclear is currently the second-largest source of low-emission electricity globally and essential for achieving net-zero targets in Europe. Its firm baseload power complements intermittent renewables, supporting grid stability and decarbonisation in hard-to-abate sectors such as industry and district heating.

Nuclear energy has not only been a significant source of electricity but also an important contributor of heat in several countries. Through nuclear combined heat and power, reactors generate electricity while simultaneously supplying usable thermal energy, making much more efficient use of the original fuel. In applications where heat itself is the primary output, drawing directly on the reactor's fission heat avoids the conversion losses associated with producing electricity first.

Many nuclear plants are modelled to provide both heat and power

Around seventy reactors worldwide currently provide some form of nuclear co-generation. The range of possible applications depends largely on the temperature of the heat delivered and the type of reactor involved. Most of today's operating reactors, whether light water or heavy water designs, can supply heat at temperatures below about 150°C. This limits their use primarily to district heating networks, desalination plants, and selected industrial processes, such as in pulp and paper production, where moderate-temperature heat is sufficient. The idea itself is far from new. Nuclear district heating has been in operation since the 1960s, when Sweden's Ågesta plant and the UK's Calder Hall began supplying both electricity and heat to nearby communities. Since then, the practice has become well established in countries including Bulgaria, the Czech Republic, Hungary, Romania, Russia, Switzerland and Ukraine, demonstrating that nuclear heat supply can be integrated into local energy systems on a long-term basis.

A wide range of new technologies are marketed

- **Technological innovation:** The emergence of SMRs and advanced Generation III+ and IV reactors offer advantages including lower upfront capital costs, modular deployment, enhanced safety and shorter construction times. The long-term cost/MW for SMRs if a widespread rollout is successful allows for targeted investments in the energy system with a wider variety of investors than large-scale plants would allow. European companies like Rolls-Royce SMR, EDF's NUWARD, and US/Japanese GE Hitachi are leading this innovation wave.

Data centres, with stable demand and a large-scale buildout, form a potential customer base for nuclear SMR capacity

- **Private sector and industrial demand:** A new and powerful driver is the surge in electricity demand from data centre, artificial intelligence, and electrification of transport and industry. Major technology companies (Amazon, Google, Microsoft, Meta) are entering into long-term power purchase agreements (PPAs) with nuclear operators and are even investing directly in SMR projects to secure clean, reliable power for their operations. This trend is particularly pronounced in Western Europe and the US, where data centre growth is explosive.
- **Financial innovation and de-risking:** New financing models such as regulated asset base (RAB), contracts for difference (CfD), green bonds, alongside government guarantees and export credit agency support, are improving project bankability. Inclusion of nuclear in sustainable finance taxonomies opens the door for ESG investors.

What are the challenges?

There are several challenges involved in the construction of any infrastructure component, and nuclear plants are no exception. Some of the most notable challenges are:

If demand does not materialise, then the nuclear investment case becomes more difficult to fund and will impact timelines

- **Structural growth in electricity demand:** Electricity demand is projected to nearly double globally by 2050, with Europe experiencing robust growth driven by electrification of transport, heating and industry, and expansion of data centre and AI infrastructure. The demand expansion may be unevenly distributed, suffer delays and may affect how much new capacity is added.

- **Decarbonisation imperatives:** Europe's net-zero targets and the phase-out of coal and natural gas necessitate new low-carbon, dispatchable generation. Renewables alone cannot yet meet demand due to intermittency and land constraints, positioning nuclear as the scalable firm option to replace retiring fossil and aging nuclear assets.

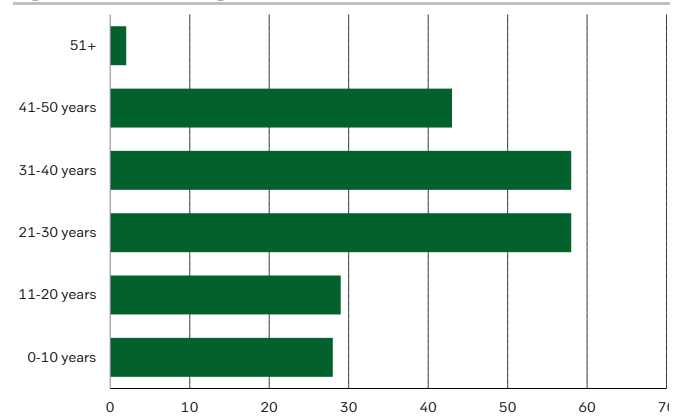
Data centres change demand profile, leading to more competition for existing generation capacity

- **Data centre and industrial loads:** Hyperscale data centres constitute a significant share of electricity consumption, up towards 20% in Ireland, requiring around-the-clock reliable power, increasingly sought through nuclear-backed PPAs. Electrification of heavy industry and green hydrogen production further amplify the demand for non-emitting, high-capacity power.
- **Policy and market signals:** European Commission support, national targets (e.g., France's plans for multiple EPR2 reactors, Sweden and Poland's considerable nuclear programmes, which include both large reactors and SMRs), and international pledges to triple nuclear capacity by 2050 indicate a strong demand pipeline.

The aging nuclear fleet, particularly in the EU and US, needs to extend operations or be replaced

- **Lifetime extensions and replacement needs:** With Europe's nuclear fleet averaging over 36 years old, twice the average age of other markets, many reactors will retire in the 2030s-40s, necessitating lifetime extensions or new builds to maintain or grow nuclear's share. The average age of reactors in the EU and US reflects the low volume of capacity additions in these markets in recent decades.

Age distribution of global nuclear retirements as of 2025



Source: IAEA-PRIS

Nuclear power is expensive but with a learning curve when scaling. Nuclear is also suited to match other generation sources

- **Cost competitiveness:** The LCOE for new nuclear in Europe is projected by Lazard at USD 75-110/MWh for large reactors and USD 110-130/MWh for SMRs by 2040. Although higher than wind and solar, nuclear competes well with other dispatchable low-carbon options and costs are expected to decline with standardised SMR technologies scaling up.
- **Value to the energy system:** Nuclear's high-capacity factor (80-90%) and firm, weather-independent power provision offer system value beyond LCOE, including ancillary grid services. Value-adjusted LCOE analyses by BNEF show nuclear is currently competitive or advantageous to renewables-plus-storage in high-renewables systems depending on conditions.

Financing is key challenge for nuclear power, with capital costs making up the vast majority of project and lifetime costs

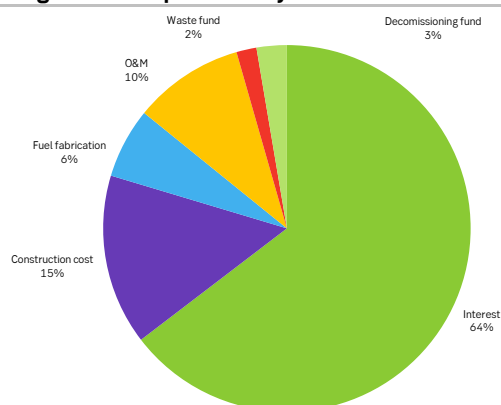
- **Financing and de-risking:** Nuclear projects are highly capital-intensive, with long construction periods and payback horizons. This creates challenges for traditional project finance, but new models, such as RAB (used for Sizewell C in the UK), CfD (Hinkley Point C), and long-term PPAs with creditworthy off-takers, are mitigating revenue and construction risks. State involvement, whether through direct ownership, guarantees, or concessional loans, remains critical, especially for first-of-a-kind projects.

Debt financing strategies

Financing option	Provider	Pros	Cons
Corporate finance (loans/bonds)	Private lenders/investor	Large companies or companies with explicit or implicit government support can benefit from a lower cost of capital by leveraging their creditworthiness; Flexibility in the use of proceeds (if the loan is for general corporate purposes).	If a company already carries substantial debt, securing additional financing may be difficult regardless of the project's risk; Typical corporate loan tenors are shorter than a nuclear project's delivery period, exposing the project to rating changes during construction; Lending capacity depends on the company's size and credit strength.
Corporate finance (green/transition loans/bonds)	Private lenders/investor	Access to a broader investor base interested in sustainable finance; Potential benefit from a green premium (greenium)	Unlikely to meet all capital needs during construction; Project-specific support will still be necessary.
Project finance	Private lenders/investor	Limits sponsors' liability to the project, protecting other assets; Can attract investment through a ring-fenced risk structure (special-purpose vehicle); Enables longer-term financing.	Requires early and highly predictable cash inflows and multiple conditions, including clear and strong government support; The pool of entities capable of arranging nuclear project finance is smaller than for corporate finance; Structuring is complex and time-consuming.
Government guarantee	Government	Lowens perceived risk for the project and other stakeholders; Can attract investors and improve bankability; May reduce financing costs.	Exposes taxpayers to potential project failure or cost overruns; Schemes can be difficult to expand and may create contingent government expenditure.
Export credit financing	ECA	Similar to a government guarantee – can lower project risk and improve bankability.	Relies on diplomatic relationships; Creates contingent government expenditure.
MDB loans	MDBs	Concessional terms typical of MDB lending, particularly for developing countries; MDB participation can add credibility and crowd in other investors.	MDB funding is limited and requires compliance with stringent guidelines; Processes can be lengthy and subject to political influence.

Source: SEB

Financing cost example: Hinkley Point C



Source: EdF, IEA, SEB est

- **Private sector appetite:** Private financial institutions are increasingly open to nuclear, especially SMRs, provided there is strong government support, predictable cash flows, and robust risk-sharing mechanisms. The emergence of green and transition bonds for nuclear (e.g., EDF, Bruce Power, Constellation) is a positive signal. However, reputational risk and policy uncertainty remain concerns for some investors.

Renewable electrons are less costly than nuclear power when compared. Renewables coupled with storage capacity competes with nuclear capacity

- **Competition with alternatives:** While renewables are cheaper on a pure LCOE basis, their intermittency and the costs of grid integration, storage, and backup capacity mean that nuclear remains essential for system reliability and decarbonisation. Gas with CCS, while a competitor, faces its own cost, technology and policy risks. In many European markets, nuclear is now seen as complementary to renewables rather than a competitor.

Key hurdles for developing new nuclear power and solutions

Construction risks and cost overruns have been the norm for nuclear projects and increasingly so over recent decades

- **Construction risk and cost overruns:** Lengthy construction times, cost overruns, and regulatory complexity have hindered projects, not only in Western Europe but across the world, especially when unproven features are introduced in reactor designs and construction risks increase accordingly, since there is less or no operating experience. The main drivers of these risks are the complexity and scale inherent in such projects. Unlike other energy-generation methods, nuclear initiatives must meet exceptionally rigorous regulatory oversight, safety protocols, and technical standards. In addition, the often-debated nature of nuclear power can create further political and regulatory challenges. Solutions include design standardisation, serial construction, modularisation (SMRs) and improved project management.

“Five years” is seldom five years when building a nuclear plant

A review of nuclear projects classified as “under construction” as of 1 July 2025 reveals a sector marked by persistent uncertainty and structural delays, despite the continued use of five-year construction timelines in many project proposals. On average, the 63 reactors currently being built have already spent more than five years under construction, and many remain far from completion. While this average is slightly lower than a year earlier, it does not signal improved performance so much as reflect the addition of recently started projects that have not yet had time to run into difficulties.

Construction time of units started and connected to grid between 2015-24 (yrs)

Country	Units	Mean time	Minimum	Maximum
China	37	6.3	4.1	10
Russia	8	16.2	8.1	35.1
South Korea	5	8.7	6.4	10.5
Pakistan	4	5.6	5.5	5.8
UAE	4	8.3	8	8.6
India	3	12.5	10.1	14.2
US	3	21.5	10.1	42.8
Belarus	2	8	7	8.9
Finland	1	16.6	16.6	16.6
France	1	17.1	17.1	17.1
Slovakia	1	38.1	38.1	38.1
Total	69	9.4	4.1	42.8

Source: WNISR

Repeat construction and lessons learned allow for cheaper and faster projects

Delays are widespread and, in many cases, systematic. All reactors under construction in at least six of the eleven building countries have experienced significant schedule slippages, often measured in years rather than months. More than one-third of all ongoing projects are formally documented as delayed, and the status of many others remains uncertain. Most reactors still described as “on time” began construction only within the past three years, and the majority of these are in China. Given the limited transparency and inconsistent public reporting on Chinese projects, it is difficult to assess whether these units are genuinely progressing according to plan or have simply not yet reached stages where delays become visible.

Supplier choice is also a geopolitical choice

Geopolitical factors further complicate the outlook. Projects based on Russian technology or involving Russian contractors face growing risks due to sanctions and disrupted supply chains. In most countries where such projects are underway, delays are already documented, with sanctions explicitly cited in several cases. Over the past year, the situation has deteriorated further, with additional delays reported across multiple countries and only a small number of projects still lacking confirmed schedule slippages.

The cheque is in the mail... soon...

The gap between planned and actual delivery is also evident in recent reactor startups. Of the reactors expected to begin operation in 2024, only half were connected to the grid on schedule, while the remainder were pushed back by at least a year. Several emblematic projects highlight the extreme duration nuclear construction can reach: some units in Slovakia and Iran trace their origins back 4-5 decades, having undergone long suspensions before resuming work. Others in India have remained officially “under construction” for more than ten years, underscoring how project status labels often obscure the true length and complexity of nuclear development.

Learning curves important along with consistency in policies

Looking beyond individual cases, historical data confirm that long construction times are no longer anomalies but the norm. While reactors completed in the 1970s and 1980s showed relatively consistent build times, the past two decades have been characterised by wide variation and frequent overruns. Reactors that started up between 2022 and 2024 took, on average, nearly 11 years to complete, and none entered operation on schedule. Even China, which performs better than most countries, rarely achieves construction times below seven years. Over the past decade, global average construction durations have stabilised around ten years, reflecting not only technical challenges but also the cumulative impact of regulatory processes, financing negotiations, licensing, and broader political and legal hurdles. From an analytical perspective, the evidence suggests that short, predictable nuclear construction timelines remain the exception rather than the rule.

Both supply chains for technologies but also for the creation of skilled labour to build and operate nuclear plants are essential to achieve new capacity

- **Supply chain and workforce constraints:** Europe's nuclear supply chain and skilled workforce have diminished over decades. The aging workforce and the decline in the number of nuclear professionals under average age present a risk for both the future construction and operation of nuclear power plants as well as the decommissioning of older nuclear plants. Scaling up requires coordinated investment in manufacturing, training, and research and development. The future of the global, but also the European, nuclear industry relies heavily on the strength and stability of its supply chains. A resilient supply chain can adapt quickly to disruptions by using flexible contingency plans and accurate forecasting across all its stages. To support this, governments should avoid “stop-and-go” policy cycles, which make it difficult for any industry, especially those requiring long-term planning, to commit to major supply-chain investments. Careful, early planning is essential, including finding the right mix of global and local suppliers for each part of the chain. France plans to hire 10,000 workers annually over the next decade, putting the spotlight on the need to

provide training and education relevant to the needs of the sector.

Long-term investments require long-term policy support

- **Regulatory and policy uncertainty:** Divergent national policies and shifting regulations create uncertainty. Streamlined and harmonised frameworks like UK planning reforms and the EU SMR Industrial Alliance are helpful to enable and reduce costs for projects.

Nuclear power is now part of the EU Taxonomy

- **Financing and bankability:** Capital intensity and long payback periods challenge commercial financing. Government support via guarantees, CfDs, RAB models, and direct investment is critical. Green and transition bonds and sustainable finance inclusion expand capital pools. The screening criteria focus on two key areas: greenhouse-gas (GHG) emissions and activities that must “Do No Significant Harm” (DNSH). For GHG emissions, nuclear energy projects are required to keep life-cycle emissions below 100 grams of CO₂ equivalent per kilowatt-hour. The DNSH criteria address safety, regulatory compliance, and the sound management of radioactive waste and decommissioning. This classification has helped support the use of green bonds, financial instruments in which the proceeds are dedicated to funding eligible green projects, by nuclear operators within the European Union. One prominent example is EDF, which has issued multiple green bonds to support activities linked to nuclear power.
- **Public acceptance:** While public opinion is improving due to energy security and climate concerns, opposition persists. Public acceptance relates both to the operations of nuclear capacity and the waste management process and subsequent storage required for spent nuclear fuel. Transparent communication, safety culture, and community engagement are vital.

Fuel supply and processing capacity for key elements of the supply chain

- **Fuel supply security:** Europe’s dependence on Russian enrichment and fuel services is a strategic risk. Diversification efforts include investments in domestic enrichment (Orano, Urenco) and development of new fuel types like HALEU for SMRs. According to the Euratom Supply Agency, the European Union now sources nuclear materials and related products from a broad and diversified set of suppliers. In 2024, total imports amounted to around 14,000 tonnes. As in the previous year, roughly one-third of these imports came from Canada, while Kazakhstan supplied about one-quarter. Russia’s share declined further in 2024, whereas imports from Australia and China increased.

Decommissioning and waste management plans must also be developed

- **Waste management and decommissioning:** Progress on geological repositories is underway in Sweden, Finland, France, and Switzerland, but long-term waste management remains a political challenge requiring robust funding and governance. Under the Radioactive Waste and Spent Fuel Management Directive, each EU member state must establish a national framework for managing spent nuclear fuel and radioactive waste. These requirements stem from the Euratom Treaty. Countries are obligated to develop and implement national programmes that ensure the safe and responsible handling of these materials. In line with Euratom’s waste-management rules, the EU periodically publishes inventory data on radioactive waste and spent fuel. All EU member states generate radioactive waste, and 17 of them also manage spent fuel within their territories. Because of the radiological characteristics and potential hazards associated with these materials, their safe management is essential at every stage of the nuclear lifecycle. This material must therefore be regulated as radioactive waste under the relevant legislative and regulatory systems overseen by the appropriate authorities.

Technology development

The evolution of nuclear reactor designs and deployment

Introduction

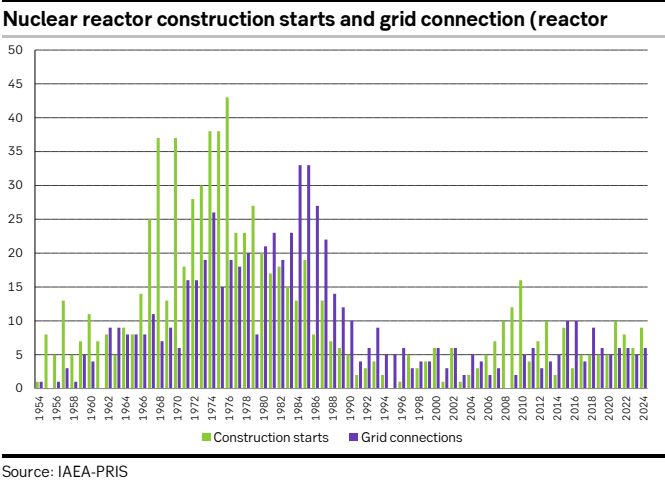
Technology development continues with new applications

Since the dawn of the nuclear age in the mid-20th century, reactor designs have undergone a profound evolution, shaped by scientific advances, safety imperatives, economic pressures and shifting public attitudes. Early reactors were experimental devices focused on demonstrating the feasibility of nuclear fission for electricity generation. Today, the nuclear industry pursues highly advanced Generation III and IV designs, as well as Small Modular Reactors (SMRs) and even fusion concepts.

Operational nuclear plant technologies by region							
	Africa	Asia	East Europe & Russia	North America	South America	West & Central Europe	Total
BWR		19		33		8	60
FNR			2				2
GCR						8	8
HTGR		1					1
LWGR			10				10
PHWR		24		17	3	2	46
PWR	2	109	41	63	2	96	313
Total	2	153	53	113	5	114	440
Source: IAEA-PRIS							

First-generation reactors: From experiment to commercialisation

The first nuclear reactors, built in the 1940s and 1950s, were largely research and military oriented. The Chicago Pile-1 (1942), designed by Enrico Fermi's team, demonstrated the world's first self-sustaining nuclear chain reaction. Shortly afterward, reactors such as the Hanford B Reactor in the US produced plutonium for weapons programmes. Civilian deployment began in earnest in the 1950s. The Obninsk reactor in the Soviet Union (1954) was the first grid-connected nuclear power plant, though it produced only 5MW of electricity. Britain's Calder Hall (1956) marked the first large-scale plant intended for both electricity generation and plutonium production. These early reactors, often categorised as Generation I, were characterised by bespoke designs, relatively low efficiencies, and limited safety features by today's standards.



PWR technology first developed for submarine technology

Generation II reactors: Standardisation and expansion

The late 1960s through the 1990s witnessed the rise of Generation II reactors, which formed the backbone of today's operating fleet. The most prominent design was the Pressurised Water Reactor (PWR), first developed for naval propulsion in the US Navy's submarine programme. Its compact design, stable behaviour under varying conditions, and closed primary loop made it attractive for commercial deployment.

Alongside PWRs, the Boiling Water Reactor (BWR), developed by General Electric, became widespread. Both designs use light water as a coolant and moderator but differ in steam generation: in a BWR, steam is produced directly in the reactor vessel, while in a PWR, steam is generated in secondary loops. Other Generation II technologies included the CANDU (Canada Deuterium Uranium) reactor, using heavy water as moderator and natural uranium as fuel, and the AGR (Advanced Gas-cooled Reactor) in the UK, which used carbon dioxide as coolant and graphite as moderator.

The era of Generation II saw rapid expansion: by the mid-1980s, hundreds of reactors were under construction worldwide, particularly in North America, Europe, Japan, and the Soviet Union. Standardisation efforts sought to reduce costs and improve reliability, but the industry also grappled with significant setbacks.

Nuclear accidents have led to further safety features being implemented

Impact of accidents on design evolution

Two major accidents profoundly influenced reactor design philosophy.

1. **Three Mile Island (1979, US):** A partial meltdown at a PWR in Pennsylvania exposed deficiencies in operator training, instrumentation, and emergency preparedness. While no deaths occurred, the incident damaged public trust and highlighted the need for redundant safety systems and human-machine interface improvements.
2. **Chernobyl (1986, USSR):** A catastrophic explosion at an RBMK graphite-moderated reactor in Ukraine underscored the risks of flawed design, inadequate safety culture, and lack of containment. The disaster accelerated international cooperation on nuclear safety and imposed stricter design standards worldwide.

Both accidents led to a re-emphasis on containment structures, passive safety features, and robust emergency core-cooling systems. They also spurred the industry to pursue new generations of reactors with enhanced safety margins.

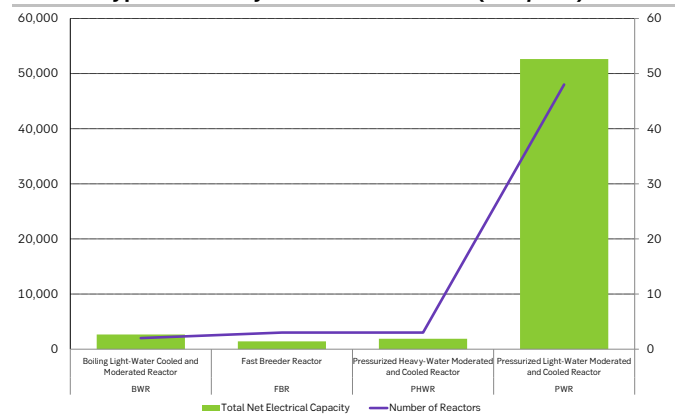
Generation III and III+ reactors: Safety and economics

Generation III reactors emerged in the 1990s and 2000s as evolutionary improvements over their predecessors. Their goals were enhanced safety, longer operational lifespans, reduced maintenance, and improved economics. Key features included:

- **Passive safety systems** that rely on natural circulation, gravity, or convection, reducing dependence on active pumps and operator intervention.
- **Standardised designs** to streamline licensing and reduce construction costs.
- **Longer fuel cycles** and higher burnup rates, improving fuel utilisation.

Examples include the AP1000 (Westinghouse), the EPR (European Pressurised Reactor, Areva/Framatome), and the VVER-1200 (Rosatom). Japan developed the ABWR (Advanced Boiling Water Reactor), which became the first-Generation III design to enter service in 1996. Despite technical advances, deployment has faced significant obstacles. Projects such as Olkiluoto-3 in Finland, Flamanville-3 in France, and Vogtle-3/4 in the US have experienced severe cost overruns and delays. These challenges highlight the tension between increasingly complex safety demands and the realities of large-scale project management.

Reactor types currently under construction (MW/unit)



Source: IAEA

Generation IV includes a sustainable approach to recycling nuclear waste

Generation IV concepts: Toward sustainability and innovation

The Generation IV International Forum (GIF), established in 2000, coordinates research into advanced reactor concepts targeting long-term sustainability, safety, and proliferation resistance. Six main designs are under active study:

1. Gas-cooled Fast Reactor (GFR)
2. Lead-cooled Fast Reactor (LFR)
3. Molten Salt Reactor (MSR)
4. Sodium-cooled Fast Reactor (SFR)
5. Supercritical-water-cooled Reactor (SCWR)
6. Very-high-temperature Reactor (VHTR)

These designs often aim to close the nuclear fuel cycle by recycling actinides (metallic radioactive elements), reduce long-lived waste, and achieve higher thermal efficiencies. The MSR has attracted renewed attention for its potential flexibility in load-following and hydrogen production. While most Generation IV concepts remain at the research or demonstration stage, their promise lies in combining low-carbon energy with long-term resource sustainability.

SMR holds promise in standardisation and capital efficiency**Small modular reactors (SMRs): A new paradigm**

In parallel with Generation IV work, the industry is focusing on SMRs. These reactors, typically producing 50-300MW each, are designed for factory fabrication, modular deployment and scalability. SMRs aim to address many of the economic and logistical barriers of large reactors:

- **Lower upfront capital costs** due to smaller size and modular construction.
- **Shorter construction times** through factory manufacturing.
- **Flexibility** in pairing with renewable energy sources or deployment in remote regions.
- **Enhanced safety** features, often relying on passive systems and underground siting.

Projects include NuScale's VOYGR design in the US, Rolls-Royce's SMR in the UK, and Russia's floating Akademik Lomonosov plant. Countries such as Canada and China are also actively pursuing SMR deployment. SMRs may redefine nuclear deployment by enabling incremental capacity additions, more manageable financing, and applications beyond electricity, such as district heating, desalination, or hydrogen production

Ongoing SMR projects

Country	Reactor developer	Reactor type	Plant Capacity (MW)	Status	Expected Start-up	Key developments
Argentina	CNEA	PWR	29	Under construction	2027	Construction started
Canada	GE Hitachi	BWR	300	FID	2030	Government working on licensing for three units
China	China Huseng	HTGR	211	Operational	2023	Started 2023
China	CNNC	PWR	125	Under construction	2026	Construction began in 2021
Romania	NuScale Power	PWR	77	Pre-investment	2030	
Russia	Rosatom	PWR	35	Operational	2019	Operational
Russia	Rosatom	PWR	55	Under construction	2027	Construction started in 2022
Russia	Rosatom	PWR	55	Pre-investment	2028	Non-nuclear construction started in 2024
Russia	Rosatom	FBR	300	Under construction	2026	Construction started in 2021
US	GE Hitachi	BWR	300	Pre-investment	2033	Application submitted
US	Kairos Power	MSR	35	Under construction	2027	Construction started Tennessee
Uzbekistan	Rosatom	PWR	55	FID	TBD	Contract for six units signed in 2024

Source: IAEA, BNEF

Nuclear fusion remains a distant prospect, but interesting

Nuclear fusion: The long-term vision

Parallel to fission development, research in nuclear fusion has long promised virtually limitless clean energy. Fusion seeks to replicate the processes powering the sun, typically by fusing deuterium and tritium at extremely high temperatures. The most visible project is ITER (International Thermonuclear Experimental Reactor) in France, a massive international collaboration aiming to demonstrate sustained fusion power by the 2030s. Complementary efforts include stellarator research in Germany (Wendelstein 7-X) and private ventures exploring compact fusion devices.

While commercial fusion remains decades away, its pursuit symbolises the nuclear industry's long-term aspiration for safer, cleaner, and more abundant energy.

Deployment trends and regional variation

The evolution of designs has unfolded unevenly across regions.

- **Europe and North America** initially led in deployment but have slowed in recent decades due to high costs, regulatory complexity, and public opposition.
- **Asia** has become the centre of new construction. China is aggressively building advanced reactors, including Hualong One and CAP1400 designs, and is exploring both SMRs and Generation IV concepts.
- **Russia** continues to expand exports of VVER reactors, positioning itself as a dominant supplier for global markets.
- **Emerging economies** in the Middle East, Africa, and South America are considering nuclear power to meet rising energy demand and climate targets.

These regional patterns reflect broader trends in financing capacity, state support, and societal acceptance.

Small modular reactors (SMR)

SMRs are in focus, specifically in the US and in Europe

Nuclear power has returned to the centre of energy policy debates as governments confront the combined pressures of climate commitments, volatile energy markets, and rapidly rising electricity demand. Within this renewed attention, SMRs are receiving particular interest. For investors, the appeal is straightforward: SMRs promise a version of nuclear power that could be more modular, predictable, and scalable than the large, bespoke gigawatt-scale units that have historically dominated the sector. While traditional nuclear development has often been restrained by concerns about cost overruns, safety incidents, and unresolved waste questions, the landscape is shifting. The Nuclear Energy Agency now tracks a large and growing pipeline of SMR designs, and billions of dollars in venture and corporate capital have been committed to move the most promising concepts towards demonstration and eventual commercial operation.

The core argument behind SMRs is that they can integrate more flexibly with wind and solar, be built and financed in smaller increments, and offer investors a pathway to replicate

standardised plants across multiple sites. These characteristics, if demonstrated, could position SMRs as a viable building block of the energy transition, particularly in markets where grid constraints, industrial heat demand, or policy frameworks call for firm, low-carbon power that complements renewables.

A changing capacity outlook to 2050

Looking ahead to 2050, nuclear capacity is expected to grow meaningfully as electrification accelerates across transport, buildings, digital infrastructure and industry. Much of the capacity slated for the 2020s and early 2030s remains tied to traditional large reactors that are already in construction or advanced planning, reflecting long project cycles. But as governments reassess their decarbonisation pathways, new nuclear options, including SMRs, are beginning to feature more prominently.

Asian nuclear capacity growth will form the centre of nuclear capacity growth while SMRs will play a more prominent role in EU and US plans

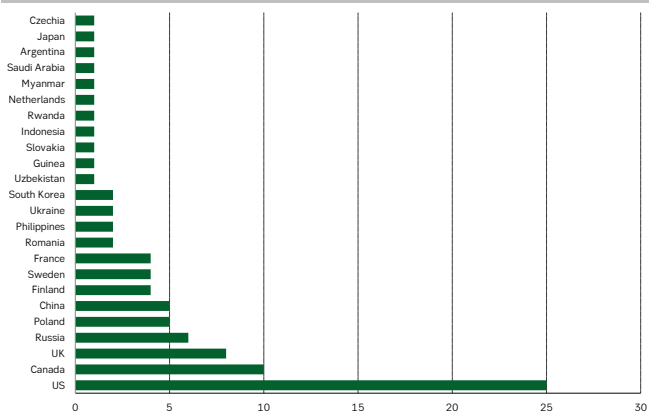
Projections suggest global nuclear capacity could reach roughly 800GW by 2050, with Asia leading the expansion. Importantly, more than a third of that potential sit in “unannounced” or early-concept stages, projects that are not yet tied to a specific design. If SMR technologies reach regulatory and supply-chain maturity early in the 2030s, a portion of these future builds could be redirected towards modular units. For investors, this means that SMRs may become a credible competitor for capital allocation in the next decade, provided early demonstrations prove deliverable on time and within budget.

Global distribution of SMR development efforts

Current development patterns offer some insight into where commercial activity may first materialise. Based on Nuclear Energy Agency (NEA) tracking, nearly half of all active SMR technologies originate in North America, where public-private models and a strong innovation ecosystem have driven early momentum. Europe accounts for nearly one-third of the tracked designs, reflecting both decarbonisation ambitions and regional energy-security concerns. Asia holds a smaller but nonetheless important share, with China in particular advancing both conventional and next-generation concepts under state-led programmes. Russia also continues to invest in modular reactors for remote locations and maritime applications.

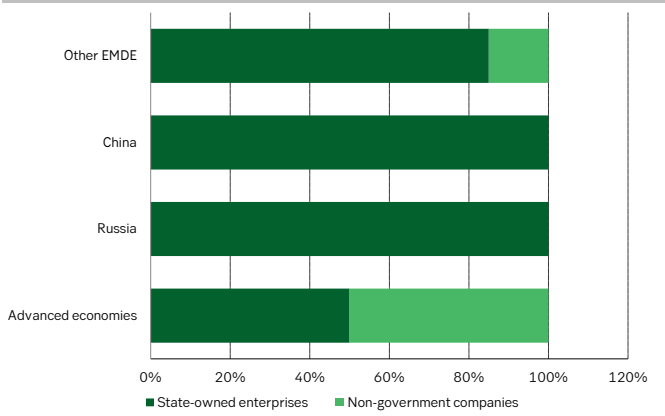
Thus far, water-cooled SMRs, those closely related to today's light-water reactors, have progressed most quickly, leveraging existing regulatory familiarity and supply-chain capability. However, incremental licensing developments in Generation IV designs, such as molten-salt, liquid-metal, and high-temperature gas-cooled reactors, indicate that advanced concepts are beginning to gain traction in select countries. A handful of gas-cooled and molten-salt projects have reached critical licensing milestones, signalling early regulatory learning in these novel systems.

SMR activity, 2025 (units)



Source: IAEA

Sponsor type by market



Source: IAEA, IEA

A broad portfolio of projects is developing in parallel

Assessing commercial readiness across projects

Twelve projects stand out with the strongest overall momentum. Russia dominates this group, driven by state-supported investment and tightly integrated fuel supply and manufacturing capabilities. China and the US also feature prominently, each with several high-performing initiatives that highlight a mix of public leadership and private-sector innovation. Canada, Romania, Argentina and Uzbekistan are likewise represented among the most competitive efforts. A clear distinction is evident in development models: Russian and Chinese projects tend to advance through centrally coordinated state programmes, while those in the US are more often propelled by private capital, strategic partnerships, and supportive policy frameworks.

Currently, large nuclear plants dominate but there is an increasing focus on SMRs

First movers and near-term candidates

Although conventional nuclear plants still dominate global construction pipelines, a handful of SMRs have already crossed the threshold into operation. Russia operates a floating nuclear power plant based on twin 35MW pressurised-water modules that have delivered power since 2019. In China, the HTR-PM, a high-temperature, pebble-bed gas-cooled design, entered commercial service in late 2023. Its ability to reach high outlet temperatures suggests long-term potential not only for electricity but also for industrial heat applications. Japan's small reactor, though not a commercial SMR in the modern sense, demonstrates long-running operational experience in compact reactor systems.

Risk sharing with governments central to developing new technologies

In North America, the real test will come over the next several years. The US hosts the world's largest pipeline of SMR developers, ranging from light-water concepts to advanced fuels and coolants. The federal government has begun reshaping regulatory and financing structures to support first-of-a-kind deployments, recognising that private investors alone cannot shoulder demonstration risk.

Canada is also moving decisively. Ontario has granted approvals for the nation's first SMR at the Darlington site, using GE Hitachi's BWRX-300 design. With a construction license secured and an operational target in sight, Darlington is widely viewed as one of the most credible candidates to deliver a replicable model in the near term. GE Hitachi has expanded its footprint with an agreement in Poland that contemplates up to 24 SMR units, an indication of growing demand among European industrial players seeking dependable, low-carbon power.

In the US, the BWRX-300 and Kairos Power's Hermes molten-salt reactor have attracted great attention. The former is advancing through the federal licensing process, with early-2030s operation envisioned at the Tennessee Valley Authority's Clinch River site. The latter is building its low-power demonstration reactor, marking the first US authorisation for a Generation IV technology in more than half a century.

NuScale, one of the earliest and best-known SMR developers, continues to promote its 77MW module, designed to be deployed in multi-unit configurations. Its partnership-based deployments in Tennessee and Romania, including a six-module 462MW project in Doicești, highlight the role of international cooperation and anchor-customer commitments in advancing SMRs towards final investment decisions.

In Asia, China's ACP100, a 125MW integrated pressurised-water design, is nearing completion after entering construction in 2021. It benefits from state backing, strong domestic supply chains, and a track record of rapid execution across multiple reactor types. Russia, through Rosatom, continues to push forward with floating RITM-200S reactors and other maritime concepts designed for remote or harsh environments.

SMRs to be considered with final investment decision in 2029-30

Europe is also gaining momentum. Sweden's Ringhals programme, with Vattenfall evaluating GE Hitachi and Rolls-Royce SMR designs, could become the first Nordic deployment by the mid-2030s. The UK is advancing Rolls-Royce's 470MW design through its national assessment process and has committed billions in government support. Interest from Poland, Czechia, and Sweden further underscores the region's appetite for SMRs as part of industrial and grid-transition strategies.

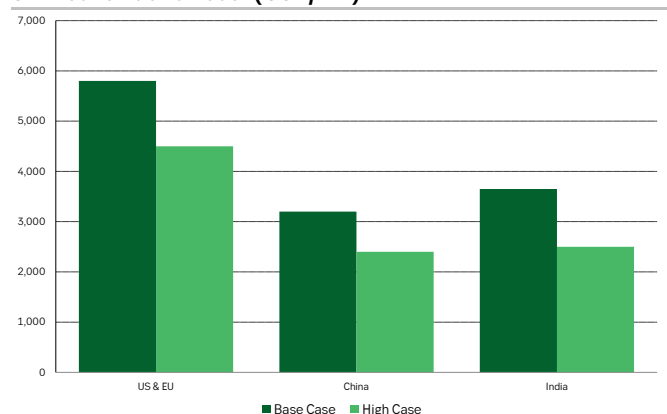
The cost challenge

Can SMRs deliver competitive, repeatable costs? First-of-a-kind units almost always face cost premiums, and the true financial promise of SMRs rests on achieving economies of series, standardised units, replicated across multiple sites, produced with factory-built modules, and built by a trained labour force that gains efficiency with every unit. Investors will require evidence that schedule discipline and cost predictability are attainable, not merely aspirational.

Learning curve and standardisation to support cost decreases in construction cost, construction time and supply chain

Two trends strengthen the investment case. Large technology companies, particularly those operating energy-intensive data centres, increasingly seek 24/7 carbon-free electricity and are prepared to sign long-term offtake agreements with nuclear developers. Second, governments are modernising incentives, through tax credits, risk-sharing mechanisms, regulated revenue models or direct funding, to reduce the cost of capital and attract private investment.

SMR construction cost (USD/kW)



Source: IEA (2025)

Nuclear power is suited to generate heat, and SMR could play a role in heating

Opportunities and remaining uncertainties

From an investor's perspective, the potential upside is significant. SMRs offer a pathway to modular, scalable, low-carbon power that can be tailored to industrial clusters, remote grids, or brownfield sites where large reactors would be impractical. Advanced SMRs can supply high-temperature heat, enabling new revenue opportunities in hydrogen,

synthetic fuels, district heating, and industrial processing. Their ability to ramp up output could complement variable renewable energy, enhancing system-level reliability.

However, key uncertainties remain. Manufacturing capacity for reactor modules and components must expand. Licensing frameworks need to evolve to accommodate standardised designs and multi-site deployment. Fuel supply chains, especially for advanced fuels such as HALEU, must scale up. Above all, SMR developers must show that a series-based model can substitute for the economies of scale that supported traditional large reactors. Without repeatable delivery at predictable costs, investors will remain cautious.

Signals to watch in the coming years

For capital providers evaluating SMR participation, several milestones will clarify the technology's trajectory:

- First-of-a-kind SMRs that meet cost and scheduling expectations, including Darlington's BWRX-300, early Kairos Hermes achievements, China's ACP100, and Europe's initial deployments.
- Progress towards regulatory standardisation and multi-unit licensing frameworks.
- Fuel-cycle investments that ensure secure supply for advanced SMRs.
- Corporate off-take agreements from data centres, industrial customers, and utilities.
- Policy and revenue structures, such as contracts for difference or regulated asset models, that reduce risk and lower financing costs.

If these elements align, SMRs could transition from isolated demonstrations to a recognised, financeable asset class, complementing renewables and storage while strengthening energy security.

SMRs could play an important role in providing stable baseload at lower risk

Bottom line

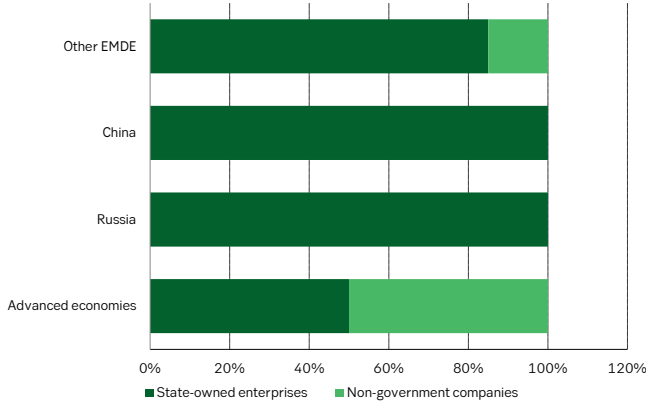
SMRs will not transform the energy system on their own, but they have a credible pathway to become a meaningful pillar of the low-carbon portfolio. The early signs – operational units in Russia and China, licensing progress across multiple countries, growing corporate interest, and advancing projects in North America and Europe – suggest the sector is maturing. The decisive test now is execution. Should the first generation of commercial SMRs deliver on time and budget, the technology could evolve into a scalable, exportable, and investable clean-energy solution throughout the 2030s and 2040s. If not, expectations may reset once again. The next several years will determine how much of the promise translates into bankable reality.

SMR construction cost (USD/kW at 2020 prices)

Region	2020	2030	2050
EU	6,600	5,100	4,500
United States	5,000	4,800	4,500
India	2,800	2,800	2,800
China	2,800	2,800	2,500

Source: IEA

Investment share by type of sponsor and region



Source: IEA (2024)

Conclusion

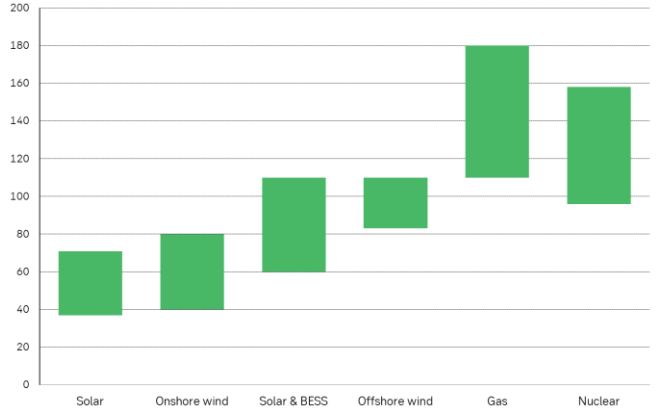
The story of nuclear reactor evolution is one of technical ingenuity shaped by external pressures. From the rudimentary Generation I prototype to today's advanced designs, each stage has sought to improve safety, efficiency, and reliability. Accidents like Three Mile Island and Chernobyl profoundly shifted design priorities, while climate change and the decarbonisation agenda now drive renewed interest in nuclear deployment. The industry stands at a crossroads: large Generation III+ reactors face economic headwinds, but SMRs and Generation IV concepts hold promises for a more sustainable, flexible, and globally deployable future. The continuing evolution of nuclear reactor designs will remain central to the debate over how best to meet the world's growing energy needs while minimising environmental impact.

Levelised cost of energy (LCOE)

Clean power became markedly cheaper from 2024 onwards, as easing supply-chain pressures and improving financing conditions pushed costs to new lows. The benchmark LCOE fell to roughly USD 36/MWh for fixed-tilt solar and USD 38/MWh for onshore wind, while four-hour standalone battery storage dropped sharply to around USD 104/MWh. This decline in storage costs was driven by battery pack prices falling to about USD 115/kWh, a 20% reduction in a single year and more than 90% below 2010 levels. These trends reflect long-established learning curves, particularly in solar, where costs have historically fallen by close to 28% with each doubling of cumulative capacity, and they also mirror the return of more favourable capital market conditions. Interest rates and subsequently return requirements have a significant impact on the viability and competitive outlook for a nuclear plant, or for that matter most long-term infrastructure investments.

Interest rates affect the overall LCOE of a project significantly, especially if there are delays

LCOE, 2025 (EUR/MWh)



Source: Lazard, BNEF

Mass production reduces costs

Manufacturing scale, especially in mainland China, remains the single most powerful force shaping global cost dynamics. Persistent oversupply pushed mainstream solar module prices to around USD 0.10/W by the end of 2024, close to or even below production cost. As a result, global fixed-tilt solar costs fell by more than 20% in one year, with little sign that oversupply will abate in the near term. Markets that can source directly or indirectly from Chinese supply chains continue to benefit most. Wind power tells a more complex story. Within China, intense competition drove turbine prices down and cut onshore wind costs significantly, but outside China higher turbine prices lifted average wind LCOEs despite the global benchmark still falling slightly. Offshore wind costs rose worldwide as expensive turbines and tight installation markets offset financing improvements. These divergences have created stark regional differences: Chinese benchmark costs for solar, wind, storage and offshore wind are often 10-60% lower than those elsewhere, while outside China average costs are roughly double. The main exception is gas-fired generation, where China lacks access to abundant low-cost fuel.

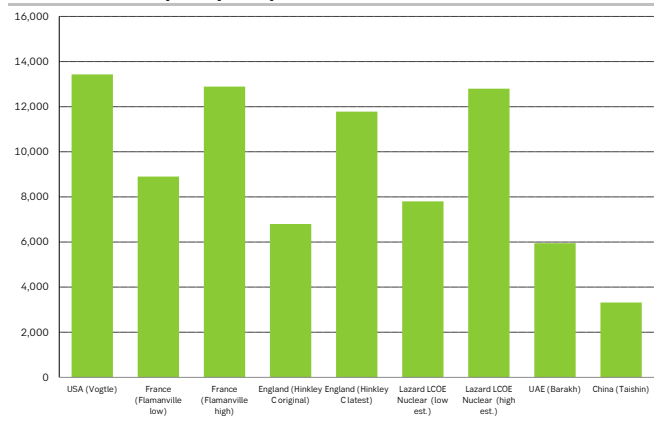
Fossil fuels-based generation is subject to greater volatility for fuel and is exposed to carbon pricing

Thermal generation costs moved in a different direction. As fuel price assumptions eased from the peaks of 2022, coal- and gas-fired LCOEs fell by around 12%. Hydrogen-fired power, however, became substantially more expensive, rising to roughly USD 425/MWh as estimates for green hydrogen costs were revised upward. In many markets that price carbon, combined-cycle gas turbines regain a short-run dispatch advantage over coal, yet their lifetime economics remain structurally worse than before the energy crisis. In Europe, new gas plants frequently require prices well above EUR 100/MWh to be viable, even before blending with hydrogen, reinforcing the view that new gas capacity is increasingly a system insurance product rather than a competitive energy source. Capacity payments, ancillary services and credible

decarbonisation strategies now matter more for bankability than headline LCOEs.

The easing of financing conditions has been pivotal in accelerating cost declines. As inflation subsided and rate-cut cycles began in 2024, the weighted average cost of capital for new solar projects fell materially. In China, state-backed developers secured extremely cheap debt, pushing capital costs down even further, while coal projects faced rising financing costs as lenders priced in stranded-asset risks. Expectations of further monetary easing in 2025 suggest additional reductions in clean energy LCOEs, particularly where falling debt costs combine with maturing technology risk. This reinforces a central lesson of recent years: for most power technologies, the cost of capital is now the dominant swing factor, and differences in financing terms increasingly explain the dispersion seen in auctions and corporate power-purchase agreements.

Estimated cost/kW (USD)



Source: Lazard (2025)

Renewables, with or without various forms of storage attached, make up the bulk of new generation capacity

Across most of the world, renewables now dominate new-build economics. Wind or solar is the cheapest source of new bulk power in systems representing roughly two-thirds of global electricity demand, and in many markets new renewables are already cheaper than operating existing coal or gas plants. India illustrates this shift clearly, where rapid solar cost declines mean new utility-scale PV can undercut the running costs of typical coal units. While some fossil assets remain competitive in specific contexts, pairing renewables with four-hour storage is increasingly decisive, allowing output to be shifted into higher-priced hours and reducing curtailment risk. In several advanced markets, combined wind-plus-storage projects are already cheaper than existing coal or gas generation.

The United Kingdom provides a clear example of how policy design can convert technology learning into bankable outcomes. After offshore wind auctions failed in 2023 due to unrealistic price caps, revised frameworks with higher caps and longer contracts restored investor confidence. Subsequent auctions secured record volumes of offshore wind at prices well below new gas generation, illustrating that stable,

long-tenor revenue arrangements can materially lower financing costs and restore competitiveness, even for capital-intensive technologies.

The LCOE does not consider system value in a volatile price environment

As renewable penetration rises, however, the distinction between LCOE and system value becomes critical. Falling wholesale prices and a growing number of negative-price hours in parts of Europe highlight that flexibility, storage, interconnection and market design increasingly determine realised revenues. Projects with contracted offtake, hybrid configurations or locational advantages command lower required returns, while merchant projects exposed solely to spot markets face rising cannibalisation risks.

Nuclear power remains sharply bifurcated. Extending the life of existing reactors is often one of the cheapest sources of low-carbon electricity, typically below EUR 40/MWh, because it is capital-light and benefits from sunk infrastructure costs. New nuclear builds, by contrast, carry substantial cost and schedule risks. Their economics are highly sensitive to financing and delivery performance, with long construction periods magnifying interest during construction. In many European and North American contexts, first-of-a-kind nuclear projects still produce electricity at multiples of renewable costs, even when compared with gas equipped with carbon capture.

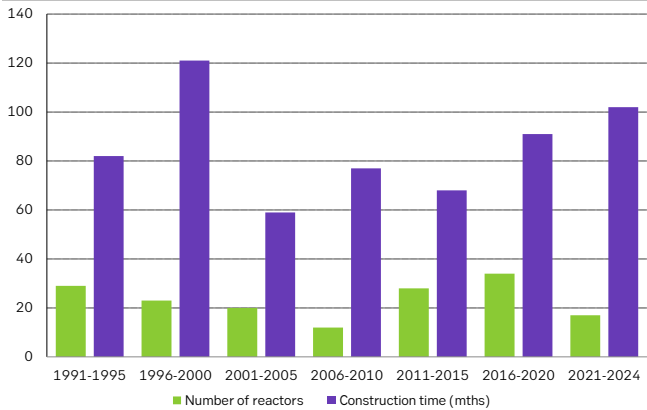
Implementation must be backed by policies and frameworks

Overall, the evidence points to a durable reshaping of global power-sector economics. Solar, wind and storage are cheaper than ever and set to become cheaper still, with new renewables increasingly outperforming both new fossil generation and the operating costs of existing plants, particularly when combined with storage. Yet low theoretical costs only translate into built projects where policy, market design and financing frameworks are aligned. For investors and lenders, the message is: renewables are the default choice for new capacity, storage is becoming essential for value capture, gas increasingly relies on capacity revenues, and nuclear competitiveness hinges on life extension and explicit risk-sharing models rather than greenfield construction alone.

Delays lead to cost overruns

The future viability of the nuclear sector hinges largely on whether both large reactors and the emerging generation of small modular reactors can be delivered on schedule and within their projected budgets. Because nuclear power is dominated by capital and financing costs, any delay or cost overrun can sharply erode its economic appeal. When investors perceive construction risks as high, the technology quickly becomes less competitive compared with other options.

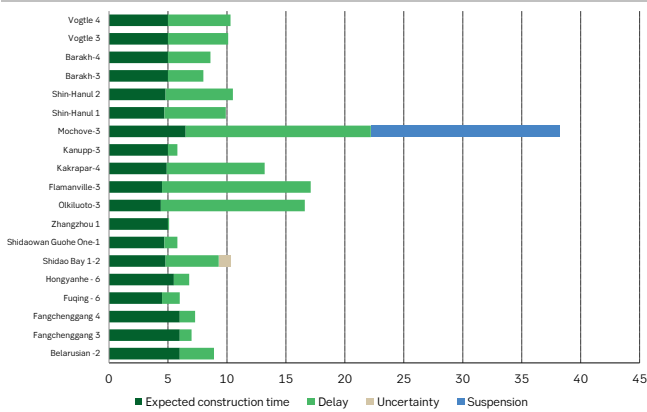
Construction time, number of reactors, 1991-2024



Source: IAEA-PRIS

Nuclear energy’s competitiveness ultimately depends not only on how much it costs to build and operate reactors, but also on how these costs compare with alternative sources of power. For technologies that provide similar services, particularly dispatchable, bulk electricity generation, the LCOE is the standard benchmark. This metric captures all major expenses over a plant’s lifetime, from construction and maintenance to fuel, carbon pricing and eventual decommissioning.

Construction time (years)



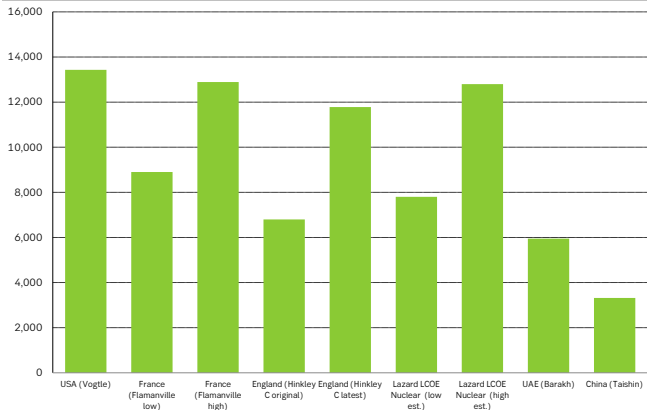
Source: WNISR, 2025

While nuclear plants require large upfront investments, they benefit from relatively low fuel costs and typically achieve high-capacity factors, often exceeding 75%. These features help keep their lifetime generating costs lower than might otherwise be expected for such capital-heavy infrastructure.

Construction cost and cost of capital are central to being able to develop competitive nuclear power

Looking ahead to 2040, projected LCOEs vary widely by region, largely depending on construction costs and the cost of financing. In China, estimates fall between USD 50-70/MWh, depending on borrowing costs. In the US and European Union, higher construction expenditures push projected LCOEs into a higher range: roughly USD 60-100/MWh in the US and USD 75-110/MWh in Europe. By contrast, extending the operational life of existing reactors remains significantly cheaper than building new ones, suggesting that lifetime extensions will continue to play an important role wherever feasible.

Cost (USD/kW)

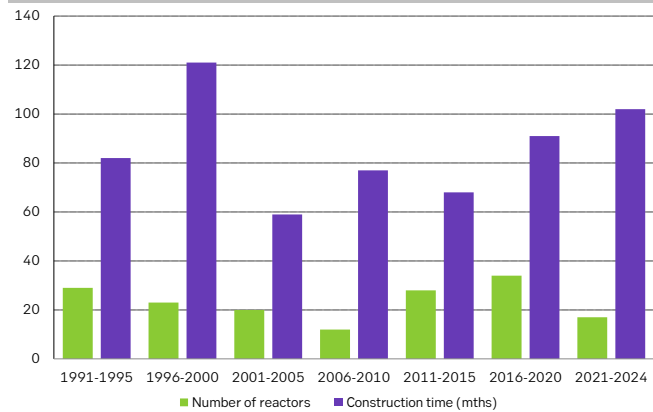


Source: Lazard (2025 est)

Construction cost is central to the viability of new nuclear plants. Although nuclear energy can provide decades of stable, low-carbon electricity with relatively low operating costs, the enormous upfront investment required to build a plant is what ultimately determines whether projects move forward, attract financing, and remain competitive with alternatives such as renewables, gas, or storage-based systems. In other words, nuclear economics are shaped far more by the capital cost, and the cost of capital, than by fuel or operational expenses. This is why governments and developers focus so intently on financing models, risk-sharing mechanisms, and project delivery discipline: construction cost is the make-or-break variable.

Delays have a significant impact on the cost of power

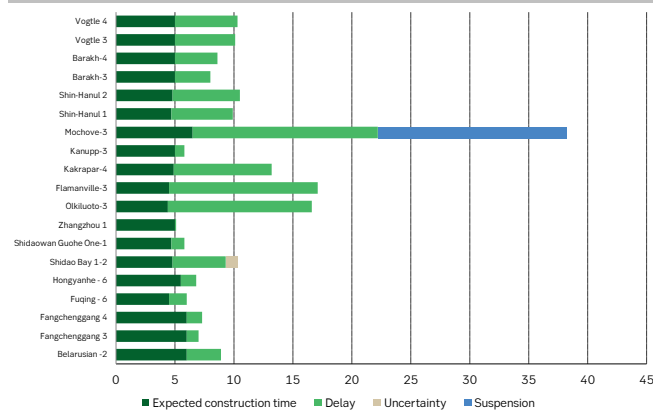
The sensitivity to construction cost arises from nuclear power’s long development timelines and the fact that capital must be tied up for many years before a plant begins earning revenue. Every year of delay compounds the financial burden, pushing up interest during construction and eroding the economic case. This is starkly illustrated by large Western nuclear projects in recent decades. Plants such as Olkiluoto 3 in Finland or Flamanville 3 in France saw costs multiply to several times their original estimates, largely because schedules stretched far beyond what was planned. Even in the UK, Hinkley Point C’s cost inflation and repeated delays have forced a complete re-evaluation of the financing model for Sizewell C. These cases show how nuclear construction risk translates almost directly into financial risk, and why investors demand predictable revenue arrangements before committing capital.

Development timeline

Source: IAEA, WNISR

Construction costs are driven by a variety of factors

What drives construction costs is a complex interplay of technical, regulatory, industrial, and political factors. One of the most important is project standardisation. When a nuclear fleet is built using a repeatable design – France's expansion in the 1970s and 80s is the classic example – learning curves, established supply chains, and experienced construction teams drive down both cost and risk. Conversely, one-off or first-of-a-kind designs tend to suffer from design changes, integration challenges, and quality-assurance issues that cascade into delays. The global resurgence of small modular reactors is, in part, an attempt to recreate the economies of replication that earlier nuclear programmes enjoyed.

Delays in recent nuclear plants (years)

Source: WNISR, IAEA (2025)

A predictable regulatory environment creates opportunities for a lower construction cost

Regulatory environments also influence cost significantly. While stringent safety oversight is non-negotiable, fragmented or unpredictable licensing processes can slow projects, require redesigns, or force developers to carry large contingencies. Countries with more coordinated regulatory frameworks, such as South Korea during its rapid nuclear build-out, have historically delivered lower construction costs. In contrast, environments where regulators, utilities and political authorities are misaligned tend to experience prolonged permitting cycles and shifting compliance requirements.

Supply-chain maturity is another major determinant. Modern nuclear construction requires highly specialised manufacturing capability, from heavy forgings to reactor coolant pumps to control systems. If the domestic supply chain has atrophied, as happened in Europe after decades of limited new build, countries must either rebuild industrial capacity or rely on imported components, both of which increase cost and risk. Workforce capability is part of this: nuclear construction requires a deep reservoir of civil engineering, welding, materials testing, and quality-control expertise, and losing these skills between build cycles leads to rising costs when new programmes eventually restart.

Government measures or support can affect financing conditions

Financing conditions interact tightly with construction risk. Since nuclear projects are capital-intensive, the cost of borrowing, driven by country risk, political confidence and policy clarity, can represent a large share of the total cost. For this reason, governments increasingly play a direct role in financing, through state loans, guarantees, or revenue-stabilisation schemes. Lowering the cost of capital can cut the LCOE by tens of percent, even if the physical construction cost remains unchanged. Conversely, high financing costs can render otherwise technically sound projects unviable.

Political stability and a predictable outlook for energy policies is central for nuclear power

Policy stability plays its own quiet but powerful role. Nuclear power requires decisions that span multiple political cycles. Uncertain future regulation, shifting energy strategies, or political opposition can raise perceived risk and push financing costs upward. Germany's abrupt post-Fukushima policy reversal is often cited as an example of how political risk affects the investment climate. Countries attempting new build today therefore focus on creating long-horizon frameworks that give investors' confidence the rules will not shift mid-construction.

Beyond economics, public acceptance can influence construction cost by shaping siting decisions, regulatory scrutiny and litigation risk. Projects that face prolonged opposition often encounter delays, redesigns, or costly mitigation measures. Conversely, where there is strong social license, permitting tends to proceed more predictably, which reduces the overall risk premium attached to the project.

All these factors combine into one overarching conclusion: construction cost is not just a technical matter but a systemic one. Delivering cost-competitive nuclear power requires stable policy, a mature supply chain, well-structured financing, and designs that can be repeated rather than reinvented. When these pieces align, construction costs fall and nuclear becomes a powerful long-term asset. When they do not, even the best engineering cannot overcome the financial drag of delays and uncertainty.

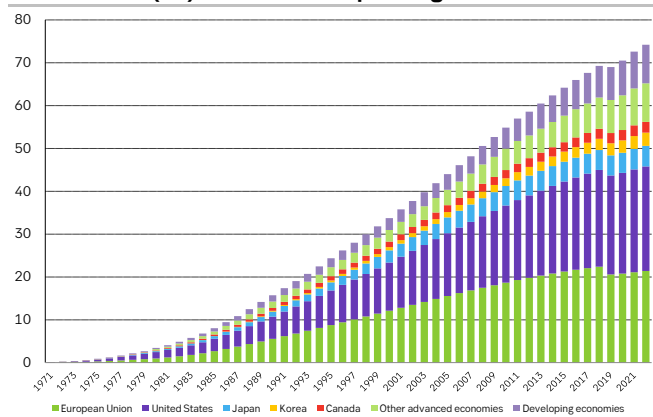
Nuclear as a sustainable investment

Nuclear power is now included in the EU taxonomy

Nuclear power and the EU taxonomy

The inclusion of nuclear energy in the European Union's sustainable finance taxonomy marks one of the most consequential, and controversial, policy debates of the energy transition. It reflects Europe's attempt to balance scientific assessment, political compromise, and market realities as it defines what can be labelled "green" for investors. The outcome has not made nuclear power fully "green" in the same way as renewables, but it has provided a carefully framed path for recognising nuclear as a legitimate, transitional tool in the EU's decarbonisation strategy.

Avoided CO₂ (Gt) due to nuclear power generation



Source: IAEA (2023)

Nuclear power has varying degree of support throughout Europe

From exclusion to cautious inclusion

When the EU Taxonomy Regulation was adopted in 2020, its goal was to guide investors by identifying which activities substantially contribute to climate and environmental objectives. Nuclear power was initially excluded pending further study, as member states were sharply divided: France and several Eastern European countries were strongly in favour, while Germany, Austria, and Luxembourg opposed its inclusion on environmental and safety grounds.

To resolve the impasse, the European Commission requested an independent scientific review. The Joint Research Centre (JRC), the Commission's in-house science body, was tasked with assessing whether nuclear could meet the taxonomy's "do no significant harm" (DNSH) principle, particularly regarding waste management and long-term environmental safety. In 2021, the JRC concluded that nuclear energy, when managed under strict EU safety and waste rules, did not cause more significant harm than other low-carbon technologies. Two additional expert groups, the Euratom Article 31 group and the Commission's Scientific Committee on Health, Environmental and Emerging Risks (SCHEER), broadly endorsed the JRC's findings.

The Complementary Climate Delegated Act

Building on this evidence, the Commission proposed the Complementary Climate Delegated Act (CDA) in February 2022, which added certain nuclear (and natural gas) activities to the taxonomy under the label of "transitional activities." This meant that nuclear power could be considered sustainable if it contributed to climate mitigation and did not undermine other environmental goals, but it would be subject to strict technical screening criteria and enhanced disclosure rules.

New technologies and lifetime extensions are included in the EU taxonomy

The act defined three eligible categories of nuclear activity:

- New Generation III+ reactors with construction permits granted before 2045.
- Lifetime extensions of existing reactors, if approved before 2040.
- Research and development of advanced technologies, such as Generation IV and small modular reactors (SMRs).

To qualify, projects must demonstrate the highest safety standards, have robust waste-management plans, and provide a credible path toward operational disposal facilities for radioactive waste. Exporting nuclear waste for disposal outside the EU is prohibited. Moreover, companies involved must publicly disclose how their nuclear activities meet these requirements, allowing investors to make informed choices.

After heated debate, the European Parliament voted in July 2022 not to block the CDA, effectively approving the inclusion of nuclear in the taxonomy. The act entered into force on 1 January 2023, with a review scheduled every three years to ensure that criteria remain aligned with technological and environmental developments.

Legal and political outcomes

The inclusion of nuclear immediately faced legal challenges. Austria and Luxembourg brought an action before the EU General Court, arguing that nuclear power cannot be considered environmentally sustainable. However, in September 2025, the Court dismissed the challenge, ruling that the Commission had acted within its powers and that the inclusion of nuclear as a transitional activity was legally justified. This decision effectively settled years of regulatory uncertainty and confirmed that nuclear energy could remain part of the EU's sustainable finance framework, at least under its current conditions.

Acceptance of nuclear power differs from country to country

Politically, the taxonomy debate underscored Europe's divide over nuclear energy. France, Finland, the Czech Republic, and several Eastern European member states viewed the outcome as a pragmatic recognition of nuclear's role in achieving net-zero targets and energy security. In contrast, Austria and Germany reaffirmed their opposition, maintaining domestic bans and calling for stronger safeguards in future revisions.

Effects on policy, investment, and permitting

Although the taxonomy is primarily a financial classification system, its ripple effects have been substantial. The inclusion of nuclear has given political and regulatory legitimacy to national programmes aiming at expanding or extending nuclear capacity, especially in countries that rely on it for low-carbon baseload generation. France, for instance, cited the taxonomy decision as supporting its plans for new reactors and its small modular reactor programme, while other member states, such as Poland, the Czech Republic, and Sweden, have referenced taxonomy alignment in justifying their nuclear strategies.

Inclusion in the EU taxonomy reduces regulatory risk for nuclear power

From an investment perspective, nuclear's taxonomy status has reduced perceived regulatory risk and clarified disclosure rules for investors. It enables nuclear projects to be included in taxonomy-aligned investment portfolios and potentially in EU Green Bonds, provided they meet all technical criteria and transparency obligations. While the taxonomy does not compel investors to finance nuclear projects, it removes the formal barrier that excluded them from "sustainable" investment categories, helping utilities and developers access a broader range of capital.

Financial markets have responded with cautious optimism. Companies linked to advanced nuclear technologies, such as Rolls-Royce SMR, GE Vernova (through GE-Hitachi), and NuScale, have seen rising investor interest, reflecting a belief that policy signals and new financing mechanisms could open the door to nuclear innovation.

There is a time limit for extension and new construction of nuclear power in the taxonomy

The taxonomy has also had practical effects on permitting and project design. Developers now align their environmental and technical documentation with taxonomy criteria, particularly on waste management, safety, and financing structure, to demonstrate eligibility and attract green capital. The time-bound conditions (permits by 2040 for extensions, 2045 for new builds) create planning pressure on member states to make decisions within this window. While the taxonomy does not directly shorten permitting times, it has raised the quality and transparency of submissions, with more emphasis on lifecycle assessments and long-term waste solutions.

Broader implications

The EU's decision ultimately reframed nuclear's role in the clean-energy transition. By recognising it as a transitional technology rather than a permanent pillar, the taxonomy strikes a delicate balance between scientific evidence, political realism, and investor demand for clarity. The 2025 court ruling has provided regulatory stability, allowing member states and companies to plan nuclear investments with greater confidence.

Still, inclusion in the taxonomy does not resolve nuclear's fundamental challenges, high capital costs, long construction times, and waste management, but it places the technology on a more even footing with renewables in EU finance law. In practice, the taxonomy has opened the door for nuclear projects to compete for sustainable finance, reinforced its policy credibility, and signalled to investors that nuclear can play a legitimate, albeit conditional, role in Europe's path to net zero.

In essence, the EU taxonomy has not declared nuclear power "green," but it has acknowledged its importance as a transitional, low-carbon technology under strict environmental and safety safeguards. This carefully balanced decision has strengthened investor confidence, clarified rules for project developers, and given member states a clearer framework for integrating nuclear power into their long-term decarbonisation strategies, while keeping the door open for future revisions as technology and policy evolve.

The fuel supply chain

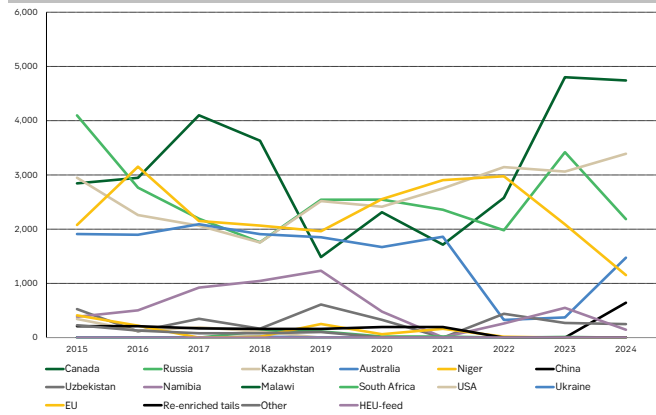
The global nuclear-fuel supply chain has become one of the most strategically sensitive parts of the energy system, and its structure today reflects both decades-old industrial geography and the intense geopolitical pressures that have reshaped fuel flows since 2022. At its core, the chain moves uranium from mined ore through refining and conversion into uranium hexafluoride, then through enrichment to raise its fissile content, and finally into fabricated fuel assemblies. Each step requires specialised facilities that exist in only a handful of countries, which is why the supply chain remains vulnerable to bottlenecks and external shocks. Conversion capacity, for example, is concentrated in Canada, France, the US, China, and Russia, and global output has recently tightened as Western converters report being sold out well into the next decade. Honeywell and ConvergDyn's Metropolis Works facility in the US, the country's only uranium-hexafluoride conversion plant, restarted after years offline but is already fully committed through 2028, highlighting how limited Western conversion availability has become.

Nuclear fuel processing and sourcing has been affected by geopolitics

Mining itself has shifted towards an even more dominant role for Kazakhstan. The country has long been the world's largest uranium producer, supplying more than 40% of global primary output in 2024, and its national champion Kazatomprom expects production to have risen further to roughly 25,000-26,500 tonnes in 2025. Although Kazakhstan's resource base is vast, the country has repeatedly emphasised supply discipline, signalling that market conditions do not yet justify a full return to previously higher extraction levels. This restraint, combined with long project lead times for new mines elsewhere, underscores why analysts expect potential structural supply deficits to emerge in the 2030s. Similar patterns play out in Canada and Australia, which remain major suppliers but are unable to match Kazakhstan's large in-situ recovery-based (as opposed to conventional open mining)

output. Utilities' procurement data from the US shows just how globally integrated the uranium trade is: in 2024, deliveries to US reactors were dominated by Canada (36%), Kazakhstan (24%), and Australia (17%), with smaller shares from Uzbekistan, Namibia, Russia, and domestic US producers.

Uranium to the EU by source (tonnes)



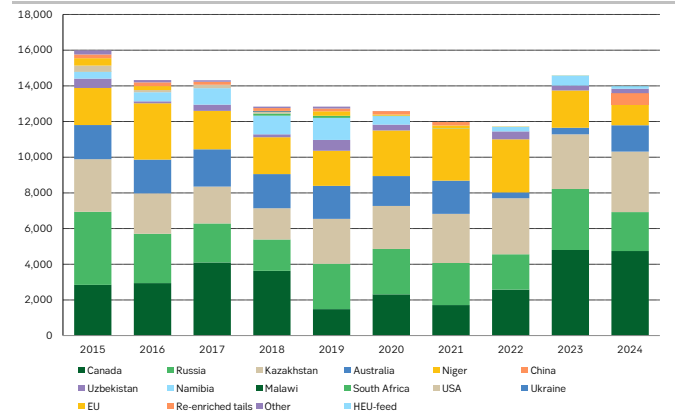
Source: WNISR

Once uranium is mined and refined, conversion and enrichment become the true chokepoints. Russia's Rosatom has historically commanded a very large share of global conversion and enrichment services, accounting in 2022 for roughly 38% of conversion capacity and 46% of enrichment capacity across world markets. Europe's dependence is particularly pronounced, as Russian entities previously supplied more than a fifth of the EU's conversion needs and nearly a third of its enrichment services. Despite the sweeping sanctions imposed on other areas of the Russian economy, nuclear fuel has been carved out of most headline restrictions because European utilities still rely heavily on these services. Even so, the political and reputational risks of this dependency have become untenable, prompting the EU to stockpile fuel and accelerate diversification efforts.

The US has taken a more forceful route. In 2024 it adopted legislation banning imports of Russian enriched uranium, subject to limited waivers that expire entirely in 2028. As a result, TENEX, the Russian enrichment exporter, issued force-majeure notices to US customers in 2024, warning that utilities would need government waivers to maintain deliveries. Washington's approach has created uncertainty in short-term supply but also catalysed major federal investment in reconstructing domestic enrichment capability. The Department of Energy (DOE) has committed USD 2.7bn to establish both LEU and high-assay low-enriched uranium (HALEU) capacity, awarding long-term task orders in early 2026 to firms including Centrus Energy, General Matter, and Orano. Centrus reached an important milestone in June 2025 by delivering 900 kilograms of HALEU to the DOE, the first meaningful commercial-scale production of HALEU in the US, underlining how far the country has come after decades without indigenous enrichment.

At the same time, Western enrichers are gradually expanding. Urenco has announced capacity additions across its US, Dutch, and German sites, including LEU+ (up to 10% U-235) capability that provides a stepping stone towards HALEU and advanced reactor fuels. Orano is also enlarging its footprint, while the EU has launched multiple research and industrial programmes, such as APIS and SAVE, and related initiatives to accelerate the provision of European-made fuel for VVER reactors in Central and Eastern Europe. These efforts were a direct response to the geopolitical shock of Russia's invasion of Ukraine, which revealed that more than 30 reactors in the EU and Ukraine relied on Russian fuel designs. Some countries, such as Czechia, have already begun receiving Westinghouse-fabricated VVER-compatible fuel assemblies, a major step towards ending the reliance on Russian supplier TVEL. The diversification process is significant for credit analysis because it reduces single-supplier concentration risk but comes with transition challenges, including licensing requirements, first-core testing, and operational adjustments that can introduce timing and inventory uncertainties.

Source of uranium by origin (tonnes)



Source: EuroAtom Agency

Laser-based technology as an emerging technology of enriching

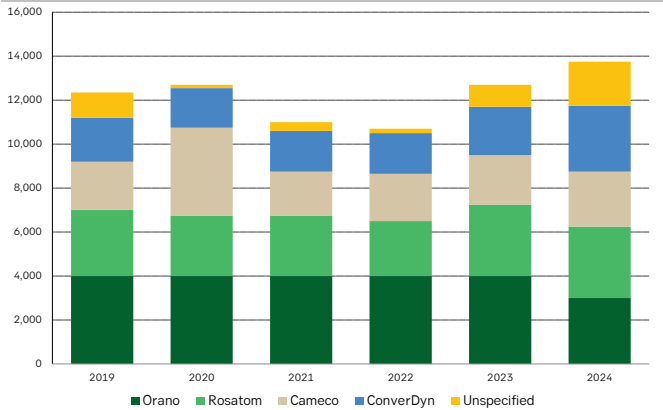
Beyond centrifuge technology, the fuel sector is also witnessing the emergence of laser-based enrichment. Global Laser Enrichment (GLE), a joint venture between Silex Systems and Cameco, completed a major demonstration of its SILEX laser-enrichment process in 2025 and advanced its licence application with the US Nuclear Regulatory Commission. The technology promises greater efficiency and the ability to re-enrich depleted uranium tails, an enormous, underutilised resource, thereby expanding Western fuel independence. While commercial viability is still several years away, laser enrichment represents a potential structural shift in enrichment economics and supply security over the long term.

All these dynamics have intersected with a pronounced tightening in the uranium, conversion, and enrichment markets. Conversion capacity is particularly constrained, with Western suppliers indicating that little or no uncommitted conversion output will be available before 2029-31, and spot conversion prices have risen accordingly. Enrichment demand has grown sharply as utilities seek to replace Russian volumes, further

elevating SWU (enrichment) prices. Uranium spot prices, meanwhile, have fluctuated between roughly the mid-USD 60s and mid-USD 80s per pound through 2024-25, buoyed by investment funds accumulating physical material and by robust long-term contracting from utilities. In the US, utilities paid an average of USD 52.71 per pound for delivered uranium in 2024, with long-term contracts dominating purchasing behaviour, an important stabilising factor for both cash-flow planning and credit risk.

Taken together, the world's nuclear-fuel supply chain is undergoing a managed but still delicate transformation. Western governments are attempting to rebuild capabilities that had atrophied over decades while utilities hedge against delivery risks by diversifying their supply books, increasing inventories, and locking in multi-cycle procurement agreements. From a credit-analysis perspective, this environment elevates the importance of counterparty quality and contract structure. Producers and enrichers with long backlogs, Western policy support, and diversified customer bases are generally positioned relatively more strongly, especially as conversion and enrichment remain seller-advantaged markets. Utilities face rising procurement costs and greater working-capital needs, but many benefit from regulated cost-recovery mechanisms, long-term fuel inventories, and a shifting supplier landscape that promises improved energy security once the ongoing expansion cycle fully matures.

Fuel suppliers EU 27+ UK



Source: EuroAtom Agency

Decommissioning and waste management

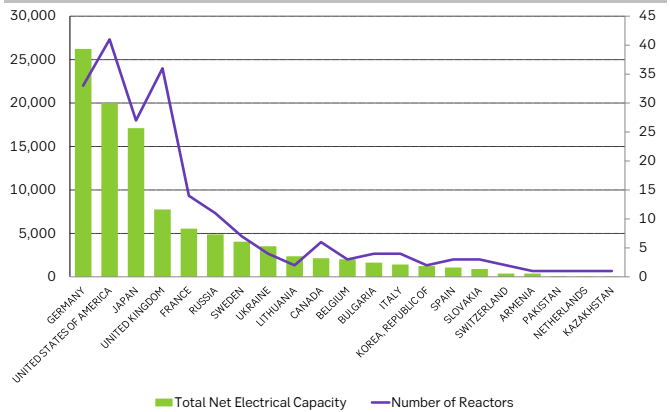
The residues of the nuclear power generation process require long-term commitment

Decommissioning is a central issue for the nuclear industry, requiring the safe handling of radioactive materials. By the end of 2023, 210 reactors worldwide had been permanently shut down, but only 23 were fully decommissioned. Over two-thirds of these reactors are in just five countries: the US, the UK, Germany, Japan, and France. Despite growing experience, global decommissioning remains limited, making future cost estimates uncertain.

Decommissioning makes up a significant part of the lifetime cost of a nuclear plant

For newly built plants, decommissioning typically represents about 15% of total lifetime investment, equating to roughly 1% of electricity costs when funds are accumulated during operation. Most countries legally require utilities to set aside these funds, with regulators approving both the mechanisms and amounts.

Reactor shutdowns by country and installed capacity (MW)



Source: IAEA (2025)

Safe disposal of spent fuel and radioactive waste remains essential to public trust in nuclear energy. Several countries are advancing deep geological repositories for long-term storage:

- Finland: Posiva applied in 2022 for an operating licence for its Olkiluoto disposal facility. Finland's Onkalo spent nuclear fuel repository is the world's first fully realised deep geological storage facility for high-level radioactive waste. Situated on Olkiluoto Island in the municipality of Eurajoki, it is designed to isolate spent nuclear fuel roughly 450 metres below the surface in exceptionally old and stable crystalline bedrock. The facility relies on the KBS-3 method, in which the waste is sealed inside copper canisters and surrounded by bentonite clay, creating multiple protective barriers intended to prevent water intrusion and ensure containment for up to 100,000 years.
- Sweden: The SFR (short-lived radioactive waste) facility under the Baltic Sea, operational since 1988, received approval in 2024 to triple its capacity.
- France: ANDRA applied in 2023 to build the Cigéo geological disposal site in northeastern France.
- Canada: In 2024, the Nuclear Waste Management Organization selected Wabigoon Lake Ojibway Nation and Ignace as host communities for its deep repository.
- Switzerland: In 2024, Nagra applied for a federal permit to build a repository and spent fuel encapsulation plant with a final federal decision awaited in 2029-30.

The emerging global nuclear decommissioning industry

As many nuclear power plants (NPPs) approach the end of their operational life, a vast new global industry dedicated to nuclear decommissioning is taking shape. Beyond power reactors, numerous other facilities involved in the nuclear fuel cycle, such as research centres, reprocessing plants, and waste treatment sites, will also need to be dismantled safely. Between now and 2050, hundreds of billions of dollars are widely expected to be spent on decommissioning activities worldwide, drawing strong interest from engineering firms, service providers, and investors positioning themselves in this growing market.

A decommissioning plan is required before construction of nuclear plants

Currently, around 420 reactors operate globally, while roughly 200 are due to enter a decommissioning phase by 2050. Given that dismantling a single plant can take two decades or more, the industry's workload is substantial and long-term. Moreover, every new nuclear project now requires a decommissioning plan before construction begins, further strengthening the future pipeline for decommissioning expertise.

The sector relies heavily on engineering, construction, waste management, and demolition companies, whose role is to safely dismantle facilities, decontaminate materials, and restore sites for future use. Environmental and sustainability principles are becoming central to this process: recovered metals, cables, and clean concrete are recycled wherever possible, and waste is carefully segregated to minimise the environmental impact. To manage this complex, highly regulated work, the industry also requires a specialised and expanding workforce to avoid future skill shortages.

To promote safety and efficiency, the International Atomic Energy Agency (IAEA) plays a pivotal role through its International Decommissioning Network, which connects organisations and experts to share knowledge, best practices, and lessons learned. The IAEA also provides technical, legal, and safety guidance, conducts reviews, and supports training and workshops worldwide.

A leading example of decommissioning expertise comes from Sellafield in the United Kingdom, one of the world's most complex nuclear sites. Built in the 1950s and home to Calder Hall, the world's first commercial nuclear power station, the site includes numerous legacy facilities in proximity. Decommissioning began in the 1980s and is set to continue well into the next century. Sellafield has developed innovative digital and robotic solutions for dismantling and waste management and shares this expertise internationally to help improve design and operational standards for future facilities.

The economic impact of decommissioning is significant. In 2021, the UK's Nuclear Decommissioning Authority, which oversees Sellafield, spent about 55% of its USD 4bn annual budget on services provided by partner companies, illustrating the substantial supply-chain opportunities associated with this field.

Globally, experience from past projects is helping to accelerate decommissioning timelines and reduce costs. In the US, recent projects aim to complete the main dismantling phase of commercial reactors in five to seven years, roughly half the historical average. This is achieved through improved project coordination, early regulatory approvals, and efficient waste recycling. Early authorisation for decommissioning, as seen in Germany, also shortens the overall project duration by enabling work to start immediately after plant shutdown. Furthermore, increasing the recycling and re-use of materials from dismantled facilities plays a key role in cutting costs and advancing circular economic principles in the nuclear sector.

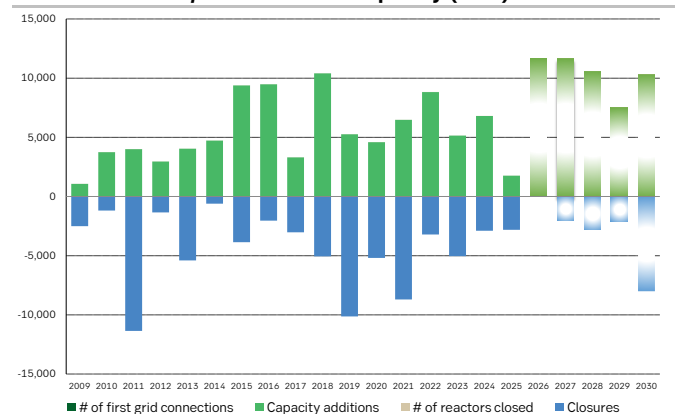
Decommissioning and waste management set to expand operations

Overall, the global decommissioning industry is entering a period of rapid growth and professionalisation. It combines high technical complexity with significant environmental and economic opportunities, transforming nuclear end-of-life management into one of the most dynamic and strategically important segments of the global energy transition.

Conclusion

The global nuclear sector is entering a decisive end-of-life phase that will shape capacity, costs, and policy for decades. Even with the growing trend toward lifetime extensions to 60 years, and in some cases 80, the age profile of the fleet implies a large, unavoidable retirement wave. Under a conservative 40-year assumption, nearly 100 additional reactors would be expected to shut by 2030, with more than 150 further closures likely by 2065, on top of units already beyond 40 years and those in long-term outage. While life-extension programmes will soften the cliff, they will not eliminate it.

Recent additions/retirements in capacity (MW)



Source: BNEF (January 2026)

Decommissioning is emerging as a strategic bottleneck. More than 200 reactors have been permanently closed worldwide, representing over 100GW of retired capacity, yet only a small subset has completed the full decommissioning cycle and even fewer sites have been returned to greenfield status. Most shut units are partway through lengthy, complex dismantling processes that frequently last two decades or more, often longer than the plants operated. Cost and schedule estimates have historically been optimistic; as the cohort of aging

reactors grows, underestimation risks solidifying into material fiscal burdens for operators, governments, and ratepayers.

The issue is most acute in Europe. It accounts for most global closures and now faces parallel decommissioning programmes at a scale without precedent. Progress is uneven: some jurisdictions are refining strategies and moving projects forward, while others are confronting funding gaps, escalating cost projections, and slippages in completion dates. The variability underscores differences in regulatory capacity, supply-chain maturity, and policy commitment to long-term dismantling efforts.

Global experience with full decommissioning remains limited and skewed towards smaller, older units. That creates a knowledge and capability gap as larger, more complex reactors approach retirement. Meanwhile, dozens of reactors remain in interim stages, ranging from warm-up to hot-zone dismantling, or in long-term enclosure, reflecting technical constraints, financing hurdles, or policy choices to defer work.

The implications are clear. First, decommissioning needs to be treated as core infrastructure policy, not a back-end afterthought: transparent provisioning, robust cost audits, and strengthened industrial capacity are essential. Second, lifetime extensions can create breathing space but must be paired with realistic end-of-life planning. Third, as retirements accelerate, countries will need coherent strategies for system reliability and replacement capacity, ensuring that the exit of aging nuclear units does not compromise decarbonisation trajectories or grid stability.

Outlook by geography

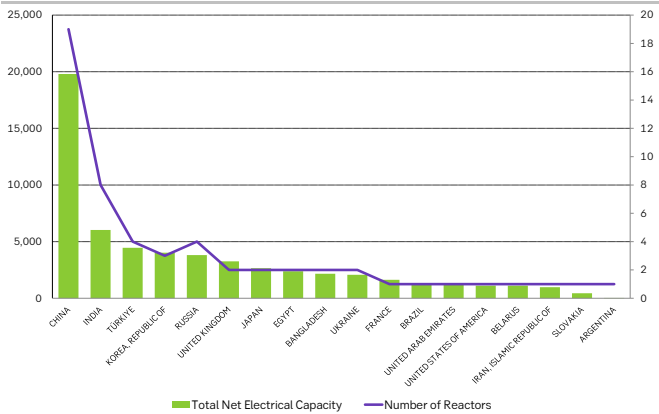
National strategies for nuclear power

Introduction

The outlook for nuclear power differs widely depending on the conditions and preferences of each country

The global role of nuclear energy is shaped not only by technology, but also by each country's policies, history, and political culture. Three examples highlight how different these paths can be: France, which developed one of the world's most nuclear-reliant power systems; Germany, which chose to completely phase out nuclear power; and China, which is rapidly scaling up its reactor fleet. These cases reflect varying priorities around energy security, public acceptance, costs, and climate goals, and together they offer insights for today's renewed global discussion about nuclear power's future.

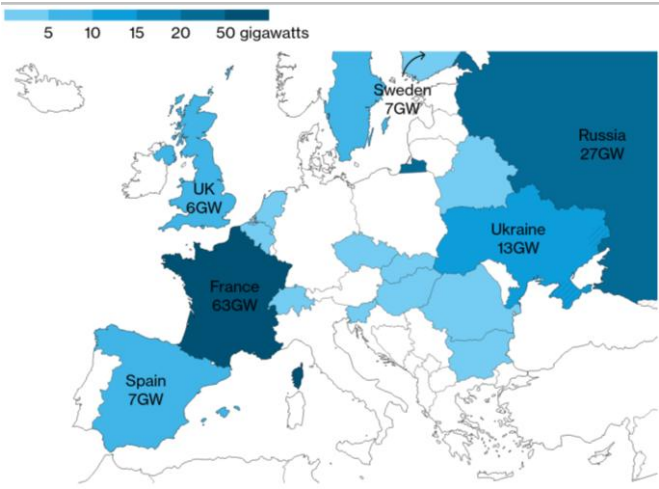
Nuclear plants under construction (MW)



Source: IAEA (2025)

European nuclear power is concentrated mainly in a few specific markets.

European nuclear capacity overview



Source: BNEF

France has a long tradition of operating nuclear power

France: The nuclear-heavy model

France is often cited as the classic example of a country betting heavily on nuclear power. Beginning in the 1970s, following the oil shocks, France adopted the Messmer Plan, a state-driven initiative to reduce dependence on imported fossil fuels. Within a few decades, France constructed a large and standardised fleet of pressurised water reactors (PWRs).

By the 1990s, nuclear provided around 75% of French electricity generation, the highest share of any major economy. This had several outcomes:

- Energy security: France achieved independence from volatile oil and gas markets.
- Low emissions: France's electricity sector became among the least carbon-intensive in Europe.
- Industrial capability: The nuclear build-out supported a domestic industry (Areva/EDF) capable of designing, building, and exporting reactors.

However, challenges have emerged:

- Aging fleet: Most reactors are now over 30 years old, requiring costly upgrades and life-extension programmes.
- Delays in new projects: The Flamanville-3 EPR reactor has been plagued by technical problems, delays, and ballooning costs.
- Policy uncertainty: Successive governments have oscillated between maintaining or reducing nuclear's share. Recent policy under President Macron reaffirmed nuclear as part of France's decarbonisation strategy, with plans for new reactors and SMR development.

Lesson: France shows that a state-led, standardised nuclear programme can deliver large-scale low-carbon electricity, but sustaining such a system requires consistent investment, technological renewal, and public trust.

Germany: The nuclear phase-out

Germany provides a stark contrast. While it initially built a fleet of 17 nuclear reactors, generating around 25% of its electricity by the early 2000s, nuclear has become politically untenable in German society.

Germany has long had a strong opposition to nuclear power

The anti-nuclear movement gained strength after Chernobyl in 1986, when radioactive fallout reached parts of Germany. Political parties such as the Greens capitalised on public fear and opposition. In 2000, the government signed an agreement to phase out nuclear power gradually.

The decision was briefly reversed in 2010, but the Fukushima accident in 2011 triggered a decisive shift. Then-Chancellor Angela Merkel announced a complete phase-out by 2022, accelerating the closure of reactors.

Outcomes of the phase-out:

- Expansion of renewables: Germany launched the Energiewende, a massive investment in wind and solar, which now supplies over 40% of its electricity.
- Continued reliance on fossil fuels: During the transition, Germany relied heavily on coal and natural gas, raising questions about climate consistency. The 2022-23 energy crisis, linked to reduced Russian gas supplies, exposed vulnerabilities.
- Cost implications: Consumers bear high electricity prices, partly due to renewable subsidies and grid integration costs.
- Climate trade-offs: Germany has cut emissions overall, but this has been slower than expected, and the nuclear exit removed a significant source of low-carbon power.

Lesson: Germany demonstrates that political and social acceptance can outweigh climate goals in shaping nuclear policy. While renewables have grown impressively, the absence of nuclear created short-term setbacks in emissions reduction and energy security.

China's demand for nuclear power makes its nuclear capacity the primary growth market

China: The expanding nuclear powerhouse

China represents the opposite trajectory, a country using nuclear to fuel rapid growth and decarbonisation. China began with imported reactor designs in the 1980s and 1990s but has since developed its own technology base, including the Hualong One and CAP1400 designs.

Key features of China's nuclear strategy:

- Scale and speed: As of the early 2020s, China operates more than 50 reactors and has around 20 under construction. It is now the world's most active builder of nuclear power plants.
- Integration with climate goals: Nuclear is framed as a complement to renewables in China's pledge to achieve carbon neutrality by 2060.
- State-driven financing and planning: Centralised governance enables coordinated long-term investment, reducing the financial risks that plague Western projects.
- Technological innovation: China is pursuing advanced reactors, including high-temperature gas-cooled reactors and SMRs, as well as heavy investment in fusion research.

Challenges remain, including managing costs, public perception, and the long-term safety of a rapidly expanding fleet. Nonetheless, China views nuclear as central to energy security, pollution reduction, and technological leadership.

Lesson: China shows that strong state support and centralised planning can overcome many of the barriers seen in liberalised Western markets. However, questions remain about replicability in other political contexts.

Contrasts and lessons

Comparing France, Germany, and China illustrates three distinct pathways:

- France: Demonstrates the benefits of a consistent, centralised approach to nuclear, achieving low-carbon electricity but now facing aging infrastructure and project management challenges.
- Germany: Illustrates how public opposition and political culture can override nuclear's decarbonisation benefits, showing that social acceptance is a critical determinant of energy strategy.
- China: Highlights the potential of state-driven expansion, integrating nuclear as part of a broader climate and industrial policy, though its model may not be directly transferable elsewhere.

From these cases, several observations can be made:

1. Policy consistency is crucial. Nuclear requires decades-long planning horizons and shifts in policy can undermine investment and public trust.
2. Public perception matters. Without social license, nuclear struggles to gain traction, regardless of technical merits.
3. Economics cannot be ignored. France's difficulties with the EPR and Germany's high renewable costs show that both nuclear and renewables face challenges that require careful management.
4. Diverse strategies exist. No single model fits all nations; geography, politics, and culture heavily shape the role of nuclear in decarbonisation.

Conclusion

France, Germany, and China embody three contrasting narratives of nuclear power, deep reliance, deliberate exit, and rapid expansion. Together, they underscore that nuclear's role in a decarbonised energy mix is not determined by technology alone but by political will, societal attitudes, and strategic priorities. For countries weighing nuclear as part of their climate strategies, these lessons highlight both the opportunities and the pitfalls of different paths.

Nuclear optimism in the Nordics

Nordic nuclear power has traditionally been concentrated in Finland and Sweden

The growing share of intermittent renewable generation in the Nordic region has increased the need for stable, dispatchable power sources. Onshore wind has expanded rapidly and in 2024 accounted for nearly 20% of electricity generation, second only to hydropower's 50%, with nuclear providing 18.6%. However, the rising contribution of variable output has exposed challenges: capture rates in Finland fell to just 33% in mid-2025, the lowest in Europe, while the number of hours with negative power prices surged to 739 in 2024 from 467 in 2023, also the continent's highest. These trends highlight the economic and operational difficulties of relying heavily on non-dispatchable sources like wind.

SMRs are being considered

Against this backdrop, nuclear energy is gaining renewed attention as a source of firm, zero-emission baseload capacity. Sweden's Ringhals site is emerging as a key test case, with Vattenfall shortlisting GE Vernova and Rolls-Royce SMR as potential suppliers of small modular reactors (SMRs). Apart from a successful deployment of SMRs in the regional power mix, we expect Nordic nuclear generation to remain broadly stable until 2050.

Swedish support scheme proposed for new nuclear power

Political and corporate interest in nuclear is also rising. Sweden has introduced a contracts-for-difference scheme to support up to 5GW of new nuclear capacity, while Finland is weighing up fleet expansion. In Norway, nuclear is receiving growing support among the political parties. Mentions of nuclear topics such as "SMR," "baseload power," and "zero-emission" in Nordic company earnings calls have increased significantly since 2022, we believe, reflecting mounting concerns about intermittency. This mirrors trends in the US, where policy support has recently put nuclear firmly back on the energy agenda.

Finland's Fennovoima nuclear project has been formally terminated following the country's decision to end its dependence on Russian energy and persistent delays by its Russian partner. In May 2022, Fennovoima cancelled its contract with Rosatom, citing the supplier's failure to meet delivery timelines. The Finnish government, emphasising energy independence in the wake of Russia's invasion of Ukraine, supported the decision and confirmed it would not issue a construction permit for the plant. The termination has since triggered a legal dispute between the two parties. A Dispute Review Board initially ruled that Fennovoima's cancellation was unlawful, but that decision is being appealed. Rosatom is seeking USD 3bn in damages, while Fennovoima has filed a USD 2bn counterclaim. Rosatom intensified the dispute in May 2025 by launching a substantial lawsuit in Moscow, demanding RUB 227.8bn, roughly USD 2.8bn, from the Finnish companies Fortum and Outokumpu. The Finnish side

has consistently maintained that it was never contractually bound in a way that would justify such claims, and although the case is moving forward within the Russian legal system, any ruling issued there would be difficult to enforce internationally. The Russian proceedings have since slowed. A hearing originally scheduled for mid-2025 has been postponed until March 2026, which means that no substantive progress is likely before then. Meanwhile, Fennovoima itself has largely wound down its normal operations. After cancelling the Hanhikivi-1 project, the company laid off almost its entire staff and now functions mainly to manage the ongoing legal processes stemming from the project's termination.

Norway and Denmark are considering potential nuclear capacity although existing Nordic markets are more likely to continue to lead on nuclear power in the Nordics

Emerging nuclear projects in the Nordics

Several Nordic countries are moving beyond debate and beginning to outline concrete steps toward new nuclear development. In Norway, the Ministry of Energy has created a nuclear power committee to examine opportunities for deployment, with a landmark report due in April 2026 that could mark the country's first formal step toward nuclear power.

Sweden has taken a more decisive stance, adopting pro-nuclear policies that allow new reactor construction and establish contracts-for-difference schemes to support future projects. This has encouraged a wave of early-stage proposals, particularly for small modular reactors (SMRs). Many of these projects are concentrated in the more populous southern regions of Norway, Sweden, and Finland, where electricity demand and prices are higher and interconnectors to continental Europe create additional pressures.

Despite the activity, the near-term prospects remain uncertain. In Norway, the emergence of companies like Norsk Kjernekraft has spurred interest in SMRs, but the absence of a regulatory framework and the very early project stage limit the likelihood of rapid progress. Sweden and Finland face a more pressing need, as much of their existing nuclear capacity is due to retire within 20 years, yet commercialisation risks remain high. Globally, SMR projects are still grappling with untested costs, delivery timelines, and technology readiness.

Two suppliers have been shortlisted for new SMRs at Ringhals

The most advanced Nordic initiative is Vattenfall's plan for an SMR at the Ringhals nuclear power plant in Sweden. The utility has shortlisted GE Vernova and Rolls-Royce SMR as potential suppliers, both seen as capable of delivering within targeted budgets and schedules. The project has been assigned a tentative start-up date of 2035, making it the strongest candidate to become the region's first operational SMR.

If realised, the Ringhals SMR would be strategically important not just for Sweden but also for the broader region. It could serve as a test case for how quickly next-generation nuclear can be deployed in Europe and provide a reference model for

other Nordic countries seeking to balance energy security with decarbonisation goals. For now, however, most analysts caution that completing the project within its proposed timeline remains unlikely.

Outlook for Nordic power and nuclear's potential

Hydropower is set to remain the backbone of the Nordic energy system for decades to come, providing a stable foundation as the region phases out fossil fuels and moves toward a predominantly renewable mix. Rising electricity demand is set to be met largely through an expansion of renewables, supported by this hydropower base.

Although interest in nuclear power has grown, the baseline outlook sees little change in Nordic nuclear generation through 2050. Most proposed projects are still at an early stage, and uncertainty remains high over how much new capacity can realistically be added within the next 25 years. Even so, if small modular reactor (SMR) technology matures and concrete projects advance, nuclear could secure a more meaningful role in the Nordics' long-term energy mix, complementing renewables and supporting the path to a low-carbon system.

Sweden has recently proposed a comprehensive state aid scheme to support investment in new nuclear energy, aiming to bolster its electricity supply, enhance energy security, and maintain competitive electricity prices. A detailed breakdown of the proposed support scheme is as follows:

Key elements of the Swedish nuclear support scheme

1. Legislative Framework:

- A new Act and accompanying ordinance are proposed to regulate the state aid process.
- The framework entered into force on 1 August 2025

2. Scope of Support:

- The aid is designed to support investments in new nuclear reactors with a total installed capacity of up to 5,000 MW, equivalent to about four large-scale reactors

3. Application Requirements:

- Companies must submit detailed applications to the government, including:
 - Company and ownership details
 - Management structure
 - Project plan with location and implementation timeline
 - Capital investment assessments by the owners

Financial Backing:

- The 2025 budget included over SEK 1bn (approx. USD 107m) for expanding fossil-free electricity production, including nuclear
- Specific allocations include:
 - SEK 100m for nuclear pilot and demonstration projects
 - SEK 30m to strengthen conditions for new nuclear power
 - Additional funding for streamlining the permitting process

4. Strategic Goals:

- Ensure long-term, stable electricity supply
- Support Sweden's climate goals (net-zero by 2045)
- Attract electricity-intensive industries
- Protect households from electricity price volatility

5. Political Backing:

- The initiative is supported by the Swedish government and the Sweden Democrats, reflecting a broader political consensus on the importance of nuclear in Sweden's energy mix

The Swedish nuclear support scheme is intended to stabilise and potentially lower electricity prices in the long term. How this would work is as follows:

1. Increased supply of baseload power:

- By supporting up to 5,000 MW of new nuclear capacity, the scheme would significantly boost Sweden's baseload electricity supply
- This would help reduce reliance on volatile imports or intermittent sources like wind and solar, especially during peak demand or low renewable output.

2. Price stability through predictable costs:

- Nuclear power has high upfront costs but low operating costs, making it suitable for long-term price stability.
- The government's financing model aims to reduce capital costs for investors, which in turn can lower the LCOE

3. Protection against price spikes:

- The government explicitly states that the scheme is designed to protect households from electricity price spikes
- This is especially relevant given recent energy crises in Europe, where price volatility has impacted both consumers and industries.

4. Support for industrial competitiveness:

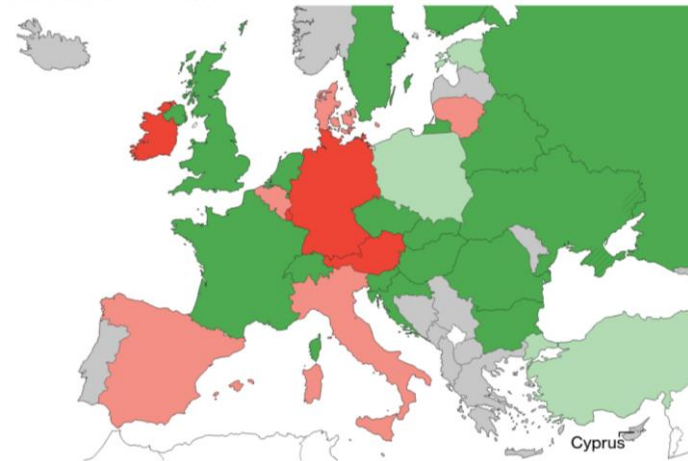
- Stable and affordable electricity is crucial for electricity-intensive industries like steel, data centre, and manufacturing.
- The scheme is intended to attract new industrial investments by ensuring a reliable and competitively priced energy supply.

5. Contract-for-difference (CfD) mechanism:

- A CfD pricing model is being considered, which guarantees a fixed price for electricity producers. If market prices fall below this level, the state pays the difference; if prices rise above it, producers pay back the surplus. This mechanism reduces investor risk and smooths out price fluctuations for consumers.

Nuclear policy overview in Europe

- Plants in operation
- Pursuing first plant
- None in operation and no ban
- Phaseout or ban in effect but being reconsidered
- Phaseout or ban in effect



Source: BNEF (January 2026)

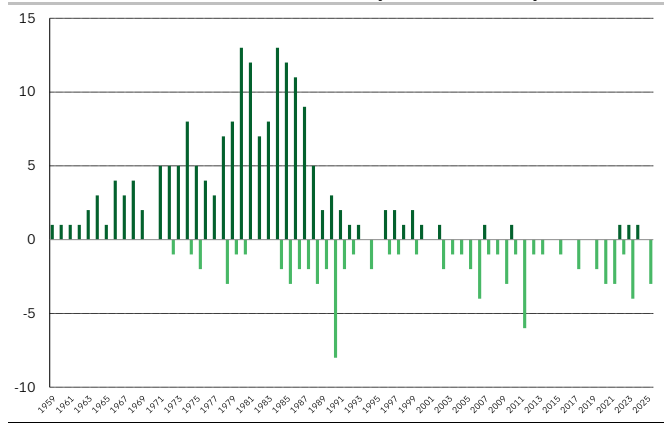
Nuclear policy overview as of mid-2025

Country	Scheme type & instruments	Who bears which risks?	Tenor / scope	Where it stands
Sweden	New Act on state aid for investments in new reactors; toolkit includes state loans and price guarantees (CfD-like); dedicated MoF financing secretariat to handle notifications. Government also mulling over the compensation law for future political shutdowns.	State de-risks financing & revenue; developers retain build/performance risk (to be set case-by-case).	Applications opened 1 Aug 2025; ordinance sets application content.	Framework adopted, now in implementation phase.
France	For fleet renewal (EPR2): state preferential loan to EDF to finance $\geq 50\%$ of six-reactor programme; separate EUR 300m state-aid for Nuward SMR R&D approved by EC.	State carries significant financing risk via cheap debt; EDF bears delivery/performance risk.	Multi-reactor programme; R&D support already cleared.	Loan plan announced; R&D aid approved.
United Kingdom	RAB model for Sizewell C (consumer levy during construction + regulated returns). Hinkley Point C preceded with a 35-yr CfD (GBP 92.50/MWh 2012 prices, GBP 89.50 if SZC proceeded). Ofgem issued economic guidance for SZC.	Construction risk partly socialised via bills; investors get regulated return; delivery risk still on project.	SZC RAB is ongoing; HPC CfD fixed for 35 years post-COD.	SZC financing package and shareholdings taking shape; govt minority stake expected.
Poland	EC-notified package for first NPP: equity injection (c. EUR 14bn), 100% state guarantees for debt (c. EUR 33bn), and a 60-yr two-way CfD with FX protection; CfD counterparty is state-owned (as with RES schemes). Beneficiary: PEJ (state-owned).	State holds financing & price risk via CfD/guarantees; PEJ bears delivery/operational risk.	Very long-tenor CfD (60y).	Terms described in EU Official Journal entry.
Czechia	EC-approved price support via long-term power-purchase contract with a state SPV for 40 years + subsidised state loan + "protection mechanism" vs unforeseeable events/policy shifts (Dukovany 5). Plans for additional units' financing to be proposed to govt.	Revenue risk shifted to state/SPV; state also provides cheap debt; CEZ subsidiary retains build/ops risks under agreed caps.	40-year support horizon.	EC approval granted, further units under study.
Romania	Support Agreement for Cernavodă 3&4 includes state guarantees to financiers, intent to implement a CfD scheme, in-kind contributions (heavy water, initial fuel), and grid works; EPCM contract signed (2024).	State covers financing backstops & price stabilisation; SNN/Nuclearelectrica retains delivery/ops risk.	Multi-stage: financing, permitting, EPCM underway.	Support Agreement in force; CfD framework in preparation.
Belgium	Life-extension package for Doel 4/Tihange 3 approved by EC after in-depth probe: shareholder loans & equity (c. EUR 2bn) and transfer of long-term waste liabilities (c. EUR 15bn) to state; aim is security of supply to 2034.	State assumes major waste liability and part of financing; operator bears ops/availability responsibilities.	Time-limited to 2034 operations.	Final EC approval 21 Feb 2025.
Finland	No bespoke state-aid scheme for new build; projects typically financed via the Mankala cost-price cooperative model (shareholders take output at cost). Longstanding legal acceptance (KHO rulings).	Private/industrial shareholders carry construction & market risk in proportion to their offtake shares.	Corporate structure rather than time-bound aid; used across energy, incl. nuclear.	Model remains the Finnish default; Fennovoima cancelled, OL3 completed under Mankala principle.
Netherlands	Govt ramp-up of budget allocations for new reactors (Borssele site), with reports of tripling funds; exploring state equity/ownership (offer to buy 70% of existing plant) and programme-level funding; separate EC-approved state-aid for a medical isotopes reactor.	Significant fiscal participation (equity) and preparatory funding to crowd-in private capital; detailed revenue model TBD.	Programmatic, multi-reactor ambition (up to four).	Funding envelopes announced/increasing; specific plant-level instruments still being designed.
Hungary	Paks II: EC 2017 state-aid approval has been annulled by the EU Court of Justice (Sept 2025) over procurement-law checks; financing hinged on a Russian state loan and state guarantees. Next steps subject to Commission follow-up/procurement compliance.	Previously, state carried bulk of financing risk; legal uncertainty now elevated pending remediation.	Two VVER-1200 units planned.	Approval annulled; project says it will proceed while working with EC.
Slovakia	Considering state-owned new unit at Bohunice ($\geq 1,000$ – $1,200$ MW); signalling state ownership/financing and partnerships (e.g., Westinghouse). Broader sector relies heavily on nuclear; recent debt financing shows investor appetite.	State likely to carry capex/financing; delivery risk allocated via EPC/vendor arrangements (TBD).	Early programme design stage.	Govt announced intent; deal architecture pending.

Source: SEB, public sources, European Commission

European nuclear capacity expansion has evolved in phases

Over the past six decades, nuclear power development in the EU27 has unfolded in distinct waves. After two relatively modest construction phases in the 1960s and 1970s, the largest expansion occurred in the 1980s, driven mainly by France. Since then, however, nuclear newbuild has slowed dramatically. Between 1995 and 2024, only 13 reactors were connected to the grid across today's EU27. Six of these were in Western Europe, five in France and one in Finland, while seven were completed in Central and Eastern Europe, notably in Slovakia, the Czech Republic, and Romania. The pace has been especially slow in the past two decades: from 2005 to 2024, only four reactors entered service. After Romania's Cernavodă-2 in 2007, Europe waited 15 years for the next startup, Finland's Olkiluoto-3 in 2022, followed by Mochovce-3 in Slovakia in 2023 and Flamanville-3 in France at the end of 2024.

EU27 reactor starts & retirements (no of reactors)

Source: WNISR (mid-2025)

Over the last 20 years, 36 nuclear plants have been closed in Europe

As of mid-2025, 100 nuclear reactors were operating in the EU27, a sharp decline from the historical peak of 136 units in 1989. This represents a reduction of more than one quarter over three and a half decades. Nuclear capacity is also unevenly distributed. Around 80% of operating reactors are concentrated in six Western European countries, with France alone accounting for 57 units, while only 20 reactors are spread across six newer member states. Reactor closures have continued in recent years, most notably in Belgium and Germany, bringing the total number of permanently shut down reactors in the EU27 to 78. More than half of these closures occurred in Western Europe, particularly in Germany. Over the last 20 years alone, 36 reactors have been retired.

Electricity production from nuclear power has fluctuated significantly. In 2024, EU27 nuclear plants generated about 617TWh net, nearly 5% more than in 2023, following a modest recovery that came after a steep 17% drop in 2022. As a result, nuclear's share of electricity generation fell to just under 22% in 2022, before edging back up to around 23% by 2024. These changes reflect both operational challenges and the long-term contraction of the fleet.

Average age of existing European nuclear reactors is 39 years

With very limited new construction, the European nuclear fleet continues to age. The average reactor age reached nearly 39 years by mid-2025. Almost nine out of 10 operating reactors have been in service for more than 30 years, and 42 have exceeded 40 years of operation. The Netherlands' Borssele reactor became the oldest in the EU, passing 51 years, in 2024, while Belgium's Doel-1 was closed shortly after its 50th anniversary in early 2025. Following the recent startups in Finland, Slovakia, and France, only one reactor remains under construction within the EU27: Mochovce-4 in Slovakia, a project whose origins date back to 1985. Once this unit finally enters service, likely in late 2025 or early 2026, no further reactor startups are scheduled for several years, meaning the fleet will continue to age and gradually shrink in the medium term.

European nuclear market needs to address challenges

In June 2025, the European Commission published its long-awaited Nuclear Illustrative Programme. It estimates that around EUR 241bn will be needed by 2050 to support lifetime extensions and limited newbuild, excluding small modular and advanced reactors. Even with these investments, the EC projects total installed nuclear capacity to rise only marginally, from about 98GW in 2025 to roughly 109GW by 2050 and could be much lower if projects face significant delays. The Commission openly acknowledges that Europe's nuclear industry faces major challenges, including limited industrial capacity, skills shortages, and an image problem that makes the sector unattractive to younger generations.

Overall, while nuclear power remains an important part of Europe's electricity system, the absence of a robust newbuild pipeline, combined with an aging reactor fleet and ongoing closures, points to a future in which nuclear capacity is likely to stagnate or decline rather than expand significantly.

Belgium

Belgium's nuclear fleet has steadily contracted over the past two decades because of a phaseout policy adopted in 2003, which capped reactor lifetimes at 40 years and banned new construction. From an original seven reactors, only four units, two at Doel and two at Tihange, with a combined capacity of about 3.5GW, remained connected to the grid as of 1 July 2025. The most recent closure was Doel-1, a 445MW reactor that was shut down in February 2025 in line with the phaseout law. In 2021, the government confirmed that all remaining reactors would close by 2025, but this trajectory changed after the Russian invasion of Ukraine in 2022 exposed energy security risks and triggered a reassessment of nuclear power's role.

Belgium extends lifetime of operational reactors

As a result, the government-initiated steps to extend the operating lives of the two youngest reactors, Doel-4 and Tihange-3, which were then around 40 years old. Negotiations with Electrabel, the Engie-owned operator, led to an interim agreement in 2023 to extend both units by 10 years. Under this deal, the Belgian state becomes an equal co-owner alongside Engie and introduces a contract-for-difference mechanism designed to balance risk and performance incentives. A central element of the agreement is a EUR 15bn lump-sum payment from Engie to the state, covering all future nuclear waste management costs for the entire Belgian nuclear fleet, including already closed reactors. The state and Electrabel also committed to financing the bulk of the EUR 2-2.5bn needed for safety upgrades and refurbishment, with the remainder coming from Luminus, a subsidiary of EDF that holds minority shares in the plants. While the legal framework enabling the lifetime extensions was adopted in April 2024, some details of the financing structure were still unresolved as of early 2025.

The European Commission approved the arrangement in February 2025 after assessing its compatibility with EU state aid rules, and the final agreement was signed in March 2025. This cleared the way for practical preparations, including fuel procurement, regulatory approvals from Belgium's nuclear safety authority, and major engineering work during planned outages in 2025. A first tranche of the waste-related payment was transferred to a newly created public agency, with the remainder due once both reactors returned to service in late 2025. Under current law, Doel-4 and Tihange-3 may operate for up to 10 additional years, but no later than the end of 2037.

Despite these extensions, nuclear output is set to decline in the short term as older reactors close. Nuclear energy still plays a dominant role in Belgium's electricity mix, generating around 42 % of electricity in 2024, although this is well below the historic peak of more than two-thirds in the mid-1980s. Natural gas, wind, solar, bioenergy, and hydro make up the remainder. With Doel-1 already closed and Tihange-1 and Doel-2 due to shut down later in 2025, total nuclear generation will fall further before stabilising around the two extended units.

At the policy level, Belgium's energy strategy has become increasingly unsettled. Long-term plans continue to emphasise carbon neutrality by 2050, largely through a major expansion of renewable energy, particularly offshore wind. However, the country missed deadlines to submit an updated National Energy and Climate Plan to the European Commission, which launched infringement proceedings in 2024. Following federal elections in June 2024, a new five-party "Arizona coalition" took office in February 2025 under prime minister De Wever. This government has signalled a sharp turn back towards nuclear power, promising major changes to the energy system, securing around 4GW of existing nuclear capacity in the short term, lifting the legal ban on new reactors, and seeking to accelerate nuclear construction and licensing, including for small modular reactors.

This political shift culminated in May 2025 with the repeal of the phaseout law and the removal of the ban on new nuclear plants. Yet significant uncertainties remain. Engie, the main operator, has publicly stated that further lifetime extensions beyond those already agreed are unrealistic and that nuclear energy no longer fits its long-term strategy. Other players, notably Luminus and EDF-linked entities, have expressed greater openness to cooperation, while also emphasising investments in renewables, storage, and flexibility.

In parallel, Belgium has seen substantial growth in renewable capacity over the past decade, especially in solar power and both onshore and offshore wind. Solar installations have more than tripled since 2015, and offshore wind capacity has expanded rapidly, although growth has slowed in recent years. New offshore projects in the Princess Elisabeth zone are now moving forward, with plans to reach nearly 6GW of offshore wind by 2030. Against this backdrop, Belgium's energy future remains contested, with nuclear power regaining political momentum even as renewables continue to expand and key industrial actors remain cautious about a full-scale nuclear revival.

Finland

Nuclear power remained a central pillar of Finland's electricity system in 2024, even though output declined. Nuclear reactors generated 31.3TWh of electricity during 2025, corresponding to approximately 40% of Finnish power generation. Finland's reactor fleet reflects a mix of technologies and suppliers: two Russian-built VVER units at Loviisa, two Swedish-designed boiling water reactors at Olkiluoto dating from the late 1970s, and the much newer Olkiluoto-3 reactor, a 1,575MW European Pressurised Reactor that represents a third-generation design.

The most recent addition to the Finnish nuclear fleet proved a challenging project, with significant delays

Olkiluoto-3 has had a particularly complex history. Built by the Areva–Siemens consortium, the project entered commercial operation more than a decade behind schedule and only reached full output for the first time in April 2023, nearly 18 years after construction began. Although joint liabilities between the operator TVO and the vendors were finally resolved in June 2025, the plant's early operational years have been marked by technical challenges. In 2024, its first annual maintenance outage took far longer than planned, and the reactor suffered a series of unexpected events later in the year, including a dropped control rod and a turbine malfunction that caused a major frequency disturbance across the Nordic power system. These issues contributed to lower-than-expected nuclear generation and raised concerns about reliability, particularly as the unit also had to reduce output at times because of high wind generation and grid constraints.

Overall electricity production in Finland in 2024 covered about 96% of domestic demand, falling just short of full self-sufficiency. Unplanned nuclear outages played a role, even though they coincided with a record year for wind power, which produced more than 20TWh. The situation was further complicated by reduced interconnection capacity after

damage to the EstLink 2 cable between Finland and Estonia, limiting the system's flexibility until the line returned to service in mid-2025. Despite these challenges, the high share of domestic generation helped stabilise the grid during a period of constrained cross-border capacity.

Operational difficulties were not limited to Olkiluoto-3. Olkiluoto-2 was taken offline in September 2024 after excessive moisture was found in its generator, leading to a lengthy repair and months of operation at reduced capacity. When the unit was being ramped back up in April 2025, the problem re-occurred, forcing another shutdown and replacement of a refurbished spare rotor. As a result, Olkiluoto-2 is expected by TVO, the operator, to continue operating below full power until its next major outage in 2026. Olkiluoto-1 also experienced a brief disruption in May 2025 due to a turbine-side malfunction, while both reactors at the Loviisa plant underwent routine maintenance and there were several shorter unplanned events, including a cooling system issue and a hydrogen leak that temporarily reduced output.

Alongside these operational challenges, Finland is actively planning for long-term nuclear operation. By mid-2025, the four reactors already subject to lifetime extensions had an average age of more than 46 years. Olkiluoto-1 and -2 have licences allowing operation until 2038, but operator TVO is now exploring further extensions to 2048 or even 2058, combined with an 8% power uprate. An environmental impact assessment submitted in late 2024 was approved by the Ministry of Economic Affairs and Employment in April 2025, enabling TVO to move forward with substantial investments aimed primarily at safety upgrades, such as modernising instrumentation and control systems. Although framed as safety-focused, these investments explicitly consider the possibility of longer operating lifetimes and higher output.

At Loviisa, the long-term commitment to nuclear power is even clearer. Both reactors were granted lifetime extensions in 2023 allowing operation until 2050, which would result in planned operating lives of around 70 years. Modernisation work scheduled from 2026 onwards is expected by the operator to slightly increase capacity, and the operator estimates that total investments of about EUR 1bn will be required by mid-century to sustain safe operation. Taken together, these developments show that while Finland's nuclear fleet has faced notable reliability issues in recent years, particularly at its newest reactor, the country remains firmly committed to nuclear power as a long-term component of its low-carbon electricity system.

Finland has taken concrete steps to diversify its nuclear fuel supply by reducing its reliance on Russia. In August 2024, Fortum loaded the first Westinghouse-manufactured fuel assemblies into the Loviisa-2 reactor, marking a significant shift away from the Russian supplier TVEL. Additional fuel deliveries are planned from 2025 onwards through Enusa of Spain, following its long-term cooperation agreement with Westinghouse. At the same time, Finland continues to navigate the fallout from the cancelled Hanhikivi nuclear project, including legal disputes triggered by Rosatom's multibillion-

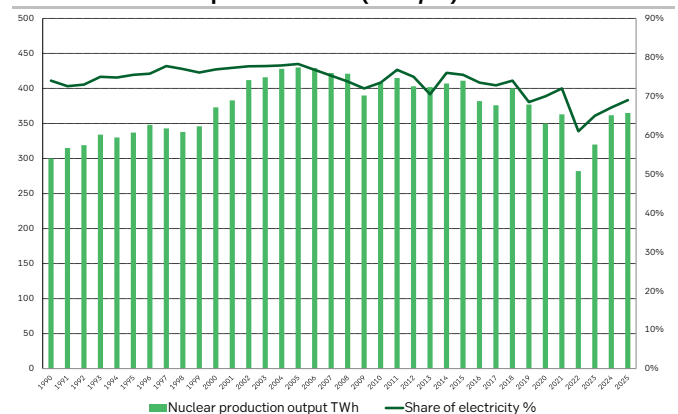
dollar lawsuit. Despite these challenges, parts of the Finnish energy sector and government remain open to new nuclear projects, provided future developers meet strict national security requirements.

French nuclear capacity recovering output fall as service backlog

France

France's nuclear sector experienced a partial rebound in 2024, with electricity generation rising by more than 40TWh from 2023 to 361.7TWh, equivalent to about 67% of national power production, and then to 367TWh in 2025. While this marked a significant improvement after the deep slump of 2022, output remained well below the levels that once defined the French nuclear model. Between 2005 and 2015, annual nuclear production typically hovered around 400TWh and peaked at 430TWh in 2005. For the ninth consecutive year, generation has stayed below that threshold, and nuclear output has declined across most of the rest of the European Union. France aims to generate over 400TWh/year of nuclear power in the future although the exact timing is uncertain.

French nuclear output and share (TWh/%)



Source: WNISR, IAEA (January 2026)

Flamanville 3 has been completed after more than a decade of delays

A symbolic milestone was reached in December 2024 with the long-awaited grid connection of Flamanville-3, France's first European Pressurised Reactor. This occurred 17 years after construction began, 12 years later than originally planned, and 25 years after the previous new reactor, Civaux-2, entered service. The start-up has been troubled. France's newly created Nuclear Safety and Radiation Protection Authority reported an unusually high number of significant safety events during commissioning, highlighting persistent operational and organisational weaknesses rather than a smooth transition to routine operation.

France harbours ambitions to expand new nuclear capacity

Despite these difficulties, the French government has doubled down on nuclear power. Legislation adopted in 2023 was designed to accelerate procedures for new reactors near existing sites and to facilitate extended operation of the current fleet. In 2022, President Emmanuel Macron announced

plans for six new EPR2 reactors, with the possibility of eight more by mid-century. Yet the EPR2 programme remains at an early and uncertain stage. Detailed designs are still incomplete, cost estimates remain provisional, and financing arrangements have not been finalised. Although EDF has already ordered major components and begun site preparations, it does not expect a final investment decision before late 2026 at the earliest.

France's EPR2 programme would signal a renewed push to expand the country's nuclear capacity and rebuild its industrial base. It involves constructing six next-generation EPR2 reactors, three identical pairs, at the Penly, Gravelines, and Bugey sites. Each unit is intended to deliver around 1,670 MW, and the shift to a fully standardised design reflects lessons learned from earlier EPR projects, including Flamanville 3, Taishan, and Hinkley Point C. The EPR2 model is deliberately more modular and "industrialised," with a simplified layout and greater use of repeatable construction methods intended to shorten build times, lower complexity, and improve reliability.

The overall programme is estimated by the French government to cost EUR 72.8bn in 2020 value, underlining both its scale and the degree of political commitment behind it. EDF has already approved funding allocations for 2026, and state-supported loans are set to play a significant role in managing the financial burden and keeping the utility's balance sheet resilient. Although initial political rhetoric suggested earlier commissioning, the schedule has since been pushed back: Construction of the first pair at Penly is now due to begin in 2027-28, with the first reactors not entering service until 2038-40. This reflects both the engineering challenge and the need to rebuild industrial and supply-chain capacity after decades with little new nuclear construction in France.

Nuclear industrial capacity expansion is already underway

Despite the long lead times, industrial mobilisation is already underway. Framatome and other suppliers have started producing key components, indicating that preparatory work is progressing and that long-lead manufacturing is being aligned with the revised timeline. Overall, the EPR2 programme signals France's attempt to re-establish itself as a leader in nuclear construction through a combination of political resolve, industrial reorganisation, and a more pragmatic, standardised approach to reactor design.

Institutional changes have also reshaped nuclear governance. At the start of 2025, the former separation between the Nuclear Safety Authority and its independent technical support body was ended through their merger into a single authority. This move, strongly contested across the political spectrum and by trade unions, reversed decades of efforts to reinforce institutional independence in nuclear oversight. At the same time, strategic decisions are increasingly centralised within the Nuclear Policy Council, chaired by the President, rather than being debated in Parliament.

Aging fleet in need of service and restoration

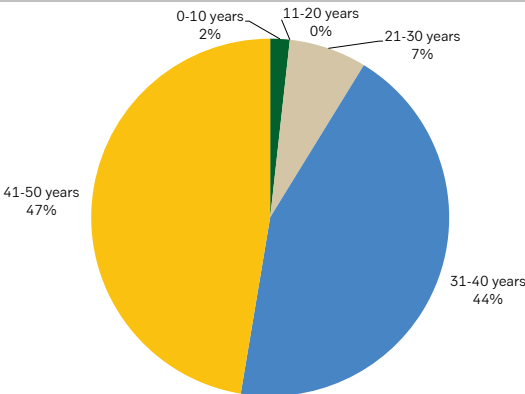
Operationally, the existing fleet of 57 pressurised water reactors showed improvement in 2024, with fewer reactors simultaneously offline and shorter cumulative outage durations than in the crisis years of 2022 and 2023. Nevertheless, unavailability remained high by historical standards, averaging about three months per reactor. Forced outages, often linked to technical issues, stayed at elevated levels. Over the five-year period from 2020 to 2024, most reactors were offline for at least a quarter of the time, underscoring the fragility of an aging fleet.

A major source of uncertainty continues to be stress corrosion cracking and thermal fatigue in safety-related piping. First identified in 2021, these phenomena have required extensive inspections, repairs, and replacements across much of the fleet. Although EDF maintains that detection and remediation techniques are now under control, new cracks were still being discovered in 2025, and safety authorities have called for additional investigations and more cautious assumptions. EDF itself expects some planned outages to be extended due to these issues.

Lifetime extension where possible to support underlying generation capacity

Lifetime extensions are therefore both essential and demanding. The average age of French reactors now exceeds 39 years, with most units already past the 40-year mark. Fourth decennial safety reviews, which are prerequisites for continued operation, involve work volumes far greater than previous inspections and have frequently exceeded planned durations by months. While the regulator has shown flexibility in allowing phased implementation of safety upgrades, this has stretched refurbishment programmes over decades and placed heavy strain on industrial capacity, skills, and finances.

Age of French nuclear fleet (2025)



Source: IAEA-PRIS

Financial pressures are significant. EDF's debt ballooned during years of low output and extended outages, then stabilised as production recovered in 2023 and 2024. However, the company remains highly exposed to electricity price volatility. Falling wholesale prices and increasing periods of very low or negative prices, driven by rapid renewable expansion, are

eroding revenues and forcing greater modulation of nuclear output. Meanwhile, the Flamanville-3 project stands as a cautionary tale, with total costs including financing and new grid connections now estimated at more than EUR 23bn and projected electricity costs by the French Court of Accounts, well above current market prices. The Flamanville 3 reactor's cost has surged from an initial EUR 3.3bn to between EUR 13.2bn and EUR 23.7bn as of early 2025, with the higher figure reflecting full financing costs. Years of delays, technical setbacks such as faulty weld repairs, and regulatory demands have significantly contributed to the overruns.

France's draft energy plan for 2030-35 reflects this tension. It assumes stable nuclear production around today's levels, no reactor closures, and continued lifetime extensions, while also projecting strong growth in solar and wind power. In practice, France is navigating a narrow path: relying on an aging nuclear fleet with unresolved technical challenges, pursuing an expensive and risky newbuild programme, and simultaneously integrating rapidly expanding renewables. The improvement seen in 2024 may have eased immediate concerns, but it has not resolved the deeper structural, financial, and industrial challenges facing the French nuclear model.

Dutch nuclear capacity consists of one nuclear plant

The Netherlands

The Netherlands today operates only one nuclear power reactor, the 482MW pressurised water reactor at Borssele, which is more than 50 years old and the oldest operating nuclear plant in the European Union. In 2024, Borssele generated about 3.4TWh of electricity, roughly 10% less than the year before, and accounted for a stable but modest 2.9% of national electricity production. This is far below the historic peak of 6.2% reached in the mid-1980s, when the country still operated a second, much smaller reactor at Dodewaard. That reactor was shut down in 1997 and is currently in a long-term safe enclosure phase, with dismantling scheduled decades after fuel removal was completed.

Despite its age, Borssele remains central to the Netherlands' nuclear debate. Plans to extend its operating lifetime beyond the current 2033 closure date, potentially by up to 20 years, are progressing but remain uncertain. Such an extension would allow the reactor to operate for as long as 80 years, a highly unusual lifespan by international standards. The government's draft Climate Fund Plan for 2026 confirms that it is investigating whether a safe and efficient lifetime extension is feasible. A pre-SALTO mission by the International Atomic Energy Agency in late 2024 identified a series of technical and organisational issues that would need to be resolved, particularly around aging management for mechanical, electrical, and civil structures, as well as the long-term availability of skilled personnel.

Borssele is operated by EPZ, which is majority-owned by a public holding company representing the province of Zeeland and local municipalities, with German utility RWE holding the remaining shares. Any lifetime extension would require amendments to the existing Borssele covenant, changes to the Nuclear Energy Act, and approval from regulators and

shareholders alike. EPZ has stated that it needs clarity by 2029 on whether the plant will continue operating or move toward decommissioning. While the government had hoped to complete the necessary legal changes by 2026 so that a licence application could follow in 2027, political instability has complicated the timeline. The collapse of the Schoof government in mid-2025 and the announcement of snap elections have added uncertainty, as has the fact that shareholders have indicated they would only support a lifetime extension if the state assumed most of the financial risk.

At the same time, Dutch nuclear policy has shifted markedly toward new construction. In recent years, successive governments have moved from cautious consideration to explicit support for nuclear expansion, even as advisory bodies warned of potential grid overcapacity. Following the 2023 elections, the new right-wing coalition committed to building four new reactors and earmarked EUR 14bn for nuclear power through 2035. However, officials have since acknowledged that this amount is likely insufficient. By early 2025, the energy minister openly stated that even two new reactors would probably require EUR 20-30bn and that the state would need to finance the early construction phases to attract private investors. As a result, earlier ambitions to have a new plant operating by 2035 can no longer be considered realistic.

Site selection and technical feasibility remain contentious. The Borssele site is often mentioned as a candidate, but expanding nuclear capacity there would require major grid reinforcements due to the already high concentration of nearby wind power. Another proposed site in Groningen faces strong local and provincial opposition. A safety review commissioned by the nuclear regulator concluded in 2025 that reactor designs from Westinghouse, EDF, and South Korea's KHNP could, in principle, be built safely in the Netherlands, though this did not guarantee licensing approval. Shortly afterward, KHNP withdrew from consideration, leaving Westinghouse and EDF as the remaining potential vendors.

Alongside large reactors, interest in small modular reactors has emerged at the regional level. Studies commissioned by provincial authorities and research institutions have identified several potentially suitable locations and suggested that SMRs could play a role in supplying both electricity and industrial heat. Nevertheless, these initiatives remain exploratory, with no binding commitments, and the national government's primary focus continues to be on large-scale reactors rather than SMRs.

Meanwhile, the Dutch electricity system is changing rapidly regardless of nuclear policy. In 2024, renewable sources generated just over half of total electricity, surpassing fossil fuels and nuclear combined. Wind and solar power have expanded at remarkable speed, with significant additions in both offshore wind and rooftop photovoltaics, making the Netherlands a global leader in solar capacity per capita. Natural gas still dominates fossil generation, but its role is shrinking. While nuclear power remains politically prominent, its actual contribution to the electricity mix is small, and environmental authorities have warned that additional policy

measures will be needed to meet near-term climate targets, regardless of whether new nuclear projects ultimately materialise.

Spain

As of mid-2025, Spain operates seven nuclear reactors with a combined capacity of about 7.1GW. In 2024, these plants produced 51.9TWh of electricity, supplying just over 19% of national electricity demand. This is a far cry from nuclear power’s peak contribution of more than 38% in 1989 and reflects both the aging nature of the fleet and the rapid growth of renewable generation. The average operating age of Spanish reactors now exceeds 40 years, placing Spain among the European countries with the oldest nuclear fleets.

Ownership of Spain’s nuclear plants is concentrated among a small group of utilities, primarily Iberdrola and Endesa, often in partnership with Naturgy and, in one case, EDP. This concentration has shaped negotiations between operators and the state, particularly around Spain’s long-standing nuclear phaseout policy. In 2019, the government reached an agreement with the main utilities as part of the country’s Integrated National Energy and Climate Plan, setting a timetable for the gradual closure of all reactors by 2035. At the same time, the agreement allowed operators to apply for lifetime extensions beyond 40 years, a more flexible approach than earlier Socialist Party proposals, which had envisaged only short deferrals of closures originally planned for the mid-2020s. The phaseout schedule was reaffirmed in Spain’s updated National Energy and Climate Plan published in 2024.

Spain nuclear fleet outlook

Reactor	Net capacity (MW)	Grid connection	Licence expiration	Expected closure
Almaraz-1	1,011	1981	2027	2027
Almaraz- 2	1,006	1983	2028	2028
Asco-1	995	1983	2030	2030
Cofrentes	1,064	1984	2030	2030
Asco-2	997	1985	2031	2032
Vandellós-2	1,047	1987	2030	2035
Trillo-1	1,003	1988	2034	2035

Source: WNISR, IAEA

Over the past five years, all Spanish reactors have received operating licence extensions beyond their initial 40-year design lives, but the closure timetable still begins later this decade. Almaraz-1 and Almaraz-2 are currently scheduled to shut down in 2027 and 2028, respectively, followed by Ascó-1 and Cofrentes around 2030, with the final reactors closing by 2035. For several units, existing licences will expire before their planned closure dates, meaning further extensions will be required if the agreed schedule is to be maintained.

The Spanish renewables market has grown rapidly

The economic environment for nuclear power in Spain has become increasingly challenging. The electricity market has seen a growing number of hours with very low or even negative prices, driven by high output from wind, solar, and hydroelectric plants. During periods of strong renewable generation, nuclear reactors have been partially or fully

disconnected from the grid because operating at market prices was no longer profitable. Such situations occurred during the Easter period in 2024 and again in spring 2025, when wholesale prices fell to exceptionally low levels. Nuclear operators and their industry association argue that a heavy tax burden, particularly fees for radioactive waste management and decommissioning, exacerbates these pressures. In mid-2024, the levy financing nuclear waste management was significantly increased following an upward revision of long-term waste cost estimates. Utilities have challenged both the revised waste plan and the higher fees in court, claiming they violate the 2019 phaseout agreement and seeking substantial compensation.

Against this backdrop, political opposition to the nuclear phaseout has intensified. Although the conservative Popular Party won the most seats in the 2023 elections, it failed to form a government, and Prime Minister Pedro Sánchez’s Socialist-led coalition remained in power into 2025 amid growing political strain. The Popular Party campaigned on reversing the nuclear phaseout and, in February 2025, succeeded in passing a non-binding parliamentary motion calling for lifetime extensions and improved economic conditions for nuclear plants. The vote exposed deep political divisions, with right-wing parties in favour and left-wing and regional parties largely opposed or abstaining.

Nuclear power plays a stabilising role

Following the vote, industry pressure mounted. Nuclear companies and suppliers, supported by firms from other sectors, issued public calls to renegotiate the 2019 agreement and revise the national energy plan to ensure continued nuclear operation. Regional leaders joined the debate, particularly in areas hosting nuclear plants. The president of Extremadura, home to the Almaraz plant, argued for extending its operation and even suggested state takeover if taxes were not reduced, while leaders in Madrid described nuclear power as essential for grid stability and economic development. Senior executives at Iberdrola warned that closing nuclear plants could raise electricity prices significantly and undermine system reliability, framing the issue as a pragmatic energy security choice rather than an ideological one.

The debate intensified further after the massive blackout that struck Spain and Portugal on 28 April 2025. The outage, the most severe in recent Iberian history, was quickly linked in public discourse to the high share of renewables in the electricity mix at the time, even before investigations were completed. A government inquiry later concluded that the blackout resulted from a complex chain of events, including miscalculations by the grid operator, inadequate responses from plants tasked with stabilising voltage, and rapid disconnections that overwhelmed the system. Renewables themselves were not identified as the root cause, though the incident highlighted vulnerabilities in grid management and coordination. In response, the government adopted urgent measures to strengthen system oversight, expand storage and flexibility, and accelerate electrification, while long-planned

investments in interconnections with France and energy storage moved forward.

Despite the controversy, Spain remains firmly committed to its long-term climate goals. The government aims for climate neutrality by 2050 and for renewables to supply more than 80% of electricity by 2030. In 2024 alone, Spain added 7.3GW of new renewable capacity, mostly solar photovoltaic, bringing total renewable capacity to over 85GW. As a result, the electricity mix is now dominated by wind and solar, with nuclear providing under 20% and fossil fuels continuing to decline. In this rapidly evolving system, nuclear power occupies an increasingly contested role, balancing aging infrastructure, economic pressures, political division, and the accelerating momentum of renewable energy.

Nuclear power is central to the Swedish energy mix

Sweden

Sweden's nuclear power output rose modestly in 2024, with the country's six remaining reactors producing 48.1TWh of electricity, about 4.4% more than the year before. Nuclear energy accounted for just under 30% of total electricity generation, a sharp contrast to the mid-1990s, when nuclear supplied more than half of Sweden's power and the fleet still comprised 12 reactors. Although nuclear remains an important component of the system, Sweden's electricity mix has changed fundamentally over the past two decades, with renewables now dominating overall production.

Aging fleet to receive lifetime extension along with potential new capacity

The Swedish nuclear fleet is aging, with an average reactor age of around 43 years by mid-2025. Four of the six reactors are already more than 40 years old, prompting operators to explore further lifetime extensions. In 2024, Vattenfall, which operates the reactors at Forsmark and Ringhals, announced a strategic intention to assess extending plant lifetimes from 60 to as much as 80 years, potentially keeping them on the grid into the 2060s. Similar plans emerged for Oskarshamn-3, whose owners, Uniper and Fortum, are studying options to continue operating the plant well beyond its current closure date in 2045. These initiatives remain under investigation, with final investment decisions expected by the operators only later in the decade and subject to stringent regulatory reviews.

Operational reliability has been a persistent challenge for Sweden's nuclear fleet in recent years. While conditions at Ringhals stabilised after severe difficulties in 2022, Forsmark-3 experienced an extended outage starting in autumn 2024 due to cracks in turbine components, delaying its return to full service until early 2025. Even after restarting, the reactor suffered an additional short outage caused by a faulty valve. At Oskarshamn-3, a planned maintenance stop in spring 2025 was repeatedly extended when cracks were discovered in critical piping, pushing the restart from early summer to late summer and eventually to September. These incidents underscored the technical risks associated with operating aging nuclear infrastructure, even as operators pursue extended lifetimes.

A reversal of long-held expectations relating to decommissioning of nuclear power

Alongside efforts to prolong existing reactors, Sweden has undergone a major policy reversal toward new nuclear construction. In 2023, the government replaced its long-standing target of "100% renewable" electricity by 2040 with a goal of "100% fossil-free" power, explicitly reopening the door to nuclear newbuild. Official plans now envision adding around 2.5GW of new nuclear capacity by 2035 and expanding to the equivalent of 10 large reactors by 2045. To coordinate this effort, the government appointed a national nuclear new-build coordinator tasked with identifying regulatory, financial, and organisational barriers and proposing solutions. Interim reports stress that licensing timelines, estimated at up to 10 years from application to grid connection, represent the most significant bottleneck, making parallel construction projects at multiple sites essential for meeting government targets.

The policy shift has been backed by new funding and legislative measures. In its 2025 budget, the government allocated dedicated resources to support fossil-free electricity generation, including funding for nuclear pilot and demonstration projects and preparatory work on permitting frameworks. Recognising that private investors are reluctant to bear the financial risks of nuclear construction, Sweden introduced a new financing model in 2025 that allows the state to cover up to 75% of project investment costs through government loans. For a hypothetical four-reactor programme, this could translate into state backing of more than SEK 300bn. The legislation also includes a contracts-for-difference mechanism that guarantees fixed revenues to operators, funded by a small levy on electricity consumers. Even so, this state aid framework still requires approval from the European Commission, and the absence of concrete projects means many details remain unresolved.

Big plans but no formal decision yet

Despite confident political rhetoric, including assertions that construction of a new nuclear plant could begin before the 2026 election, scepticism is widespread. No Swedish utility has yet committed to building a new reactor or chosen a final technology supplier. Vattenfall focuses on SMR-technology while Fortum, which holds stakes in existing Swedish reactors, has stated that while it remains interested in nuclear as a long-term option, new demand in the Nordic electricity market over the next decade will likely be met more quickly and cheaply by wind, solar, storage, and lifetime extensions of existing plants rather than by new nuclear builds.

Critics argue that Sweden's nuclear focus risks distracting from a highly successful renewable energy transition. Environmental groups warn that delays or failures in nuclear construction could undermine climate targets if renewable expansion falters. Indeed, permitting obstacles have slowed onshore wind development, and security concerns led to the cancellation of numerous offshore wind projects in 2024. At the same time, subsidies for solar power were reduced even as

photovoltaic capacity continued to grow rapidly, surpassing an installed capacity of 5GW by 2024.

In practice, Sweden's electricity system is already overwhelmingly renewable. In 2024, nearly 70% of electricity generation came from renewable sources, led by hydropower and wind, with nuclear contributing just under 30% and fossil fuels playing only a marginal role. This reality highlights the central tension in Sweden's energy debate: whether expanding nuclear power is a necessary complement to a fossil-free system or a costly and uncertain diversion from a transition that has already delivered impressive results through renewables.

Switzerland

Switzerland's nuclear power output declined slightly in 2024, with gross production falling by about 1.5% y/y to 23TWh. Because overall electricity generation reached a record 81TWh, driven largely by exceptionally strong hydropower output, nuclear power's share of the mix dropped to 28.4%. This is well below the historical peak of 43% recorded in the mid-1990s and reflects the long-term shift in Switzerland's electricity system toward renewables, particularly hydro and, increasingly, solar power.

An aging Swiss nuclear fleet drives the need to review policy decision

Switzerland operates one of the oldest nuclear fleets in the world. The average age of its four operating reactors was over 49 as of mid-2025, making the fleet second only to the Netherlands in age. Beznau-1, at 56 years, is the oldest commercial nuclear reactor still in operation globally, with Beznau-2 close behind at nearly 54 years. The Mühleberg reactor was shut down in 2019 and is currently undergoing decommissioning, leaving Beznau, Gösgen, and Leibstadt as the remaining nuclear sites.

Unlike many countries, Switzerland grants nuclear operating licences for an unlimited period, meaning reactors may continue operating if they meet safety requirements set by the regulator. In practice, operators of all remaining plants have sought to extend operation to at least 60 years, despite ongoing public and expert debates over safety issues, particularly at Beznau-1 and Leibstadt. Concerns have focused on material degradation and other age-related vulnerabilities that have been discussed extensively in past assessments.

In March 2024, Axpo launched a formal study on the technical feasibility of operating the Beznau plant beyond 60 years. Later that year, the company announced a firm decision to shut down Beznau-1 in 2033 and Beznau-2 in 2032, extending their operating lives to 64 and 61 years, respectively. To support this extended operation, Axpo committed to invest an additional CHF 350m in safety upgrades, on top of the CHF 2.5bn already spent over the plant's lifetime. The Swiss nuclear safety regulator acknowledged the operator's plans and confirmed that the next periodic safety review is scheduled for 2027, although procedural timelines mean documentation will likely need to be submitted earlier. Even with this planned

extension, there are limits to how long Beznau can operate. Experts have pointed to flaws in the reactor pressure vessel steel of Beznau-1, first identified in 2015, as a key constraint on further lifetime extensions. In addition, the plant's reliability has come under scrutiny. Unplanned outages have become more frequent, and Beznau-1's load factor dropped sharply in 2024. Technical issues unrelated to the nuclear core, such as valve failures, cable repairs, and follow-up checks on steel components, nonetheless led to prolonged shutdowns. During hot summer periods in 2024 and 2025, both Beznau units also had to reduce output or shut down temporarily to protect the Aare River from excessive thermal stress.

Possible lifetime extension and a review of new capacity

At the other Swiss nuclear plants, lifetime extension discussions remain less advanced. As of mid-2024, the operators of Gösgen and Leibstadt had not carried out in-depth studies on operating beyond 60 years, citing political uncertainty as a major obstacle. Current plans still foresee closures in 2039 for Gösgen and 2044 for Leibstadt, corresponding to 60 years of operation, although some political voices have already begun advocating for 80-year lifetimes. Beyond plant lifetimes, Switzerland is also grappling with renewed political debates over nuclear newbuild. The Energy Strategy 2050, approved by referendum in 2017, introduced a ban on constructing new nuclear power plants. However, concerns about energy security following Russia's invasion of Ukraine have revived calls to lift this ban. These efforts culminated in a popular initiative known as "Stop the Blackout," which collected enough signatures in early 2024 to force political consideration. While the Federal Council declined to put the initiative directly to a referendum, it proposed an indirect counter-proposal, including legislative changes that would repeal the newbuild ban.

Political divisions regarding nuclear power will set near-term focus on renewables

Consultations on revising the Nuclear Energy Act took place between late 2024 and spring 2025, revealing strong political divisions. Centre-right parties largely supported reopening the option of nuclear construction, while Greens and other left-leaning parties opposed it. Even if the ban were lifted, expert assessments suggest that a new nuclear plant in Switzerland would be unlikely to come online before around 2050, due to long planning, licensing, and construction timelines. Throughout the debate, the government has continued to emphasise that accelerated deployment of renewable energy remains the top priority.

Hydro power and renewables form the bulk of Swiss power generation

In practice, Switzerland's electricity system is already dominated by renewables. In 2025, hydropower alone supplied nearly 60% of electricity generation, benefiting from favourable hydrological conditions. Nuclear power contributed just over 28%, while solar power continued its rapid rise, increasing its share to more than 7%, with further growth

projected by the Swiss Federal Office of Energy in the coming years. Wind power remains marginal, and fossil fuels play only a minor role. Against this backdrop, Switzerland's nuclear future is defined less by expansion than by managing the risks and costs of an aging fleet during a period of rapid renewable growth.

Bulgaria

Bulgaria's nuclear sector is deeply rooted in its Soviet-era energy infrastructure. The country launched its nuclear programme in the 1970s with VVER-440 reactors and later added two larger VVER-1000 units at the Kozloduy nuclear power plant, which remain the backbone of Bulgarian nuclear generation today. Earlier ambitions to expand capacity further, notably through the Belene project, were repeatedly derailed after the collapse of the Soviet Union and subsequent political and economic instability.

In 2025, the two operating reactors at Kozloduy generated about 16TWh of electricity, slightly less than the year before. Despite this modest decline in output, nuclear power's share of Bulgaria's electricity mix increased to nearly 42%, mainly because coal generation fell sharply. As a result, nuclear energy has strengthened its position as Bulgaria's single most important source of electricity, underscoring its role in ensuring system stability during a period of structural change in the power sector. Historically, Bulgaria's energy system was highly dependent on Russia, not only for nuclear fuel but also for fossil fuels, especially natural gas. Before Russia's invasion of Ukraine, more than 90% of Bulgaria's gas imports came from Russia. Since then, this dependence has been dramatically reduced. Russian gas supplies for domestic consumption have stopped entirely, and Bulgaria now meets its gas demand primarily through liquefied natural gas imports via Greece and Türkiye, supported by long-term access agreements with the Turkish company BOTAŞ. At the same time, Bulgaria continues to act as a transit country for Russian gas to other parts of Europe through the Balkan Stream pipeline, which reached record transit volumes in 2024. This role, however, has brought limited financial benefit to Bulgaria, as transit fees are low and much of the economic value accrues to Gazprom and intermediaries.

Fuel diversification to reduce dependency

Fuel diversification has also become a priority in the nuclear sector. Although Kozloduy still relies on Russian-designed equipment, Bulgaria has moved decisively to reduce its dependence on Russian nuclear fuel. A long-term contract with Westinghouse was signed to supply fuel for Unit 5, with the first deliveries arriving in 2024 and further reloads in 2025. The transition to alternative fuel is scheduled to be completed by 2027. In parallel, Kozloduy has contracted Framatome to supply fuel for Unit 6 and initiated licensing work to allow Western fuel designs to be used safely in both reactors.

The two VVER-1000 units at Kozloduy are also undergoing lifetime extension programmes. Originally licensed for 30 years, they have already been refurbished to allow operation for up to 60 years. Regulatory changes adopted in 2024

replaced fixed-term operating licenses with indefinite ones, provided that periodic safety reviews are successfully completed. This effectively allows Units 5 and 6 to remain in operation if they meet evolving safety standards, reinforcing nuclear power's long-term role in Bulgaria's energy system.

New plant planned but timing uncertain

At the same time, Bulgaria is pushing forward with plans to build new nuclear capacity at Kozloduy. A project to construct two AP1000 reactors with Westinghouse technology has advanced through feasibility studies, engineering contracts, and vendor selection, with Hyundai Engineering & Construction chosen as the main construction partner. Financing discussions are ongoing, including potential support from the US Export-Import Bank, while cost estimates and final investment decisions remain unresolved. The target date for commissioning the first new unit is the mid-2030s, though timelines remain ambitious.

Meanwhile, the long-troubled Belene project continues to cast a shadow over Bulgarian nuclear policy. Although officially cancelled in 2023, debates over selling its equipment to Ukraine, repurposing the site, or even reviving the project in a new form have kept it politically alive. Against this complex backdrop, Bulgaria's long-term energy strategy positions nuclear power as a cornerstone of energy security, alongside a gradual coal phaseout and a significant expansion of solar and wind power, as the country seeks to balance decarbonisation, independence, and economic stability.

The Czech Republic

The Czech Republic relies heavily on nuclear power as a cornerstone of its electricity system. It currently operates six Russian-designed reactors at two sites: four VVER-440 units at Dukovany and two larger VVER-1000 units at Temelín. The Dukovany reactors entered service between 1985 and 1987, while both Temelín units were connected to the grid in 2000. Operating licences for all reactors are unlimited in duration but depend on successful periodic safety reviews conducted every 10 years. In 2025, nuclear power generation reached about 32.1TWh, slightly higher than the previous year, and accounted for just over 40% of total electricity production, underscoring nuclear energy's central role in the Czech power mix.

Growth for nuclear capacity planned

This importance is reinforced in the country's updated National Energy and Climate Plan, published in late 2024. The plan foresees nuclear power expanding its share of electricity generation to around 44% by 2030 and potentially peaking at close to 68% by 2040, before declining again to about 46% by mid-century. These projections reflect a strategy built on three pillars: long-term operation of existing reactors, the construction of new large nuclear units, and the eventual deployment of small and intermediate modular reactors. Existing plants are scheduled to operate for at least 60 years, subject to rigorous safety reviews, while new capacity is

intended to compensate for aging infrastructure and support decarbonisation goals.

Fuel diversification to reduce dependency on a single supplier

Reducing dependence on Russia has become a key theme in Czech nuclear policy, particularly in the fuel cycle. Historically, Dukovany and Temelín relied on Russian fuel and related services, but this began to change following geopolitical tensions and the war in Ukraine. Fuel delivery contracts with Russia expired in 2024, and alternative supplies were introduced in 2025. Westinghouse delivered the first non-Russian fuel assemblies to both nuclear plants, marking a significant milestone, with mixed-origin fuel initially loaded into the reactor cores. ČEZ, the state-controlled utility operating the plants, has also expanded on-site fuel storage capacity to improve resilience against supply disruptions. Additional diversification steps are underway. ČEZ has agreements with Framatome to develop alternative VVER-1000 fuel designs and signed a long-term contract with Urenco for uranium enrichment services extending into the mid-2030s. In 2025, ČEZ also secured supplies of natural uranium from Kazakhstan's Kazatomprom, covering a substantial portion of Temelín's future needs for Westinghouse fuel. Together, these measures aim to establish a more secure and diversified nuclear fuel supply chain, though some options, such as Framatome's fuel fabrication in Germany under a Russian licence, remain subject to regulatory uncertainty.

Plans for new large reactors have been evolving for more than two decades, particularly at the Dukovany site. After a lengthy tender process, bids were submitted by Westinghouse, EDF, and Korea Hydro and Nuclear Power. In mid-2024, ČEZ selected KHNP's APR-1000 design as the preferred option, and the Czech government endorsed the choice. The project expanded from an initial plan for one unit to two units, with the possibility of additional reactors later, to reduce costs through scale. The announced price for the Dukovany expansion was roughly CZK 400bn for two units, with completion targeted for 2036 and 2037.

The decision, however, triggered a series of legal and political challenges. EDF and Westinghouse contested the tender process, raising concerns about intellectual property rights, alleged state subsidies, and fair competition. These disputes led to temporary injunctions and scrutiny by Czech courts and European institutions. Although initial complaints were dismissed and a US–Korean settlement resolved longstanding intellectual property conflicts, the European Commission requested a pause to examine potential foreign subsidies. After further legal proceedings, the Czech Supreme Administrative Court cleared the way in June 2025, allowing the EPC contract with KHNP to be signed. The government subsequently increased its stake in the Dukovany project company to 80%, signalling strong state backing and an ambition to secure a high level of domestic industrial participation.

ČEZ to work with Rolls Royce on new SMR capacity

Beyond large reactors, the Czech Republic is also positioning itself in the emerging field of small modular reactors. ČEZ plans to deploy up to 3 GW of SMR capacity by 2050, with the first unit envisaged at the Temelín site sometime in the mid-2030s. Progress accelerated in late 2024 when ČEZ entered a strategic partnership with Rolls-Royce SMR, acquiring a significant equity stake and advancing environmental assessments for potential sites. While timelines remain optimistic and dependent on regulatory and technological readiness, SMRs are presented as a flexible complement to large nuclear units.

In the broader power system, nuclear energy remains a stabilising force. In 2025, the Czech Republic generated about 71.4TWh of electricity, with fossil fuels still the largest contributor but in marked decline compared to five years earlier. Nuclear output remained stable, while solar generation grew significantly, albeit from a low base, and hydro and wind played smaller roles. Overall electricity production has fallen since 2019, balanced by reduced exports and consumption. Within this evolving landscape, nuclear power is set to remain a central pillar of Czech energy policy for decades to come.

Hungary

Hungary's nuclear sector is centred on a single site, the Paks nuclear power plant, where four Soviet-designed VVER-440 reactors remain in operation. In 2025, these units produced about 16TWh of electricity, supplying roughly 47% of the country's power generation. While output was essentially unchanged from the previous year, nuclear power's share declined slightly as other sources, particularly solar, expanded rapidly. Nuclear's contribution had peaked a decade earlier, in 2014, when it exceeded 50% of electricity generation.

Fuel diversification more aligned with EU efforts

Historically, Hungary has been fully dependent on Russia for nuclear fuel at Paks. This began to change in late 2023, when Parliament amended national nuclear policy to allow alternative fuel sourcing. A major step followed in October 2024, when the government signed a long-term fuel supply contract with Framatome, scheduled to take effect in 2027. This move aligns Hungary with broader European efforts to reduce reliance on Russian nuclear services, even as Russian fuel continues to be used in the short term.

The existing reactors, known collectively as Paks I, are currently licensed to operate for 50 years following earlier lifetime extensions. In December 2023, the government notified the European Commission of its intention to extend operations by another 20 years, potentially allowing the reactors to run into the 2050s. Planned investments of around EUR 1.5bn would be required to modernise electrical and control systems. Unlike earlier plans that envisaged shutting down Paks I once new reactors came online, the government now expects Paks I and the planned Paks II units to operate in parallel, raising environmental and technical concerns.

One particularly sensitive issue is thermal pollution of the Danube River, which provides cooling water for the plant. A regulatory limit of 30°C is intended to protect aquatic ecosystems, but Paks I has already faced output reductions during hot periods and has reportedly exceeded this threshold in the past. In 2024, the government introduced emergency rules allowing continued full-power operation above the limit if energy security is deemed at risk, and in 2025 an advisory body recommended raising the limit by 2°C. How this framework would function once Paks II is operating, effectively doubling nuclear capacity at the site, remains unclear, especially as alternatives such as cooling towers are considered prohibitively expensive.

Rosatom to supply two reactors. No public tender submitted by Rosatom

Plans for Paks II have been under discussion for more than 15 years. After Parliament approved expansion in principle in 2009, the project took a decisive turn in 2014 when Hungary awarded a contract without a public tender to Russia's Rosatom to build two VVER-1200 reactors, financed largely by a Russian state loan. The project has since been mired in legal, financial, and political controversy, including challenges over state aid and procurement. While the European Commission approved the project, Austria challenged the decision in court, and in early 2025 the Advocate General of the EU Court of Justice recommended overturning a prior ruling that had upheld the aid approval. A final judgment is still pending.

Continuation of Paks II despite sanctions

Despite EU sanctions on Russia, Hungary has secured exemptions to keep Paks II moving forward. Key milestones were reached in 2024 and 2025, including delivery of major components, approval of the preliminary safety analysis report, and partial authorisation to resume construction after a structural incident at the site. Uncertainty persists over the supply of control systems, particularly the role of Siemens Energy, which has faced export licensing hurdles in Germany but is now seeking to deliver equipment through its Hungarian operations.

Alongside nuclear, Hungary has experienced a dramatic surge in solar power. Installed solar capacity grew from negligible levels a decade ago to 8.4GW by 2025, enabling solar to supply about a quarter of Hungary's electricity, one of the highest shares globally. This rapid growth has reshaped the power mix and reduced nuclear's relative dominance. Hungary's energy strategy envisions continued nuclear operation combined with expanding renewables, though grid constraints and unresolved issues around Paks II mean the long-term balance between these sources remains uncertain.

Romania

Romania's nuclear power sector is centred on a single site at Cernavodă, where two Canadian-designed CANDU heavy-water reactors with a combined capacity of about 1.3GW are in operation. In 2024, these units produced electricity of roughly 10.4TWh net, corresponding to c. 20% of

national generation, slightly below the previous year and just under the historical peak reached in 2020. Nuclear power thus remains a major pillar of Romania's energy system, though it now operates alongside a rapidly changing mix dominated by renewables.

Nuclear is second-largest source electricity generation

Hydropower remains Romania's leading electricity source, supplying more than a quarter of total generation. Nuclear followed as the second-largest contributor, ahead of natural gas and coal. Among non-hydro renewables, wind power played the most significant role, while solar photovoltaic generation, though still smaller, grew rapidly. The solar share nearly doubled between 2023 and 2024 as more than 1.7GW of new capacity was added, and further large additions are expected in the coming years. Romania is also rolling out battery storage at scale, targeting up to 5GW by the end of 2026 to improve system flexibility. Romania's updated National Energy and Climate Plan sets out an ambitious trajectory towards a low-carbon electricity system. By 2030, renewables are expected to account for three-quarters of electricity generation, supported by major expansions of solar, wind and hydro capacity, and a complete phase-out of coal by 2032. Within this framework, nuclear energy is also set to expand. The plan assumes continued operation and modest uprating of the existing Cernavodă reactors, the deployment of SMRs, and the construction of two additional large reactors at the same site, which together would raise total nuclear capacity to more than 3GW by the mid-2030s.

SMR strategy proceeding

Romania's SMR programme has attracted international attention as a potential first-of-a-kind deployment in Europe. The project, based on NuScale's 77MW modules, targets a former coal plant site at Doicești and envisions six modules delivering a total of 462MW. Preparatory studies and licensing work have been ongoing for several years, supported by US government institutions. In 2024 and 2025, design and engineering activities advanced further, and financing discussions continued. However, a final investment decision has not yet been taken, and timelines have gradually slipped. While official statements still point to first operation around the end of the decade, many uncertainties remain, including the fact that NuScale has not yet built a reactor anywhere and that design licensing in Romania is still incomplete.

At Cernavodă itself, maintaining the existing nuclear fleet is a major priority. Like other aging CANDU reactors worldwide, Unit 1 requires extensive refurbishment to allow operation beyond its original 30-year design life. After years of preparation, Romania signed a major engineering, procurement, and construction contract in late 2024 with an international consortium to carry out the refurbishment. The project, estimated to cost several billion euros in total, is scheduled to involve a multi-year outage from 2027 onward, after which the unit would return to service for up to another 30 years.

Meanwhile, long-running plans to add new large reactors at Cernavodă have regained momentum. Preliminary contracts for managing construction of Units 3 and 4 were signed in late 2024, supported by substantial international financing commitments and state guarantees. Although political leaders have reiterated ambitions to bring both units online in the early 2030s, the long construction schedules and unresolved financial and contractual steps suggest that achieving these dates will be challenging. Overall, Romania is pursuing an expansive nuclear agenda alongside fast-growing renewables, but the pace of implementation remains uncertain across all major projects.

Nuclear power makes up 61% of Slovakian power generation

Slovakia

Slovakia remains one of the most nuclear-dependent countries in the world, with nuclear power providing well over half of its electricity for more than two decades. This reliance deepened further after the commissioning of the third reactor at the Mochovce nuclear power plant, which helped lift the nuclear share of electricity generation to more than 61% in both 2023 and 2024. Nuclear output remained broadly stable in 2024, reinforcing Slovakia's position as the country with the second-highest nuclear share globally, behind only France.

Electricity generation from nuclear power is concentrated in two sites operated by Slovenské elektrárne. The utility is now jointly owned by the Czech-based EPH group, holding a two-thirds stake, and the Slovak state, which owns the remaining third. At Jaslovské Bohunice, two VVER-440 reactors that entered service in the mid-1980s continue to operate after extensive modernisation and capacity uprates completed over the past two decades. At Mochovce, three reactors of similar design are connected to the grid, providing a combined national nuclear capacity of around 2.3GW, while a fourth unit is in the final stages of testing ahead of commissioning.

The long and costly Mochovce expansion has shaped Slovakia's nuclear trajectory. Construction of Mochovce-3 and -4 began in the 1980s, was suspended in the early 1990s, and resumed only after Italy's Enel acquired a controlling stake in Slovenské elektrárne in 2006. Mochovce-3 eventually entered commercial operation in 2023, more than a decade later than previously planned, after years of delays and cost overruns that drove total project costs to more than double initial estimates. Mochovce-4 progressed through final testing phases in 2024-25 with a delayed start until 2026, with the utility emphasising that completing this unit remains its most urgent short-term priority.

A revival of new nuclear plans

Looking ahead, Slovakia plans to operate its existing reactors well into mid-century. Bohunice Units 3 and 4 are expected to operate until the mid-2040s, while Mochovce Units 1 and 2 could remain online until close to 2060. At the same time, the government has revived plans for new nuclear capacity at Bohunice. A state-owned project envisions a large, roughly 1200MW reactor, with construction potentially starting in the

early 2030s and commissioning towards the end of the decade or later. Cost estimates have risen sharply, and the procurement process remains politically contentious, with opposition parties criticising the lack of transparency and warning against sidelining competitive tendering.

Alongside large reactors, Slovakia is also exploring small and advanced modular reactors. Several feasibility studies are under way, backed by US government grants, to assess potential sites and technologies that could repurpose former coal plants. Partnerships with Western vendors such as Westinghouse, EDF, and new entrants developing advanced reactor designs highlight Slovakia's ambition to position itself as a future hub for modular nuclear technologies, although timelines remain vague and no final investment decisions have been taken.

Despite these ambitions, Slovakia's energy policy is marked by contradictions. The country remains heavily dependent on Russian energy, including nuclear fuel, gas and oil, even as it pursues diversification agreements with Western suppliers starting later this decade. Political choices under the current government have reinforced these dependencies, complicating efforts to meet European Union energy and climate objectives. Meanwhile, renewables play only a minor role in Slovakia's power mix, with hydropower accounting for a modest share and solar and wind lagging well behind EU averages. This imbalance underscores how decisively Slovakia continues to place nuclear energy at the centre of its electricity system, even as broader decarbonisation challenges remain unresolved.

Nuclear power output corresponds to 34% of Slovenian power generation

Slovenia

Slovenia's electricity system is built around a relatively balanced mix of nuclear power, hydropower and fossil fuels, though nuclear energy remains a central pillar. In 2024, nuclear power accounted for about 34% of electricity generation, hydropower for roughly 31%, and coal for just over 20%, while solar photovoltaics continued to grow, reaching nearly 9%. Although fossil fuels still represented almost a quarter of the power mix, their role is expected to diminish significantly as Slovenia moves toward its long-term climate goals.

The single operational nuclear plant has seen lifetime extension

The country operates a single nuclear power plant at Krško, jointly owned with Croatia. The 688MW pressurised water reactor, built by Westinghouse, has been in operation since 1983 and supplies electricity equally to both countries. In 2024, Krško generated around 5.8TWh of electricity, slightly increasing its absolute output compared to the previous year but accounting for a marginally smaller share of total generation due to growth in other sources. After refurbishment and safety upgrades, the plant's operating license was extended to 60 years, allowing operation until 2043. International safety reviews by the IAEA have supported

long-term operation, while also recommending further improvements in aging management and documentation of critical systems.

Alongside extending the life of Krško, Slovenia is pursuing plans for a second nuclear unit at the same site, known as JEK2. Discussions about a new reactor date back almost two decades, with project concepts evolving from multiple large units to a single reactor of flexible size. In recent years, nuclear newbuild has been framed as a key tool for decarbonising the power sector and replacing coal, which Slovenia plans to phase out completely by 2033. Updated energy and climate strategies call for a major increase in renewable energy, combined with a decision on new nuclear capacity by the late 2020s.

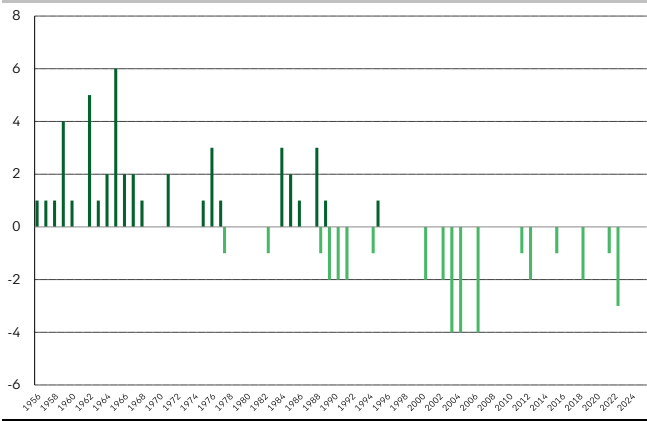
Political debate around JEK2 has been intense. While Parliament adopted a resolution in 2024 emphasising nuclear energy’s long-term role, a planned consultative referendum was later cancelled due to legal concerns and calls for clearer public information. The government has committed to holding a referendum before any final investment decision, likely before 2028. Meanwhile, feasibility studies are underway with EDF and Westinghouse as the remaining potential technology suppliers, after a Korean vendor withdrew.

Looking ahead, Slovenia aims to reach climate neutrality by 2050, with renewables supplying most of the electricity by 2030. Nuclear power is expected to complement this transition by providing stable, low-carbon generation, even as costs, public acceptance, and long timelines continue to shape the country’s energy debate.

United Kingdom

Since the election of a new government in the summer of 2024, the United Kingdom has reshaped its energy and climate policy framework with the stated aim of improving energy security while accelerating decarbonisation. Central to this approach has been a renewed political commitment to nuclear power. The government has made additional public funds available for the construction of two large reactors at Sizewell C, for the development of SMRs, and for nuclear fusion research. This renewed backing comes despite the continuation of long-standing problems in the sector, notably chronic cost overruns, prolonged construction delays, and mounting expenses associated with maintaining an aging reactor fleet.

UK reactor units’ startups and closures

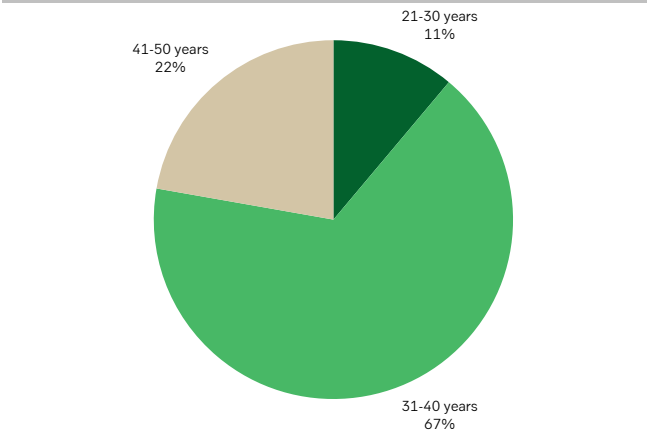


Source: WNISR

Aging British nuclear fleet

By mid-2025, the UK operated only nine nuclear reactors, with a combined capacity of roughly 5.8GW. All but one of these units are Advanced Gas-Cooled Reactors (AGRs), a design unique to Britain, built in the 1980s. The sole pressurised water reactor, Sizewell B, entered service in 1995. The average age of the operating fleet exceeded 38 years, highlighting the maturity of existing assets. Since 2018, closures have steadily reduced capacity, including the shutdown of Dungeness B, Hunterston B, and Hinkley Point B. Altogether, 36 reactors have been closed, leaving nearly 8GW in various stages of decommissioning.

Age of UK nuclear fleet



Source: IAEA-PRIS (2025)

Nuclear generation in 2024 amounted to around 40TWh, providing just over 14% of electricity supply, half the contribution of wind and far below nuclear’s historical peak share of 28% in the late 1990s. Eight AGR units continue to operate at Heysham, Hartlepool and Torness, while EDF, which owns the fleet, has pushed for limited lifetime extensions. In late 2024, EDF announced plans to operate several AGRs into the late 2020s and early 2030s, though it acknowledged that uncertainty over graphite core degradation makes future output unpredictable. Hartlepool was placed under enhanced regulatory scrutiny in 2025, reflecting concerns about safety management and aging infrastructure. Sizewell B is licensed

until 2035, and EDF is considering a potential extension to 2055, contingent on cost certainty and regulatory approval.

At the same time, the wider UK electricity system has been transformed by renewables. In 2024, wind generation reached record highs, briefly supplying more than two-thirds of national demand and producing over 80TWh across the year, almost matching gas-fired generation and double the output from nuclear. 2025 was the first coal-free year for UK power generation, and power-sector carbon intensity reached a historic low. These developments reinforced the view that renewables, rather than nuclear, are driving near-term decarbonisation.

Government targets clean energy including nuclear power

The new Labour government, elected in mid-2024, placed energy reform at the heart of its agenda, pledging to make the UK a “clean energy superpower” and to deliver a near-zero-carbon electricity system by 2030. It established a publicly owned company, Great British Energy, capitalised with more than GBP 8bn, to help finance clean energy projects. Although initially envisioned as a vehicle for renewables, it has also become a conduit for nuclear investment. Planning rules were altered to speed up grid connections, lift restrictions on onshore wind, and open new sites for nuclear development.

Nuclear newbuild remains dominated by two EPR projects. Hinkley Point C, under construction since 2016, has become emblematic of nuclear’s difficulties in Britain. Costs have escalated from an initial estimate of GBP 18bn to well over GBP 40bn in today’s money, while completion has slipped from 2025 to the early 2030s. A long-term contract guaranteeing inflation-linked prices for 35 years has further undermined the project’s economic appeal relative to rapidly falling renewable costs.

Delays and cost overruns have made reactors more costly than budgeted

Sizewell C, conceived as a replica of Hinkley Point C, has also faced mounting cost expectations. Although originally promoted as cheaper thanks to design reuse and a new financing model, reports in early 2025 suggested total costs approaching GBP 40bn. The government has gradually taken an increasingly dominant financial role, committing tens of billions of pounds through equity and subsidies. A final investment decision was taken in July 2025, confirming strong state backing but leaving taxpayers exposed to further overruns.

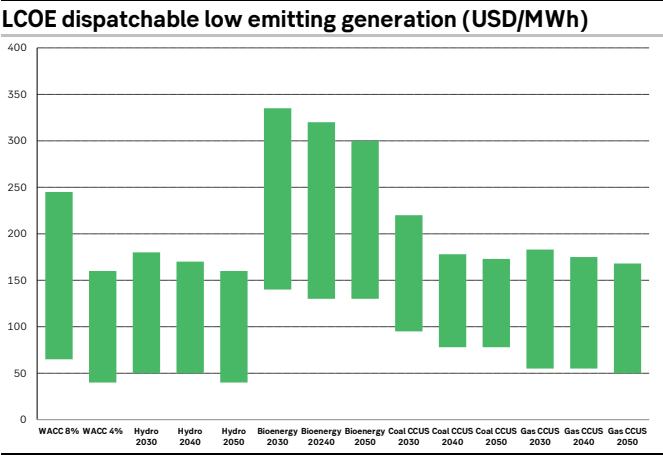
Alongside large reactors, SMRs are being promoted as a future option. Rolls-Royce’s 470MW design has received the bulk of public support and is advancing through the regulatory approval process, though it remains unproven and dependent on export markets to justify investment. Additional SMR designs from GE-Hitachi and Holtec are also under review. The 2025 spending review allocated GBP 2.5bn for SMR deployment and a further GBP 2.5bn for fusion research, underlining the scale of public commitment.

Overall, while 2024-2025 marked a politically favourable period for the nuclear industry in the UK, the sector remains defined by aging assets, expensive newbuild projects, and competition from rapidly expanding renewables. Even with unprecedented levels of public funding, nuclear power is expected to play a diminishing role in Britain’s electricity mix over the next decade, as wind and solar become the dominant sources in a largely decarbonised power system.

Our summary of nuclear power today and tomorrow in Europe

Global nuclear pathways reflect politics and institutions at least as much as technology or cost curves. France, Germany and China remain archetypes: France demonstrates how a centralised, standardised programme can deliver ultra-low-carbon power at scale, but also how aging fleets, supply-chain erosion, and megaproject risk can accumulate without steady reinvestment and operational renewal. Germany shows that social license can trump climate arithmetic: an orderly nuclear exit paired with rapid renewables growth drove decarbonisation, but at the expense of near-term system resilience, higher consumer costs, and dependence on fossil backstops during shocks. China underscores what coordinated state planning, long-term finance, and domestic industrialisation can achieve: fast build-out, integration with the national climate strategy and a pipeline of advanced designs, albeit in a governance context that is hard to replicate in liberalised markets.

The Nordic power system’s deepening renewables share (wind c. 20%, hydro c. 50%) is stressing price formation and capture, with record negative-price hours and very low capture in Finland. That’s pushing a pragmatic rethink: nuclear is being re-evaluated as firm, zero-emission capacity to stabilise a system dominated by variable generation. Sweden’s proposed state-aid framework (state loans, CfD-like guarantees, up to 5GW) is the boldest European financing innovation outside the UK’s RAB model, explicitly targeting investor risk compression and long-run price stability. Vattenfall’s Ringhals SMR shortlist (GE Vernova, Rolls-Royce SMR) is the region’s bellwether, though 2035 COD looks optimistic given global first-wave SMR uncertainties (licensing, cost discovery, supply chain). Finland’s direction of travel is clear: lifetime extensions and fuel diversification (Westinghouse/Enusa) to harden energy security post-Russia, even as OL3’s teething issues show reliability risk isn’t only an “old fleet” problem. Norway and Denmark are kicking the tires on nuclear but lack frameworks and near-term bankable projects. Net: Nordic nuclear volumes are likely flat to 2050 in baseline cases, but upside exists if SMR programmes clear cost/permits and if aging assets are credibly extended.



The EU27’s fleet’s average age (c. 39 years) and the closure overhang (36 retirements in 20 years) highlight a replacement gap that incremental life-extension alone won’t fully cover. New starts have been sparse since the 1990s; after Mochovce 4 (expected c. 2025/26), Europe could face a multi-year lull. The Commission’s Nuclear Illustrative Programme projects modest capacity growth to c. 109GW by 2050, contingent on life-extensions and selected new build, while acknowledging industrial bottlenecks (skills, manufacturing, QA culture). This is why national financing models are converging on heavier state roles: the UK’s RAB (socialising construction risk), France’s preferential sovereign loans plus EPR2 standardisation, Poland’s 60-year CfD with state guarantees, Czechia’s long-tenor price support and protection clauses, and Sweden’s loans/CfD-like toolkit. The design logic is consistent: compress WACC and revenue risk while leaving performance and delivery risk with developers; add clawbacks, transparency, and time limits to mitigate overcompensation.

Design element

Design element	Purpose / rationale	Typical approaches / variations
Revenue / price support mechanism	To reduce market / merchant risk by guaranteeing a revenue floor or adjusting payments	Contracts for Difference (CfDs, either one-way or two-way), guaranteed minimum price, power price guarantees, or top-up payments on PPAs
Risk sharing / liability allocation	To ensure the investor retains some risk and avoid overcompensation	The state may provide loans, equity injections, guarantees, but still require the investor to bear construction risk, cost overruns, performance risk, etc. Also, arrangements on nuclear waste liability (state vs operator)
Eligibility criteria & conditionality	To ensure that only projects meeting policy objectives are supported	Criteria on emissions, safety standards, technology maturity, project readiness, cost-efficiency, local content or supply chain requirements
Proportionality / minimal necessary support	To limit distortion of competition	The aid must be no more than necessary to make the investment viable; the design often involves benchmarking, claw-back clauses, or semi-competitive procedures
Time limits / durations	To limit the exposure	Support limited to defined periods (e.g. until plant is operational or for some years thereafter)
Transparency & monitoring	To allow oversight and evaluation	Reporting obligations, audits, periodic reviews, performance monitoring
Coordination with EU / state aid control	To ensure compliance with EU law	Prior notification to the Commission, assessment of compatibility with state aid rules, use of block exemptions or special guidelines
Support for supply chain and infrastructure	To strengthen industrial base	Aid may also cover manufacturing of components, R&D, plant standardisation, modular construction facilities, enabling infrastructure

Source: World Nuclear News, IAEA, SEB

For decision-makers, this means addressing the following:

Put financing at the centre

Nuclear projects rise or fall on the cost of capital just as much as on engineering performance. Making them bankable requires predictable revenue frameworks, such as RAB models, long-term CfDs, or state-backed loans, combined with strong counterparties and proven, standardised designs.

Shift from bespoke builds to repeatable programmes

One-off mega-projects are slow, risky and expensive. The path to lower costs and shorter schedules lies in standardised designs, modular construction, and a build-programme mindset that allows supply chains and workforces to develop experience through repetition.

Plan for integration with renewable-heavy power systems

As wind and solar dominate generation, nuclear’s value increasingly comes from providing firm, low-marginal-cost output that stabilises prices and grid frequency. This requires co-optimising nuclear with storage, interconnectors, and demand-side flexibility rather than treating it as a standalone asset.

Strengthen resilience across the fuel cycle

Diversifying enrichment and fuel fabrication and qualifying non-Russian alternatives for legacy VVER and CANDU reactors is now a strategic necessity. A more resilient supply chain reduces geopolitical exposure and ensures long-term operational continuity.

Build and maintain public trust

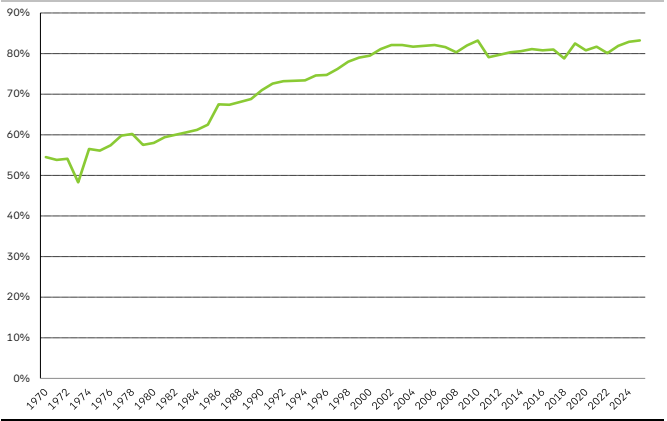
Nuclear programmes succeed only when they carry strong social license. Transparent safety oversight, credible waste-management strategies, and realistic project schedules are essential to avoid political reversals and maintain long-term support.

Better maintenance

If nothing else, the current fleet in use could increase its availability, particularly in Western Europe, some of which is related to France’s low but increasing nuclear capacity availability.

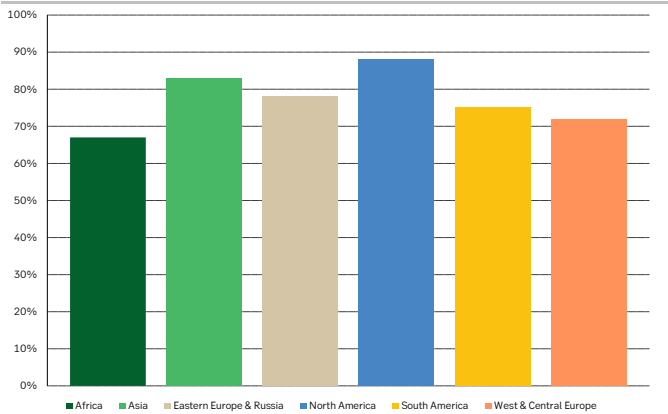
The same can be said for the Swedish reactors, which are often comparable to the Finnish reactors but have had a lower capacity factor than the latter. An overall increased capacity factor would be a comparatively low-cost strategy to add base load power to the energy system, where many of the reactors are already depreciated.

Global average capacity factor for nuclear (%)



Source: IAEA

Nuclear capacity utilisation in % by region in 2024



Source: IAEA

Concluding remarks

In our view, the nuclear power sector will be in the spotlight going forward, and markedly different from that of the last two decades. The drivers are energy security concerns sharpened by Russia's invasion of Ukraine, the need to decarbonise power systems on credible timelines, and structurally rising electricity demand from electrification and the expansion of data centres. What's different in this cycle is the emphasis on delivery certainty, regulatory clarity, and financing architecture, rather than on technology promises alone.

In the near term, the centre of gravity is life extension, not new-build capacity. Europe's reactor fleet is old by global standards, with many assets operating into their fourth and fifth decades. Extending their lifetimes is typically the cheapest and quickest form of carbon abatement available to many systems. It is also the most attractive from a credit and investor standpoint: these are known assets with established operating teams, low incremental capex, and immediate revenue continuity. The French "Grand Carénage" programme has become a reference point for comprehensive refurbishment at scale, and a growing list of countries are following with similar measures. The risks are already known: Outage duration, scope creep in maintenance, and materials surprises, as France's stress-corrosion episode reminded markets. Even so, the LCOE for life-extended units generally sits below market clearing prices in a fossil-importing Europe and below most new-build alternatives.

New nuclear construction in Europe is re-emerging, but its viability now depends far more on delivery certainty than on technological claims. Past projects like the early EPRs and AP1000s showed that delays are the biggest economic threat, as each missed month raises financing costs and strains balance sheets. As a result, the most credible new-build projects rely on risk-sharing frameworks such as the UK's combination of CfDs and the RAB model or Sweden's state-supported financing structures, which help stabilise revenues and shift part of the construction risk away from utilities. SMRs could become important in the next decade, yet they require proven schedules, replicable licensing, and secure fuel supply before they can be modelled confidently. Their success will likely hinge on long-term PPAs with strong counterparties and continued public support to reduce funding costs. Meanwhile, evaluating nuclear solely through LCOE comparisons with renewables is increasingly misleading. As renewable-heavy systems face negative prices and grid constraints, nuclear's reliable, high-capacity output offers system value that regulated regimes like CfDs or RABs can monetise, enabling stable, bond-like cash flows even when headline costs appear higher.

In terms of technology choice, we believe that small modular reactors (SMRs) are designed to complement intermittent renewables by offering flexible, rapidly deployable and reliable power. The ability to ramp output up or down helps stabilise the grid when wind or solar generation fluctuates. Due to modular construction, if successfully realised, SMRs can be installed quickly, integrated into hybrid systems, and even replace retiring fossil-fuel plants using existing infrastructure.

Although Europe's nuclear strategies are evolving in different ways, all generally point towards renewed support for the sector. France is pushing ahead with its EPR2 programme and regulatory simplification, though its short-term financial performance depends heavily on how well its aging fleet holds up during refurbishment cycles. The UK is reinforcing its commitment through new funding for Sizewell C and a structured SMR roadmap, even as it phases out older reactors. In Central and Eastern Europe, Czechia, Slovakia, Hungary, Romania and particularly Poland, nuclear is seen as essential for replacing coal and strengthening energy security, supported by state credit, export-credit agencies, and long-term off-take agreements. The Nordics present another blend: Sweden is preparing for several gigawatts of new capacity and potential early SMR deployment at Ringhals, while Finland highlights the stabilising effect of OL3. Switzerland, meanwhile, continues to debate whether to lift its long-standing ban on new builds. Alongside these national strategies, fuel diversification, credible waste solutions, stronger decommissioning funds, rebuilt supply chains, and standardised construction approaches are becoming central to ensuring nuclear's long-term viability, with life extensions and a few well-structured new builds offering the strongest near-term prospects.

The ambition to revive the fortunes of the nuclear sector throughout Europe is not a false dawn, in our view, nor do we think it is a repeat of the golden era enjoyed 40-50 years ago. We believe that Europe's nuclear future is unlikely to be a single renaissance. Rather, it will be a patchwork of life-extensions, a few standardised large units, and, if the technology is successful and supply chains can meet requirements, also SMR deployments. The winning programmes will look more like infrastructure finance plus industrial policy than pure merchant power. With an aging reactor fleet, decarbonisation in parallel to electrification lifetime extensions provide an injection into the sector, as will expanded recruitment and training efforts, but if policy implementations and supportive funding conditions do not materialise, the comeback may fizzle out. The markets with dedicated nuclear sector programmes, existing and future, will in our view be most successful in growing their nuclear capacity. The longevity of the programmes requires political staying power, while the initiation of the same require political unity to avoid "stop-go", increasing uncertainties, causing delays and driving up costs, rendering projects unviable.

This report has been compiled by SEB Large Corporates & Financial Institutions, a division within Skandinaviska Enskilda Banken AB (publ) ("SEB") to provide background information only.

Opinions, projections and estimates contained in this report represent the author's present opinion and are subject to change without notice. Although information contained in this report has been compiled in good faith from sources believed to be reliable, no representation or warranty, expressed or implied, is made with respect to its correctness, completeness or accuracy of the contents, and the information is not to be relied upon as authoritative. To the extent permitted by law, SEB accepts no liability whatsoever for any direct or consequential loss arising from use of this document or its contents.

The analysis and valuations, projections and forecasts contained in this report are based on a number of assumptions and estimates and are subject to contingencies and uncertainties; different assumptions could result in materially different results. The inclusion of any such valuations, projections and forecasts in this report should not be regarded as a representation or warranty by or on behalf of the SEB Group or any person or entity within the SEB Group that such valuations, projections and forecasts or their underlying assumptions and estimates will be met or realised. Past performance is not a reliable indicator of future performance. Foreign currency rates of exchange may adversely affect the value, price or income of any security or related investment mentioned in this report. Anyone considering taking actions based upon the content of this document is urged to base investment decisions upon such investigations as they deem necessary.

In the UK, this report is directed at and is for distribution only to (I) persons who have professional experience in matters relating to investments falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (The "Order") or (II) high net worth entities falling within Article 49(2)(a) to (d) of the Order (all such persons together being referred to as "relevant persons"). This report must not be acted on or relied upon by persons in the UK who are not relevant persons. In the US, this report is distributed solely to persons who qualify as "major U.S. institutional investors" as defined in Rule 15a-6 under the Securities Exchange Act. U.S. persons wishing to effect transactions in any security discussed herein should do so by contacting SEBEI.

The distribution of this document may be restricted in certain jurisdictions by law, and persons into whose possession this documents comes should inform themselves about, and observe, any such restrictions.

This document is confidential to the recipient, any dissemination, distribution, copying, or other use of this communication is strictly prohibited.

Skandinaviska Enskilda Banken AB (publ) is incorporated in Sweden, as a Limited Liability Company. It is regulated by Finansinspektionen, and by the local financial regulators in each of the jurisdictions in which it has branches or subsidiaries, including in the UK, by the Financial Services Authority; Denmark by Finanstilsynet; Finland by Finanssivalvonta; and Germany by Bundesanstalt für Finanzdienstleistungsaufsicht. In Norway, SEB Enskilda AS ('ESO') is regulated by Finanstilsynet. In the US, SEB Securities Inc ('SEBEI') is a U.S. broker-dealer, registered with the Financial Industry Regulatory Authority (FINRA). SEBEI and ESO are direct subsidiaries of SEB.